HW2: The Economics and Geography of Grocery Shopping

Joseph Matatyaou and Lloyd Todaro

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0. Introduction

This report looks into how grocery prices, grocery basket composition, store availability, household resources, and transportation affect the cost and accessibility of grocery shopping in a simulated scenario. This uses nine substantive CSV datasets and an accompanying data dictionary.

All substantive numerical observations are simulated. Store banners and ownership groups are fictional, and the constructed tract identifiers do not represent official Census tracts. Real California county names and FIPS codes serve as geographic labels only.

The price observations represent April 15, 2024 and April 15, 2025. County context and tract measures represent a simulated 2024 profile, where the alternative food environment indicators represent 2022. The study store roster is undated, and travel values describe hypothetical shopping alternatives. These different reference periods and definitions will be distinguished in the analysis.

Original CSV files will be retained and unchanged. Comparisons will use compatible measurement units, appropriate observational units, and explicit eligibility rules and denominators. Missing prices will not be treated as zero, and basket costs will be compared only when every required product has an observed price at the same store and date.

1. What do we actually know, and what is the unit of each dataset?

Dataset Summary

The nine substantive files describe different observational units. The data dictionary provides metadata and is not a substantive dataset. All numerical observations below are simulated data.

Table 1: Dimensions of the nine original substantive datasets.

Dataset	Rows	Columns
01 Stores	145	10
02 Banners	12	4
03 Products	40	5
04 Prices	8980	8
05 Basket items	56	4
06 County context	24	9
07 Food environment	24	6

Table 1: Dimensions of the nine original substantive datasets. *(continued)*

Dataset	Rows	Columns
08 Tract access	134	7
09 Travel	670	7

Now the units, geographic levels, and references period for each are:

- Stores: One physical store per row, identified by `store_id`. Each store has a county location. The roster is undated and the opening year is a store characteristic, not the roster's observation date.
- Banners: One banner per row, identified by `banner_id`. This is a business level table with no geographic unit or reference date. A banner may be used by multiple stores.
- Products: One generic comparison product per row, identified by `product_id`. This table is not geographic and has no reference date.
- Prices: One store-product-date observation per row. The substantive key is `store_id`, `product_id`, and `observation_date`. `observation_id` is also a candidate row key. Geography is obtained through the store. The two dates are April 15, 2024 and April 15, 2025.
- Basket items: One required product within a basket per row, identified jointly by `basket_id` and `product_id`. These are fixed scenario definitions with no geographic level or date.
- County context: One county per row, identified by `county_fips`. The demographic and socioeconomic measures represent a simulated 2024 profile.
- Food environment: One county per row, identified by `county_fips_raw`, which is padded to make `county_fips` in the working table. Its indicators and population denominator refer to 2022.
- Tract access: One constructed tract per row, identified by `tract_id`. The first five characters identify its county. Its population and access measures refer to 2024.
- Travel: One recorded tract-store alternative per row, identified jointly by `tract_id` and `store_id`. Its geographic unit is a link between a tract and a store, potentially across county boundaries. Travel values are hypothetical scenario inputs without an explicit observation date.

Key cardinality check

A key must identify one row at the intended observational level. For a composite key, uniqueness applies to the combination of fields, not necessarily to each field separately.

Table 2: Audit of the proposed keys listed above.

Dataset	Rows	Distinct	Repeated	Missing
01 Stores	145	145	0	0
02 Banners	12	12	0	0
03 Products	40	40	0	0
04 Prices	8980	8980	0	0
05 Basket items	56	56	0	0
06 County context	24	24	0	0

Table 2: Audit of the proposed keys listed above. *(continued)*

Dataset	Rows	Distinct	Repeated	Missing
07 Food environment	24	24	0	0
08 Tract access	134	134	0	0
09 Travel	670	670	0	0

Here, Distinct counts different key combinations, Repeated counts rows beyond the first occurrence of a key, and Missing counts rows missing at least one key component. Each proposed key has as many distinct values or combinations as rows, with no repeated or missing keys.

This supports the proposed observational units. This does not prove that every possible join is appropriate.

Missing values and incomplete coverage

Table 3: Missingness in selected fields (percentages use rows in each file).

Measure	Rows	Missing	Percent
Shelf price	8980	0	0.0
Package quantity	8980	0	0.0
County population	24	0	0.0
2022 low-access share	24	0	0.0
Tract low-access count	134	0	0.0
Bus round-trip time	670	178	26.6
Bus round-trip fare	670	178	26.6

The only fields with missing entries are bus round-trip time and fare. Each is missing for 178 of the 670 travel alternatives, or 26.6%. All other fields here in the nine substantive files have no missing entries.

The missing bus values occur on the 178 alternatives without bus service. Among the 492 alternatives with service, 91 have zero fare. These zero fares represent available free fare trips.

No missing entries in the price table does not mean complete price coverage. An unrecorded store product date combination has no row at all. Its price is unknown and has neither a zero price nor proof that the product was unavailable.

How the tables connect

This table describes the intended relationships. Many to one means that multiple rows in the starting table may refer to one row in the lookup table.

Table 4: Intended relationships between datasets.

From	To	Key	Relationship
Prices	Stores	store_id	Many-to-one
Prices	Products	product_id	Many-to-one
Stores	Banners	banner_id	Many-to-one
Stores	County context	county_fips	Many-to-one
Basket items	Products	product_id	Many-to-one
Food environment	County context	county_fips	One-to-one
Tracts	County context	county_fips	Many-to-one
Travel	Tracts	tract_id	Many-to-one
Travel	Stores	store_id	Many-to-one

For a concrete depiction within this scenario, price observation `PR0000001` records product `P01` at store `S0001` on April 15, 2024. Matching `store_id` connects the observation to the store Valley Basket - Yolo 1. That store record supplies banner VAL, simulated owner unit OWN-VALLEY, and raw county code 6113.

Matching banner VAL to the banner table supplies the banner name Valley Basket and parent-group label Valley Retail Holdings. Padding 6113 to 06113 connects the store to Yolo's simulated 2024 county context. The owner unit ID comes from the store table where a parent group label is not always an individual owner. Here, "Independent operators" groups unrelated owners.

The route from a tract to a candidate store is different. For example, constructed tract `06113000001` is linked to store `S0001` in the travel table. Matching the tract ID supplied tract characteristics, while matching the store ID supplies store characteristics. The recorded pair supplied the travel assumptions. Sharing a county code alone does not create a candidate shopping trip.

Why the files cannot be merged automatically

CSV is a file format, not a shared observational unit. Joining stores to tracts only by county would create every within county store tract combination, rather than identify recorded shopping alternatives.

This would produce 830 combinations, compared with 670 recorded travel alternatives. Therefore introducing unrecorded within county pairs and omit cross county alternatives. Travel analysis should use the recorded tract store links and county analysis should aggregate stores and tracts before joining their summaries.

2. What does a grocery price mean when package sizes differ?

A shelf price measures the cost of an entire package. Let P_{sjt} denote the advertised package price for product j at store s on date t, and let Q^*_{sjt} denote its package quantity converted to the product's comparison base unit. The standardized unit price is

$$U_{sjt} = \frac{P_{sjt}}{Q^*_{sjt}}$$

Depending on the product, U_{sjt} is measured in dollars per ounce, fluid ounce, or item. Comparison will use the same product and compatible units

Standardizing quantities and prices

All 8,980 recorded prices have positive quantities and prices and compatible package and comparison units. The joins preserve the original number of observations. Now, a 2 pound package is 32 ounces, while a 1 liter package is 33.8140227 fl oz. Mass, volume, and count prices are separate.

Shelf price summaries by date

These summaries have all recorded prices on each date, with promotions. The unit of analysis is a store product date observation.

Table 5: Coverage and promotional prevalence by date.

Date	Observations	Products	Stores	Promotion (%)
2024-04-15	4392	40	145	8.70
2025-04-15	4588	40	145	9.74

Table 6: Shelf price summaries, in dollars per package.

Date	Mean	Median	SD	Q1	Q3	IQR	P90
2024-04-15	5.39	4.31	3.83	2.76	6.76	4.00	10.30
2025-04-15	5.63	4.46	4.07	2.84	7.17	4.33	10.89

There are 4,392 observations in 2024 and 4,588 in 2025, covering 40 products and 145 stores on each date. Mean package prices are $5.39 and $5.63 respectively. In both years, the mean exceeds the medians indicating a right skewed price distribution. The difference between these means is not an estimate of price inflation since the observed product coverage and package sizes can differ between dates.

Package size and standardized prices for two products

Here white rice is selected (P01, oz) and whole milk (P11, fl. oz). Each comparison includes all recorded prices for that product on the specified date.

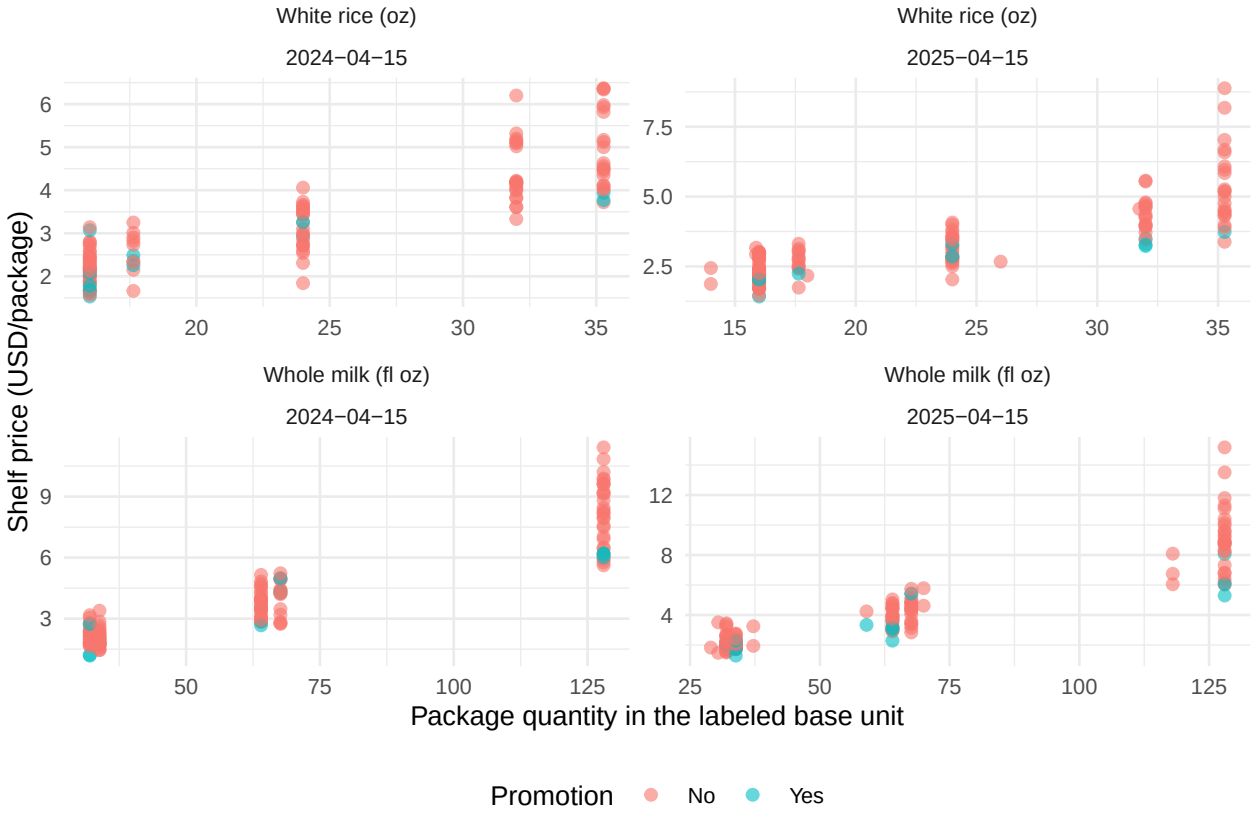
Table 7: Product specific unit prices (USD/base unit).

Product	Date	Base unit	n	Mean	Median	IQR
White rice	2024-04-15	oz	121	0.136	0.132	0.033
White rice	2025-04-15	oz	126	0.142	0.138	0.040
Whole milk	2024-04-15	fl_oz	128	0.062	0.061	0.020
Whole milk	2025-04-15	fl_oz	132	0.066	0.064	0.018

The median rice price is about \$0.132/oz in 2024 and \$0.138/oz in 2025. The median milk price is about \$0.061/fl oz and \$0.064/fl oz respectively.

Larger packages generally have higher shelf prices

All recorded white rice and whole milk prices, by date



Simulated teaching data. Rice: n = 121 (2024), 126 (2025); milk: n = 128 (2024), 132 (2025).

For both products and dates, larger packages generally have higher shelf prices. Advertised price variation thus partly reflects package quantity. Prices also differ among similarly sized packages, so quantity does not explain all variation. Standardizing by Q^*_{sjt} makes quantities comparable within each product.

Promotions by product category

Table 8: Promotion percentages use recorded prices within each category and date.

Category	n 2024	n 2025	Promo 2024 (%)	Promo 2025 (%)
Bakery	333	340	5.7	13.5
Dairy	901	942	9.1	11.5
Pantry	1090	1135	9.3	9.1
Produce	1085	1160	8.7	9.4
Protein	983	1011	8.7	8.0

Promotions account for 8.7% of recorded prices in 2024 and 9.74% in 2025. Prevalence varies by category. In 2024 it ranges from 5.7% for Bakery to 9.3% for Pantry, in 2025 it ranges from 8% for Protein to 13.5% for Bakery. These describe observed price records, not the percentage of stores offering any promotion or size of promotional discounts.

A higher shelf price can mean a lower unit price

Table 9: White rice on April 15, 2024 (both records are nonpromotional).

Store	Quantity (oz)	Shelf price (USD)	Unit price (USD/oz)
S0001	16	1.61	0.1006
S0011	24	1.84	0.0767

Store `S0011` charges $1.84 for 24 ounces of rice, while `S0001` charges $1.61 for 16 ounces. Although `S0011` has the higher shelf price, its unit price is lower (about $0.0767/oz versus $0.1006/oz). This offers more rice per dollar but requires a larger initial purchase.

An overall mean shelf price cannot fairly rank stores since it mixes different products, package sizes, and observed product coverage. A store can appear expensive just because it sells bigger packages or has more expensive product categories represented. Fair comparisons need matched products with standardized quantities, or a complete basket with the same quantities.

3. When are two grocery stores genuinely comparable?

Here we compare Valley Basket - Yolo 1 (store a, `S0001`) with Daylight Discount - Yolo 4 (store b, `S0004`). They are in the same county and have different banners. Here we restrict prices to April 15, 2025 and match products by `product_id`.

Using store b as the baseline, define

$$\Delta_j = U_{ajt} - U_{bjt}, \qquad R_j = 100\frac{U_{ajt} - U_{bjt}}{U_{bjt}}.$$

A positive R_j means Valley Basket is more expensive for product j and a negative value means it is cheaper.

Product overlap

Table 10: Product overlap on April 15, 2025.

Measure	All records	No promotions
Products at a	38.0	37.0
Products at b	30.0	26.0
Shared products	28.0	23.0

Table 10: Product overlap on April 15, 2025. *(continued)*

Measure	All records	No promotions
Only at a	10.0	14.0
Only at b	2.0	3.0
Shared / products at a (%)	73.7	62.2
Shared / products at b (%)	93.3	88.5
Shared / union (%)	70.0	57.5

The stores share 28 products which is 73.7% of Valley Basket's 38 products and 93.3% of Daylight Discount's 30. Their union contains 40 products, giving a shared union overlap of 70%

Products recorded only at Valley basket are P02, P03, P06, P12, P13, P17, P21, P26, P29, P36. Those recorded only at Daylight Discount are P09, P25.

Matched price differences

Table 11: Matched percentage differences (store b is baseline).

Measure	All records	No promotions
Matched products	28.00	23.00
Mean R (%)	16.01	11.17
Median R (%)	11.25	10.13
IQR of R (percentage points)	25.70	16.45
Store a cheaper (%)	25.00	26.09
Store b cheaper (%)	75.00	73.91
Equal prices (%)	0.00	0.00

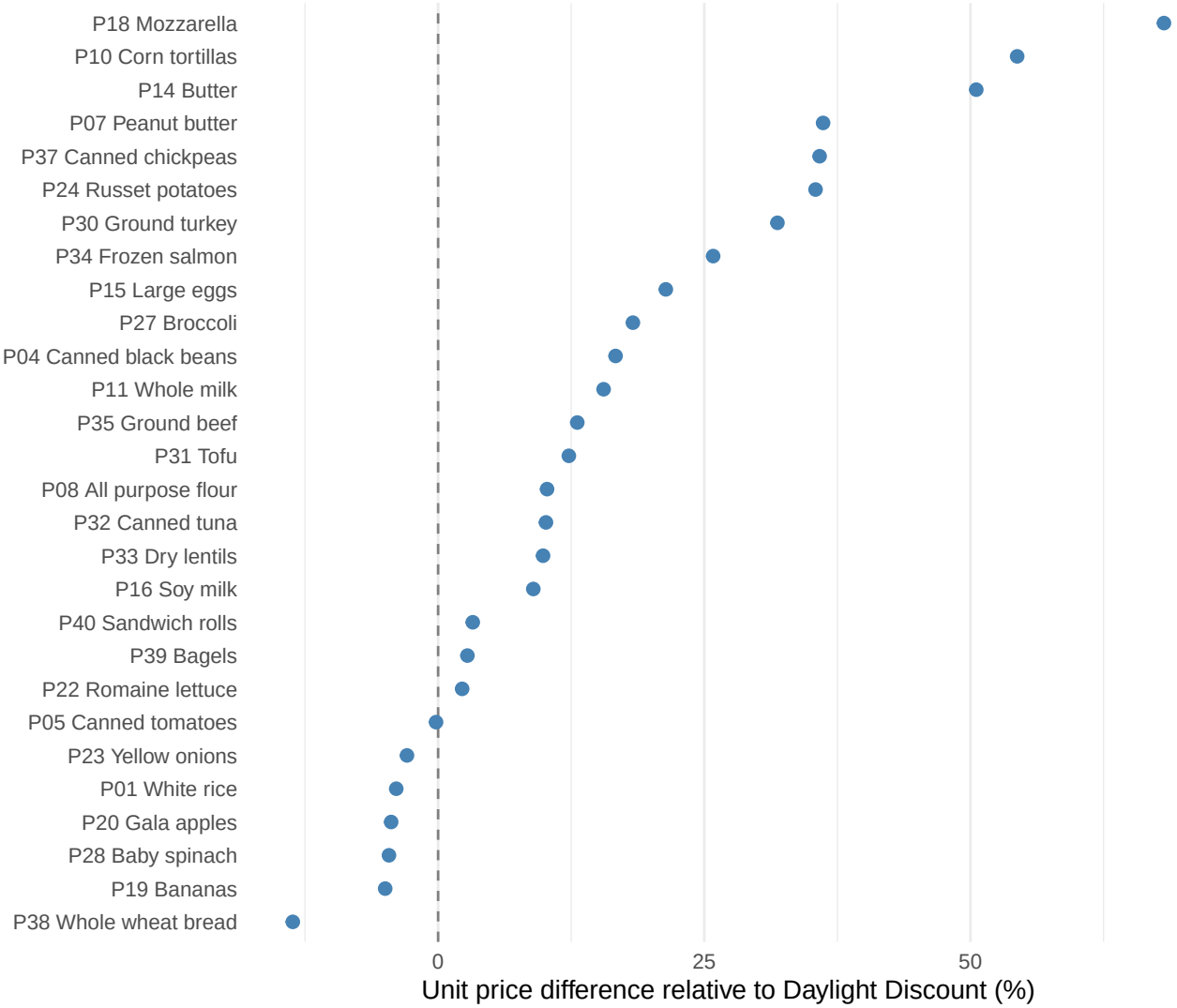
Across the 28 shared products, Valley Basket's unit prices are 16.01% higher on average, with a median difference of 11.25% and an IQR of 25.70 percentage points. Valley Basket is cheaper for 7 of 28 products (25%), while Daylight Discount is cheaper for 21 (75%) with no ties.

For white rice, the unit prices are \$0.106875/oz at Valley Basket and \$0.111250/oz at Daylight Discount.

So, Δ_j is -0.004375 dollars per ounce (Valley Basket cheaper for this product). Dollar differences across incompatible base units are not averaged.

Daylight Discount is cheaper for most shared products

Valley Basket versus Daylight Discount, Yolo (April 15, 2025)



Simulated data. All 28 shared products with promotions included.

Excluding promotional records reduces the shared set from 28 to 23 products and shared union overlap from 70% to 57.5%. The mean percentage difference falls to 11.17% and the median to 10.13%. Daylight Discount is cheaper for 17 of 23 shared products (73.9%). The general conclusion is still the same although both the prices represented and the comparison sample depend on promotions.

Being cheaper for most products does not guarantee a cheaper complete basket. For a common complete basket with required quantities q_{bj}, its cost difference depends on $\sum_{j \in J_b} q_{bj} \Delta_j$, not the number of products with positive or negative differences. Quantities purchased and the size of each price difference can outweigh a majority of small savings.

4. Does the cheapest store depend on the shopping basket?

Here we compare the three baskets in Yolo County on April 15, 2025, including recorded promotional prices. The basket cost is

$$B_{sbt} = \sum_{j \in J_b} q_{bj} U_{sjt}$$

provided every required product in J_b has an observed unit price. Coverage is

$$C_{sbt} = \frac{\#\{j \in J_b : U_{sjt} \text{ observed}\}}{|J_b|}.$$

A basket receives a cost and rank only when $C_{sbt} = 1$. Otherwise, its cost remains as `NA`.

The essentials, balanced, and expanded baskets require 12, 19, and 25 distinct products respectively. The calculation uses fixed quantities in base units rather than rounding purchasing up to whole packages.

Coverage and eligible stores

Table 12: Observed/required products and coverage percentage, Yolo, April 15, 2025.

Store	Essentials	Balanced	Expanded
S0001	12/12 (100.0%)	19/19 (100.0%)	25/25 (100.0%)
S0002	12/12 (100.0%)	19/19 (100.0%)	25/25 (100.0%)
S0003	12/12 (100.0%)	16/19 (84.2%)	21/25 (84.0%)
S0004	8/12 (66.7%)	12/19 (63.2%)	17/25 (68.0%)
S0005	10/12 (83.3%)	16/19 (84.2%)	21/25 (84.0%)
S0006	8/12 (66.7%)	15/19 (78.9%)	18/25 (72.0%)
S0007	7/12 (58.3%)	12/19 (63.2%)	18/25 (72.0%)

Three of Yolo's seven stores have complete essentials coverage. Those are `S0001` (Valley Basket), `S0002` (GreenGrocer), and `S0003` (Harvest Foods). Only `S0001` and `S0002` have complete balanced and expanded baskets. Incomplete coverage indicates unrecorded prices, not that a store is expensive or cannot sell those items.

Complete costs and rankings

Table 13: Rankings among stores with complete recorded baskets.

Basket	Store	Cost (USD)	Rank
essentials_12	S0001	56.19	1
essentials_12	S0003	57.22	2
essentials_12	S0002	66.95	3

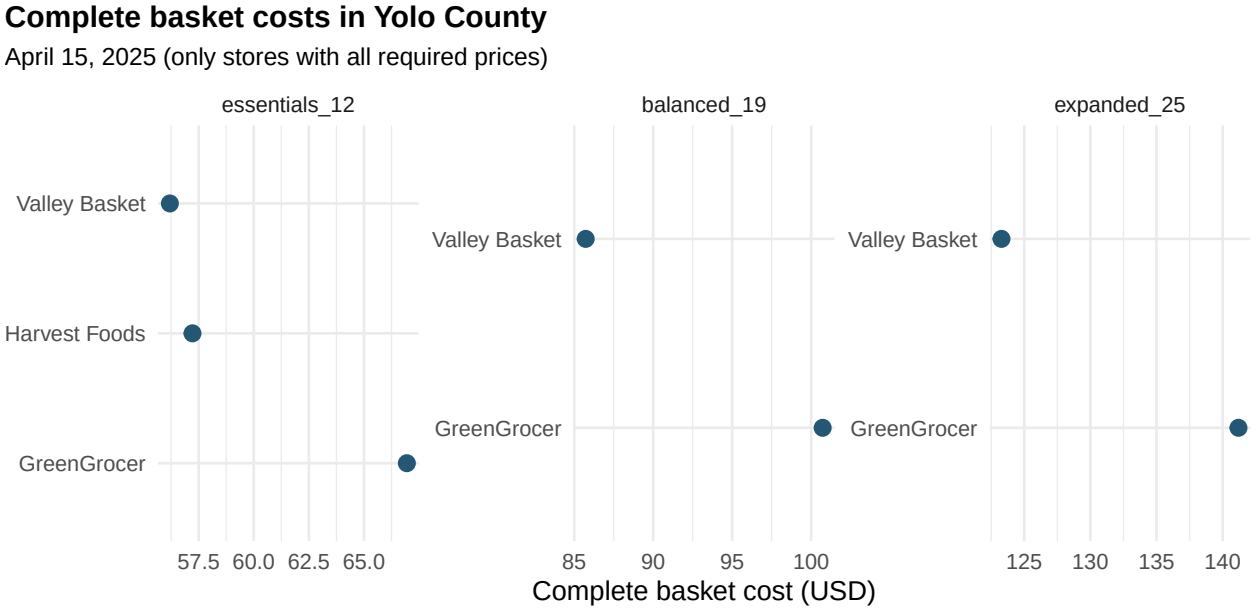
Table 13: Rankings among stores with complete recorded baskets. *(continued)*

Basket	Store	Cost (USD)	Rank
balanced_19	S0001	85.73	1
balanced_19	S0002	100.73	2
expanded_25	S0001	123.28	1
expanded_25	S0002	141.19	2

Table 14: Within-basket ranges; percentage gaps use the cheapest eligible cost.

Basket	Stores	Min (USD)	Max (USD)	Gap (USD)	Gap (%)
essentials_12	3	56.19	66.95	10.75	19.13
balanced_19	2	85.73	100.73	15.00	17.49
expanded_25	2	123.28	141.19	17.91	14.53

Essentials costs range from $56.19 at Valley Basket to $66.95 at GreenGrocer (Harvest Foods is between at $57.22). Balanced costs range from $85.73 to $100.73, and expanded costs from $123.28 to $141.19. Valley Basket is cheapest and GreenGrocer most expensive in each eligible comparison.



Can basket composition reverse the ranking?

The ranking does not reverse here. Valley Basket is still cheaper than GreenGrocer across all three baskets. Harvest Foods cannot be ranked for the larger baskets since its coverage is incomplete.

A reversal is possible since the cost difference between stores a and c is

$$B_{abt} - B_{cbt} = \sum_{j \in J_b} q_{bj}(U_{ajt} - U_{cjt}).$$

For some hypothetical sample, store a could be \$1 cheaper per base unit for one product but \$2 more expensive for another. Quantities of 3 and 1 give a basket difference of $3(-1) + 1(2) = -1$ dollar, while quantities of 1 and 3 give $1(-1)+3(2) = 5$ dollars. Changing the products or quantities can thus reverse which store is cheapest.

5. What makes a numerical summary useful, or misleading, for shopping decisions?

Different summaries answer different shopping questions. Using the April 15, 2025 comparison from question 3, store a is Valley Basket (`S0001`) and baseline store b is Daylight Discount (`S0004`).

- Frequency: Daylight Discount is cheaper for 21 of 28 shared products (75%). This counts products equally no matter their quantities purchased or the size of the savings.
- Location: The mean and median of R_j are 16.01% and 11.25%. This describes relative price differences across matched products.
- Dispersion: The IQR of R_j is 25.70 percentage points. This shows that the differences vary across products a good amount.
- Fixed basket: Valley Basket's complete essentials basket costs \$56.19. Daylight Discount has only 8 of the 12 required prices, so its basket cost is `NA` and cannot be compared.

Differences by product category

Table 15: Matched category comparisons (Daylight Discount is baseline).

Category	Shared products	Median R (%)
Bakery	3	2.75
Dairy	5	21.39
Pantry	6	13.45
Produce	7	-2.92
Protein	7	13.07

Valley Basket's median relative difference is 21.39% for Dairy but -2.92% for Produce. So, the overall comparison can hide differences relevant to shoppers who buy mainly from particular categories.

Distribution of complete essentials-basket costs

Table 16: Yolo essentials-basket costs on April 15, 2025; all monetary summaries are USD.

Stores	Minimum	Median	Mean	Maximum	Range
3	56.19	57.22	60.12	66.95	10.75

In these eligible Yolo stores, costs range from $56.19 to $66.95, a spread of $10.75. The median is $57.22 and the mean is $60.12. The highest cost raises the mean above the median. This describes only the three stores with complete recorded essentials prices.

The essentials basket panel in question 4 shows these three eligible store costs. GreenGrocer's higher cost pulls the mean above the median, while Valley Basket and Harvest Foods have similar costs.

Can the majority-product comparison disagree with a basket?

The chosen pair cannot establish an actual reversal because Daylight Discount's essentials basket is incomplete.

For some hypothetical sample, say three products each have a required quantity of one base unit. Store A charges $1, $1, and $10, while store B charges $2, $2, and $5. Store A is cheaper for two of three products, but its complete basket costs $12 versus $9 at store B.

For a shopper purchasing the specified quantities, B_{sbt} is more directly relevant than the fraction of products that are cheaper. A single average hides product specific differences, dispersion, coverage, and the quantities a shopper intends to buy.

6. How did prices change between April 2024 and April 2025?

Here we match observations by store and product across April 15, 2024 and April 15, 2025. Using 2024 as the baseline, define

$$\Delta_{sj} = U_{sj,2025} - U_{sj,2024}, \qquad R_{sj} = 100\frac{U_{sj,2025} - U_{sj,2024}}{U_{sj,2024}}.$$

Absolute changes keep each product's base unit. Percentage changes are dimensionless and can be summarized across the matched pairs, with each pair getting equal weight.

Matching and coverage

Table 17: Coverage of store product pairs across the two price waves.

Measure	Value
Potential pairs: 145 stores x 40 products	5800.00
Pairs observed in at least one wave	5342.00

Table 17: Coverage of store product pairs across the two price waves. *(continued)*

Measure	Value
Pairs observed in both waves	3638.00
2024 only pairs	754.00
2025 only pairs	950.00
Matched / potential pairs (%)	62.72
Matched / observed union (%)	68.10

There are 3,638 matched pairs out of 5,800 potential combinations (62.72%) and 5,342 pairs observed in at least one wave (68.1%). The unmatched records comprise 754 pairs observed only in 2024 and 950 only in 2025. Their changes are unknown, so they are excluded from change calculations. The potential pair denominator does not imply that every store stocks every product.

The data was reshaped from long to wide, with separate 2024 and 2025 columns for each store product pair. Checks confirmed that the reshape preserved all 5,342 distinct pairs and that 3,638 of them have observed unit prices in both years.

Distribution of matched changes and promotions

Table 18: Unit-price changes from April 15, 2024 to April 15, 2025.

Measure	All matched pairs	No promotions
Matched pairs	3638.00	3002.00
Mean change (%)	5.64	5.46
Median change (%)	4.90	4.91
SD (percentage points)	11.43	8.38
Q1 (%)	-1.43	-0.22
Q3 (%)	12.28	11.09
IQR (percentage points)	13.72	11.31
Increasing: count	2562.00	2211.00
Increasing (%)	70.42	73.65
Decreasing: count	1052.00	767.00
Decreasing (%)	28.92	25.55
Unchanged: count	24.00	24.00
Unchanged (%)	0.66	0.80

Across all matched pairs, the mean increase is 5.64% and the median is 4.9%. The standard deviation is 11.43 percentage points, showing a notable amount of variation. Prices increase for 2,562 pairs, decrease for 1,052, and remain unchanged for 24.

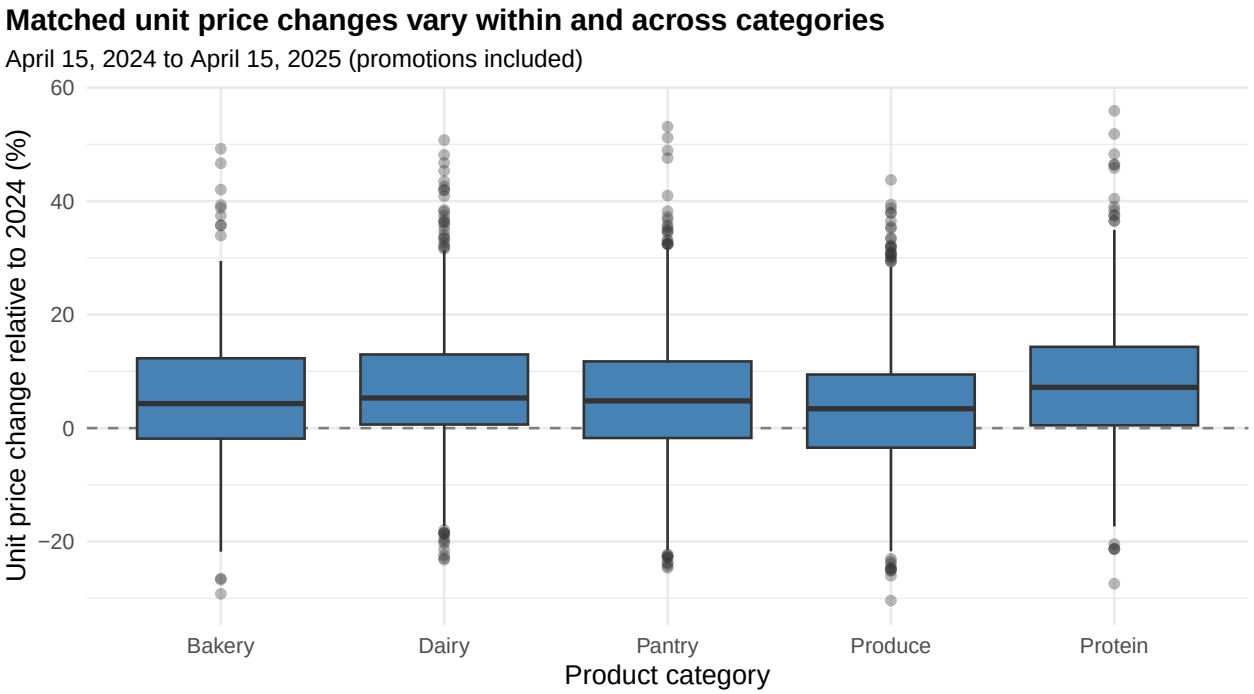
Excluding promotions at either date leaves 3,002 pairs. The mean increase becomes 5.46%, the median 4.91% and the standard deviation 8.38 percentage points. The overall increase is still apparent, but variability is lower.

Category comparisons

Table 19: Matched percentage changes by product category.

Category	Matched pairs	Mean (%)	Median (%)
Bakery	274	5.36	4.32
Dairy	760	6.86	5.31
Pantry	898	5.13	4.80
Produce	916	3.39	3.42
Protein	790	7.77	7.18

Category means range from 3.39% for Produce to 7.77% for Protein, compared with 5.64% overall. The overall mean weights categories by their number of matched pairs, not an equally weighted average of category means.



Simulated data. n = 3,638 matched store product pairs. Category sample sizes are reported in the table.

Package sizes and interpretation

Table 20: White rice at S0009: package-price and unit-price changes.

2024 oz	2025 oz	2024 USD	2025 USD	Shelf change (%)	Unit change (%)
24	26	2.54	2.67	5.12	-2.97

At S0009, the rice package grows from 24 to 26 ounces while its price rises from $2.54 to $2.67. The shelf price increases 5.12%, but the standardized price decreases 2.97%. So, package size changes can reverse the direction of the comparison.

Subtracting the means of all records from the two waves would compare different sets of store product observations. For example, cheaper products appearing only in 2025 could lower that wave's mean even if every matched price increased. Pooling unit price levels also mixes incompatible units. Matching fixes the comparison set, while percentage changes provide a common scale.

These are matched nominal changes in simulated prices, not an inflation index. The promotion sensitivity check changes the comparison sample and does not isolate a promotion effect.

7. Does the number of stores equal the number of distinct banners or owning units?

Here we join stores to banners and use `ownership_unit_id_sim` to count owners.

County counts and shares

Table 21: Undated simulated store roster (shares use stores within each county).

County	Stores	Banners	Owners	Owners/store	Top banner (%)	Top owner (%)
Alameda	9	7	7	0.78	22.2	22.2
Butte	6	5	5	0.83	33.3	33.3
Colusa	4	3	4	1.00	50.0	25.0
Contra Costa	7	6	4	0.57	28.6	42.9
El Dorado	6	5	4	0.67	33.3	33.3
Glenn	4	4	4	1.00	25.0	25.0
Lake	4	3	3	0.75	50.0	50.0
Marin	6	5	5	0.83	33.3	33.3
Mendocino	4	3	4	1.00	50.0	25.0
Napa	6	6	6	1.00	16.7	16.7
Nevada	4	4	4	1.00	25.0	25.0
Placer	5	5	3	0.60	20.0	40.0
Sacramento	9	8	6	0.67	22.2	33.3
San Benito	6	6	6	1.00	16.7	16.7
San Francisco	7	6	5	0.71	28.6	28.6
San Joaquin	6	5	4	0.67	33.3	33.3
San Mateo	6	5	5	0.83	33.3	33.3
Santa Clara	9	8	7	0.78	22.2	33.3
Solano	6	5	5	0.83	33.3	33.3
Sonoma	6	6	6	1.00	16.7	16.7
Stanislaus	6	5	5	0.83	33.3	33.3
Sutter	6	5	5	0.83	33.3	33.3
Yolo	7	7	5	0.71	14.3	28.6
Yuba	6	5	4	0.67	33.3	50.0

The owner unit ratio is the number of distinct simulated owners divided by the county's store count. The largest banner and largest owner shares are percentages of county study stores belonging to the most frequent banner or owner.

Equal store counts, different composition

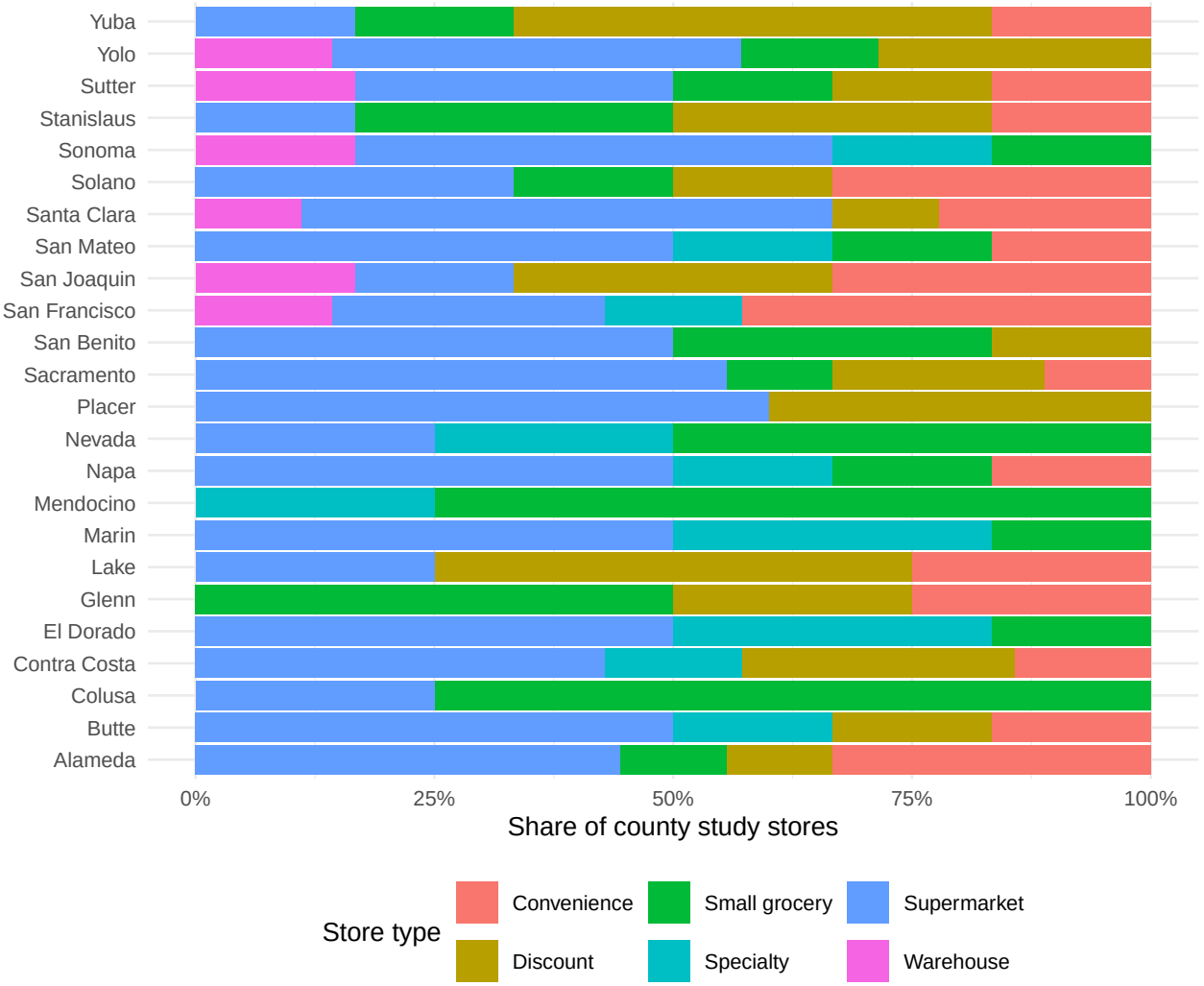
Table 22: Reproducible comparison (all counties with Yolo's store count).

County	Stores	Banners	Owners
Contra Costa	7	6	4
San Francisco	7	6	5
Yolo	7	7	5

Contra Costa, San Francisco, and Yolo each have seven stores. Their banner counts are 6, 6, and 7, while their owner counts are 4, 5, and 5. So, equal store counts do not imply equal banner or ownership diversity. Contra Costa's largest owner accounts for 42.9% of its stores, compared with 28.6% in San Francisco and Yolo.

Store type composition varies across counties

Undated study roster (denominator is each county's store count)



Simulated data. 145 stores in 24 counties.

The equal-count counties also differ in store types. Convenience stores account for 3/7 (42.9%) of San Francisco's stores, 1/7 (14.3%) of Contra Costa's, and 0/7 of Yolo's. This describes study roster, not actual stores in these counties.

Simulation SNAP authorization

Table 23: Simulated SNAP status (percentages use all study stores in each county).

County	Stores	Yes	No	Yes (%)	No (%)
Alameda	9	8	1	88.9	11.1
Butte	6	4	2	66.7	33.3
Colusa	4	3	1	75.0	25.0
Contra Costa	7	6	1	85.7	14.3
El Dorado	6	5	1	83.3	16.7
Glenn	4	4	0	100.0	0.0
Lake	4	2	2	50.0	50.0
Marin	6	5	1	83.3	16.7
Mendocino	4	3	1	75.0	25.0
Napa	6	6	0	100.0	0.0
Nevada	4	3	1	75.0	25.0
Placer	5	4	1	80.0	20.0
Sacramento	9	7	2	77.8	22.2
San Benito	6	4	2	66.7	33.3
San Francisco	7	6	1	85.7	14.3
San Joaquin	6	4	2	66.7	33.3
San Mateo	6	5	1	83.3	16.7
Santa Clara	9	7	2	77.8	22.2
Solano	6	3	3	50.0	50.0
Sonoma	6	5	1	83.3	16.7
Stanislaus	6	5	1	83.3	16.7
Sutter	6	5	1	83.3	16.7
Yolo	7	7	0	100.0	0.0
Yuba	6	6	0	100.0	0.0

Overall, 117 of 145 stores (80.7%) are labeled SNAP-authorized and 28 (19.3%) are not. Authorization can differ within a banner like how SaveMore store `S0005` in Yolo is labeled Yes and SaveMore store `S0012` in Sacramento is labeled No. So, authorization must be taken from individual store records.

Counting "Independent operators" as one corporation would understate ownership diversity since the label groups unrelated operators. More distinct owners do not imply lower prices. These counts do not establish a price relationship nor causation.

Shares of study stores are not real market shares or evidence of market power. The simulated roster does not contain sales or expenditure shares. The simulated SNAP authorization does not measure household participation or purchases. As well as this the data is simply not exhaustive enough to look into market shares and market power.

8. Where are stores numerous, and where are they numerous relative to population?

Let S_c be the number of study stores in county c and N_c is simulated 2024 population. Store density is

$$D_c = 10{,}000 \frac{S_c}{N_c}.$$

This measures study stores per 10,000 simulated residents. The store roster is undated and it has denominator `population_2024_sim`.

County summaries

Table 24: County level summaries across 24 simulated counties.

Statistic	Stores	2024 residents	Stores per 10,000
Mean	6.04	28530.79	2.28
Median	6.00	29959.50	2.11
SD	1.49	9338.14	0.62
Q1	5.75	23993.00	1.90
Q3	6.25	33692.00	2.74
IQR	0.50	9699.00	0.83
Min	4.00	10743.00	1.40
Max	9.00	42850.00	3.72

Across 24 counties, the median store count is 6 and the median density is 2.11 stores per 10,000 residents. Counts range from 4 to 9, while densities range from 1.4 in San Mateo to 3.72 in Lake.

Counts, denominators, and rankings

Table 25: County rankings by stores count and stores per 10,000 residents.

County	Stores	2024 population	Density	Count rank	Density rank
Lake	4	10743	3.72	20	1
Colusa	4	12516	3.20	20	2
Santa Clara	9	30287	2.97	1	3
San Francisco	7	23810	2.94	4	4
Mendocino	4	13624	2.94	20	5
Nevada	4	13964	2.86	20	6
Sacramento	9	33419	2.69	1	7
Contra Costa	7	27569	2.54	4	8
Yuba	6	24054	2.49	7	9
Alameda	9	37022	2.43	1	10
Sonoma	6	25292	2.37	7	11

Table 25: County rankings by stores count and stores per 10,000 residents. *(continued)*

County	Stores	2024 population	Density	Count rank	Density rank
Napa	6	28207	2.13	7	12
Glenn	4	19038	2.10	20	13
Yolo	7	33746	2.07	4	14
Sutter	6	28994	2.07	7	15
Solano	6	29632	2.02	7	16
Marin	6	30732	1.95	7	17
San Joaquin	6	30897	1.94	7	18
Stanislaus	6	33674	1.78	7	19
Placer	5	31559	1.58	19	20
El Dorado	6	39016	1.54	7	21
San Benito	6	41695	1.44	7	22
Butte	6	42399	1.42	7	23
San Mateo	6	42850	1.40	7	24

Rank 1 denotes the highest value. Ties receive the same minimum rank, so the next ranks my skip positions. Alameda, Sacramento, and Santa Clara tie for the largest count with 9 stores each. Lake ranks first by density despite only have 4 stores (10,000(4/10,743) = 3.72 stores per 10,000 residents).

Table 26: Five largest absolute rank changes (ties in selected are reslved alphabetically).

County	Count rank	Density rank	Change
Lake	20	1	19
Colusa	20	2	18
San Mateo	7	24	-17
Butte	7	23	-16
Mendocino	20	5	15

Positive changes indicate movement toward rank 1 after population adjustment. Lake moves from tied rank 20 to rank 1, while San Mateo moves from tied rank 7 to rank 24. The denominator changes which counties appear well supplied.

Metro and nonmetro comparison

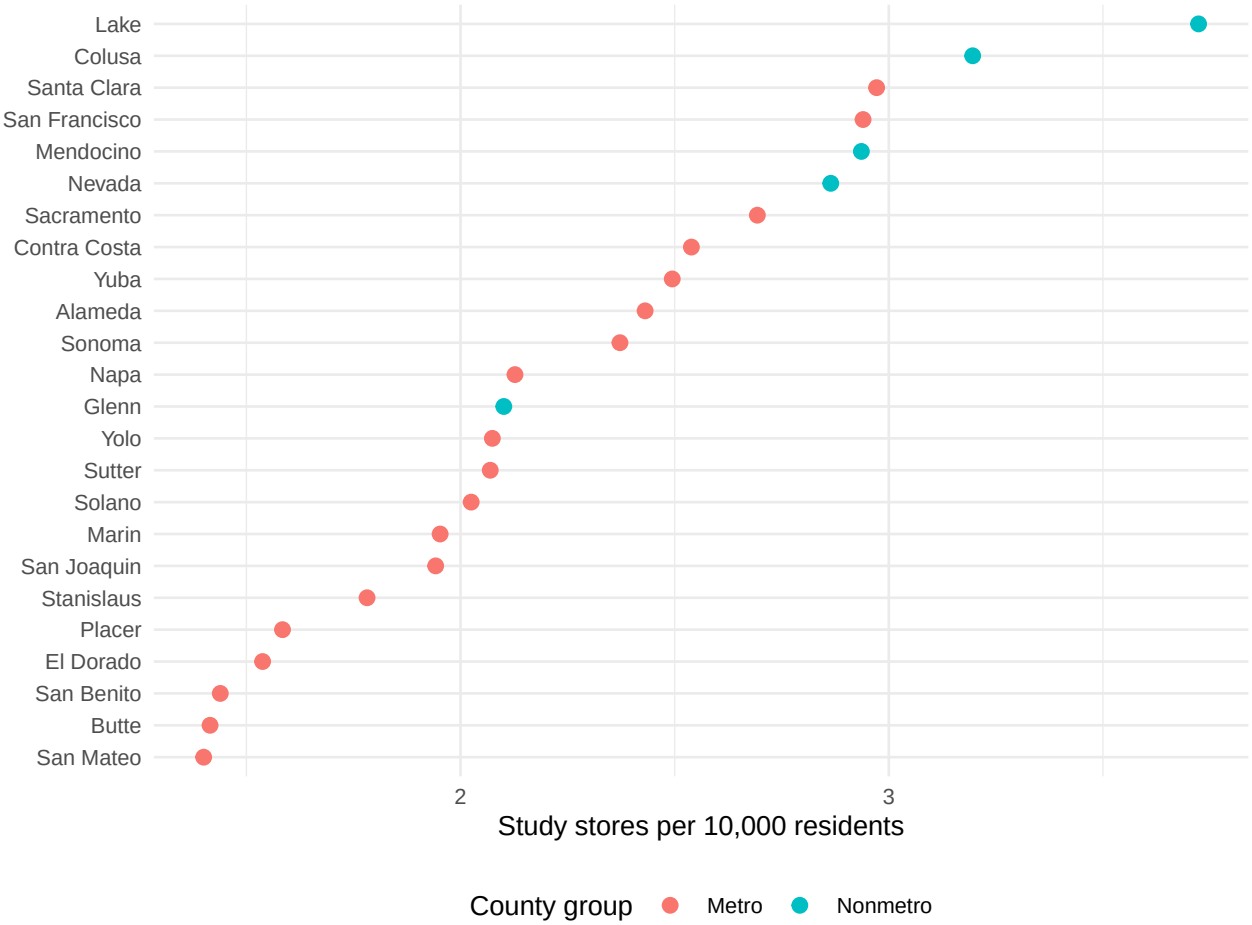
Table 27: County level summaries by simulated metro classification

Group	Counties	Mean stores	Mean density	Median density	Density IQR
Metro	19	6.58	2.09	2.07	0.78
Nonmetro	5	4.00	2.96	2.94	0.33

The 19 metro counties average 6.58 stores, compared with 4 among the 5 nonmetro counties. However, mean county density is lower in metro counties (2.09 versus 2.96 stores per 10,000 residents). These are equally weighted averages of county densities, not rates from pooled group populations.

Population adjustment changes the comparison of store availability

Undated study store counts divided by simulated 2024 population



Simulated data. All 24 counties, 145 study stores.

A high county density does not say that a particular household has a nearby or affordable store. Stores may cluster far from some residents, and density does not measure prices or household resources. Assessing proximity requires stores and household locations, within county population distribution, travel routes, and transit availability. The store roster is a teaching sample.

9. What if two measures of the food environment disagree?

Here we compare the undated study store roster with the alternative 2022 food-environment indicators. The study density D_c uses simulated 2024 population, while the alternative density uses `population_2022_est_sim`. The counts also represent different retailer definitions, so their differences are not changes over time.

Matching counties and comparing measures

Table 28: Alternative minus study measures (percentage differences use study baseline).

County	Study n	2022 n	Count diff.	Count (%)	Rate diff.	Rate (%)
Alameda	9	11	2	22.2	0.492	20.2
Butte	6	11	5	83.3	1.240	87.6
Colusa	4	4	0	0.0	-0.084	-2.6
Contra Costa	7	11	4	57.1	1.418	55.8
El Dorado	6	6	0	0.0	-0.054	-3.5
Glenn	4	6	2	50.0	1.122	53.4
Lake	4	6	2	50.0	1.788	48.0
Marin	6	4	-2	-33.3	-0.631	-32.3
Mendocino	4	5	1	25.0	0.926	31.5
Napa	6	5	-1	-16.7	-0.251	-11.8
Nevada	4	3	-1	-25.0	-0.610	-21.3
Placer	5	4	-1	-20.0	-0.247	-15.6
Sacramento	9	8	-1	-11.1	-0.241	-9.0
San Benito	6	10	4	66.7	1.057	73.5
San Francisco	7	8	1	14.3	0.527	17.9
San Joaquin	6	9	3	50.0	0.934	48.1
San Mateo	6	5	-1	-16.7	-0.160	-11.4
Santa Clara	9	14	5	55.6	1.535	51.7
Solano	6	5	-1	-16.7	-0.377	-18.6
Sonoma	6	4	-2	-33.3	-0.732	-30.9
Stanislaus	6	7	1	16.7	0.314	17.6
Sutter	6	7	1	16.7	0.397	19.2
Yolo	7	10	3	42.9	0.997	48.0
Yuba	6	4	-2	-33.3	-0.734	-29.4

All 24 counties match. Count differences are measured in outlets, rate differences are measured in outlets per 10,000 residents. Positive values mean the alternative measure is larger. Percentage differences divide each difference by the corresponding study measure.

The supplied 2022 rates agree with rates calculated from their 2022 counts and populations within rounding tolerance. This checks internal consistency without assuming agreement with the roster.

Association and ranking disagreements

Table 29: Five largest density rank disagreements (selection ties resolved alphabetically).

County	Study rate	2022 rate	Study rank	2022 rank	Change
Butte	1.415	2.655	23	11	12
San Benito	1.439	2.496	22	12	10
Nevada	2.865	2.255	6	15	-9
Sonoma	2.372	1.640	11	20	-9
Yuba	2.494	1.760	9	18	-9

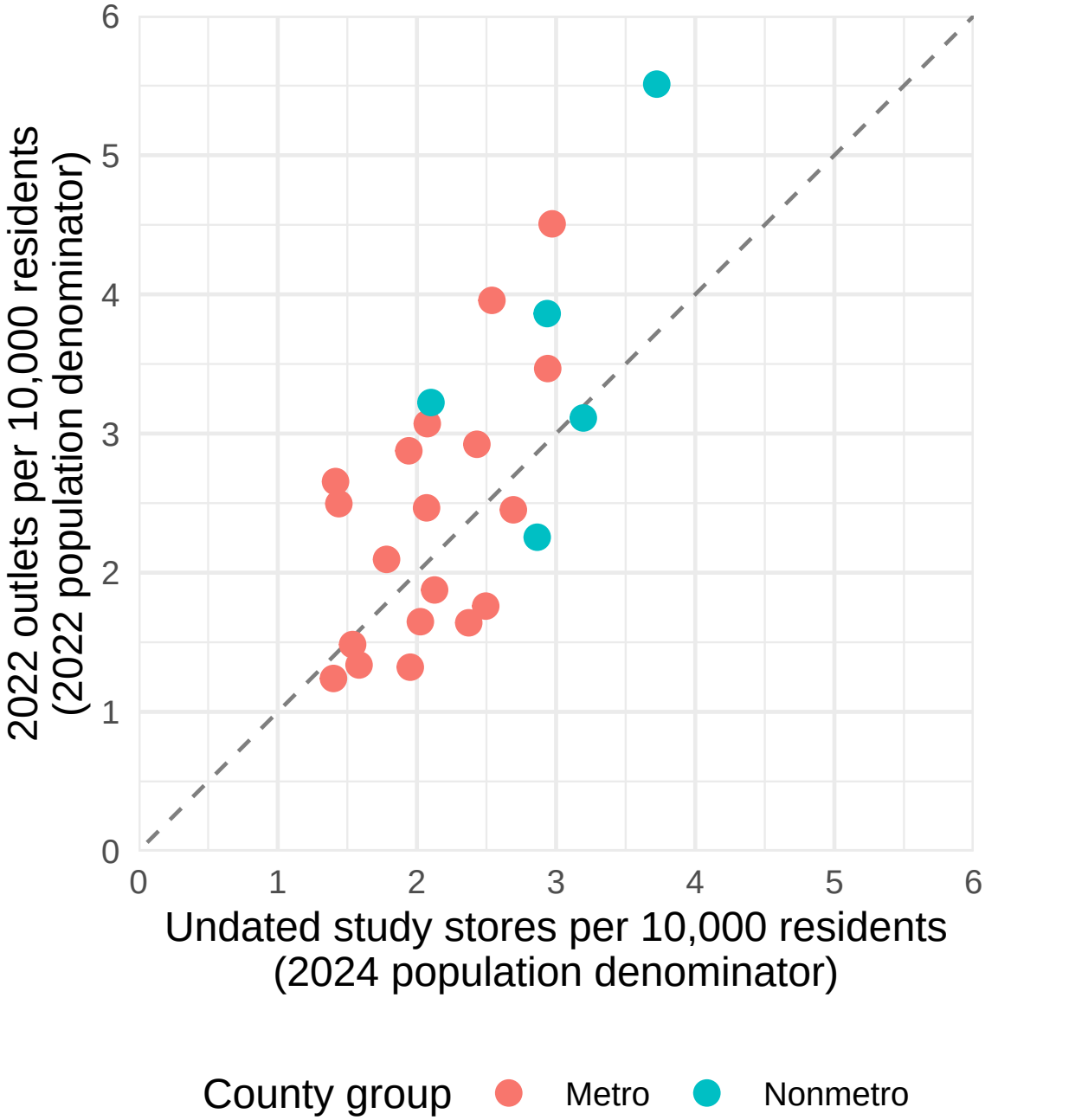
Spearman correlations are 0.669 for counts and 0.641 for densities, using 24 counties. Spearman

correlation is appropriate since the question concerns whether counties retain similar relative positions. It measures rank association, not equality between the measures.

Rank 1 denotes the highest density, with ties assigned their minimum rank. Butte moves from rank 23 under the study measure to rank 11 under the alternative, while Nevada moves from 6 to 15. So, positive overall association still allows substantial disagreement for individual counties.

County outlet densities differ between measures

Study roster versus alternative 2022 indicator



Simulated data. n = 24 matched counties (dashed line indicates equality).

Both measures give higher mean county densities in the five nonmetro counties than in the 19 metro counties. The study means are 2.96 versus 2.09 and the alternative means are 3.59 versus 2.38 outlets per 10,000 residents. These are equally weighted county means, not pooled population rates.

Interpreting discrepancies

A discrepancy is not proof of a faulty join. Incorrect county codes, duplicate keys, expanded row counts, or an incorrect denominator would indicate coding problems. The code checks above verify normalized keys, county matching, join cardinality, and the alternative rate's population denominator.

Even after those checks passed, the measures can differ because of retailer definitions, study coverage, and reference periods. The undated study count is not a 2024 census of outlets just because its density uses 2024 population. So, these differences cannot be described as growth or decline between 2022 and 2024.

10. How should tract-level food-access information be summarized for a county?

Let A_k be the low-access residents, N_k tract population, and K_c the number of tracts in county c. The low-access summaries are then

$$L_c^{\text{weighted}} = 100\frac{\sum_{k\in c} A_k}{\sum_{k\in c} N_k}\%, \qquad L_c^{\text{unweighted}} = \frac{1}{K_c}\sum_{k\in c} 100\frac{A_k}{N_k}\%.$$

The weighted percentage describes county residents and the unweighted percentage gives each tract equal weight.

County totals and reconciliation

Table 30: Summed simulated 2024 tract counts by county

County	Tracts	Residents	Households	Low access	Low income + low access
Alameda	6	37022	14373	6927	1524
Butte	6	42399	17385	11227	3396
Colusa	4	12516	4779	4480	1446
Contra Costa	6	27569	11413	8423	2201
El Dorado	6	39016	15772	12803	4022
Glenn	4	19038	8093	8483	2665
Lake	4	10743	3955	4096	1607
Marin	6	30732	12455	8402	2592
Mendocino	4	13624	5459	6277	2096
Napa	6	28207	11344	5537	1360
Nevada	4	13964	5323	6408	2104
Placer	6	31559	13226	4908	1308
Sacramento	6	33419	13990	10400	2974
San Benito	6	41695	16125	9533	2182
San Francisco	6	23810	9803	5533	1980
San Joaquin	6	30897	12052	7922	1724
San Mateo	6	42850	16234	8627	2997
Santa Clara	6	30287	11940	6154	1216

Table 30: Summed simulated 2024 tract counts by county *(continued)*

County	Tracts	Residents	Households	Low access	Low income + low access
Solano	6	29632	12136	6874	1922
Sonoma	6	25292	9714	8830	3326
Stanislaus	6	33674	13322	10220	3648
Sutter	6	28994	11282	5938	1735
Yolo	6	33746	13031	7958	2232
Yuba	6	24054	9290	6534	1958

The 134 constructed tracts have 684,739 residents across 24 counties. Summed tract populations and household counts exactly match the corresponding 2024 county totals in all 24 counties. No missing counts were excluded or replaced.

Table 31: Low-access summaries (absolute gaps are percentage points).

County	Weighted (%)	Unweighted (%)	Gap (pp)	Low income + low access (%)
Alameda	18.71	19.39	0.68	4.12
Butte	26.48	26.05	0.43	8.01
Colusa	35.79	38.54	2.75	11.55
Contra Costa	30.55	30.45	0.10	7.98
El Dorado	32.81	30.62	2.19	10.31
Glenn	44.56	44.61	0.05	14.00
Lake	38.13	39.44	1.31	14.96
Marin	27.34	26.87	0.47	8.43
Mendocino	46.07	43.02	3.05	15.38
Napa	19.63	18.69	0.94	4.82
Nevada	45.89	46.45	0.56	15.07
Placer	15.55	17.82	2.27	4.14
Sacramento	31.12	28.65	2.47	8.90
San Benito	22.86	23.10	0.24	5.23
San Francisco	23.24	24.56	1.32	8.32
San Joaquin	25.64	24.55	1.09	5.58
San Mateo	20.13	21.13	1.00	6.99
Santa Clara	20.32	21.90	1.58	4.01
Solano	23.20	23.12	0.07	6.49
Sonoma	34.91	32.66	2.25	13.15
Stanislaus	30.35	31.90	1.55	10.83
Sutter	20.48	21.69	1.21	5.98
Yolo	23.58	24.23	0.64	6.61
Yuba	27.16	26.66	0.51	8.14

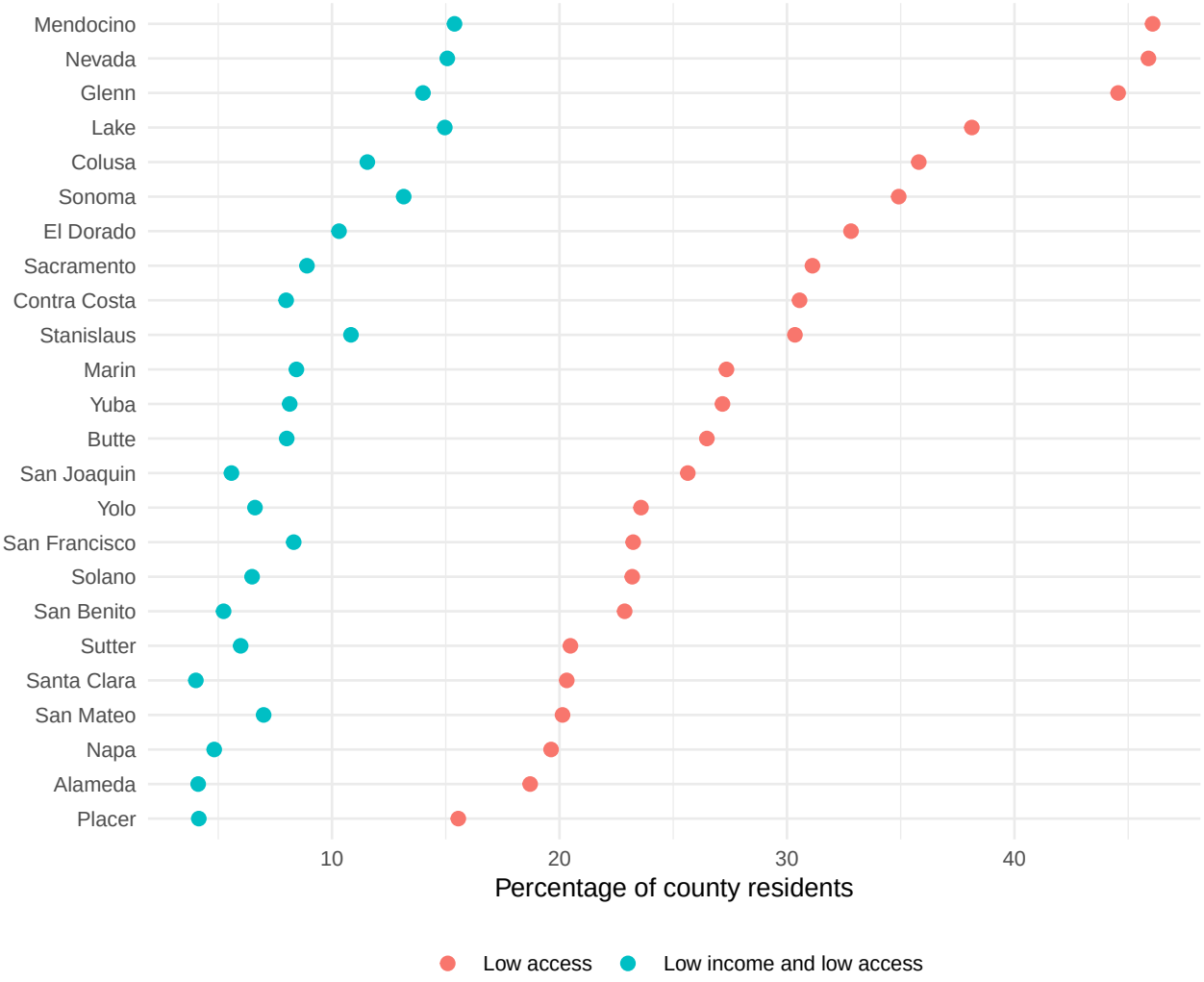
The weighted low-access and low-income + low_access percentages use summed tract residents as their denominators. The unweighted statistic averages tract percentages over the number of tracts reported for each county. Low income + low access residents are a subset of low access residents.

Reshaping access indicators for comparison

The reshape produces 268 rows, exactly two for each of the 134 original tracts. Population is repeated once per indicator, so percentages are calculated separately within each indicator. Summing population across both indicators would double count residents.

County resident shares with limited food access

Simulated 2024 counts (denominator is summed county tract population)



Simulated data. 134 constructed tracts across 24 counties.

Why weighting matters

Colusa provides an example. Its four tract populations are 1,493, 1,913, 3,776, and 5,334, with low-access percentages of 63.5%, 25.77%, 27.15%, 37.76% respectively. Their unweighted average is 38.54%, while the resident weighted percentage is

$$100\frac{948+493+1025+2014}{1493+1913+3776+5334}=100\frac{4480}{12516}=35.79\%$$

The smallest tract has the highest low-access percentage. It receives 25% of the weight in the unweighted average but only about 11.9% of the resident weight, making the weighted result lower.

So, L_c^{weighted} answers the question about what share of county residents have low access. The unweighted mean answers the question of what is the average low-access share across tracts. These are different questions even when the county totals are the same.

11. How are income, poverty, and access to a vehicle related to grocery accessibility?

For some county, c, let H_c be the households, V_c be the households with no vehicle, I_c be the median household income, and P_c be the poverty rate. Let A_k and N_k be the low-access residents and population of tract k, respectively. The measures then are:

$$\text{NoVehicle}_c = 100\frac{V_c}{H_c}\%, \qquad L_c = 100\frac{\sum_{k\in c} A_k}{\sum_{k\in c} N_k}\%$$

Income I_c and poverty rate P_c are used as reparted in the county profile. For each predictor $X \in \{I_c, P_c, \text{NoVehicle}_c\}$, the association with accessability is summarized by the correlation across the 24 counties,

$$r_X = \text{cor}\left(X_c,\ L_c\right).$$

Table 32: County-level summarries across 24 counties.

Measure	Mean	Median	SD	Q1	Q3	Min	Max
Meadian Income (USD)	81887.50	78350.00	14958.47	72950.00	93025.00	59400.00	119100.00
No-vehicle households (%)	11.29	10.89	1.88	10.03	11.71	8.64	16.12
Poverty Rate (%)	17.95	18.60	3.10	15.07	19.80	12.90	23.80
Low-access residents (%)	28.52	26.82	8.75	22.27	33.34	15.55	46.07

Across 24 counties we see a mean substantially higher than the median for median household incomes, indicating a right skew. This dataset also appears to have high variance. No missing values were dropped or imputed.

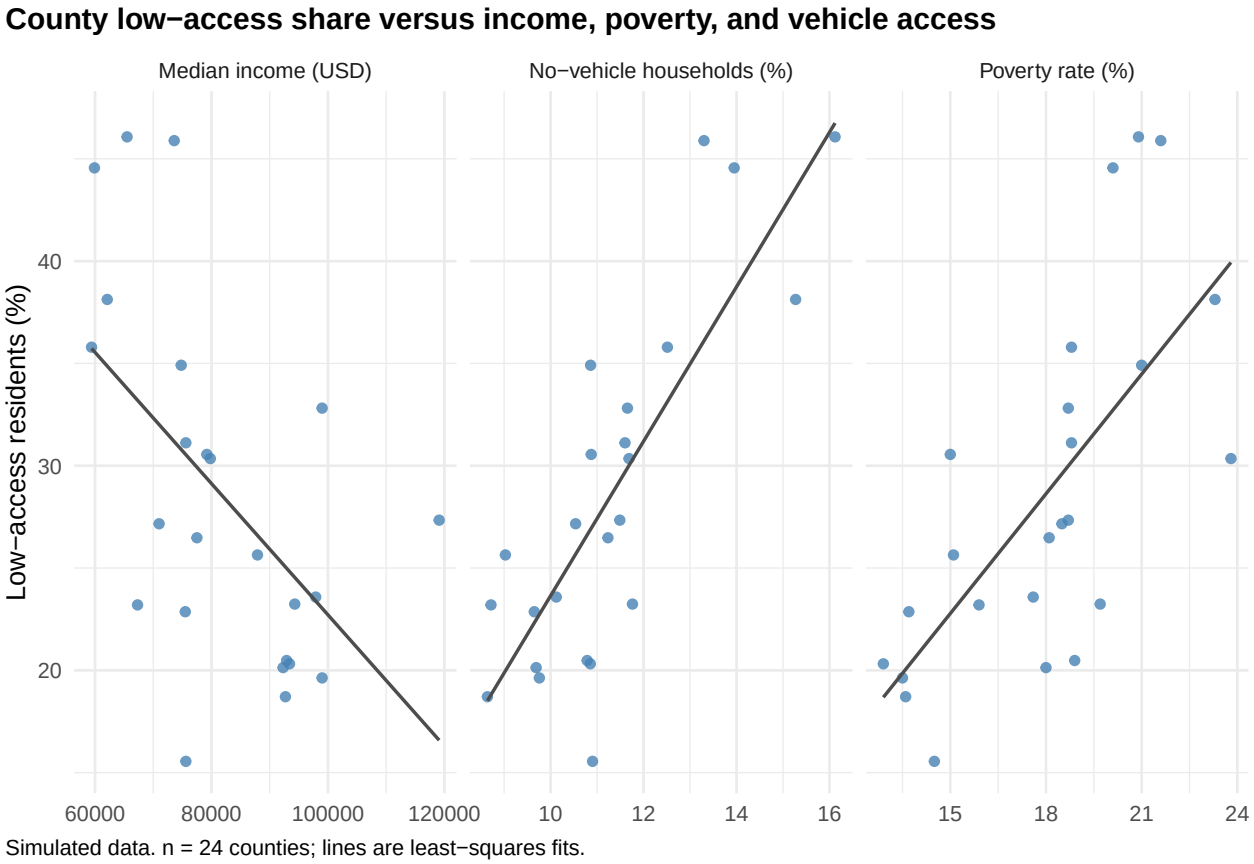
Correlation Values Across Different Predictors

Table 33: Association with county low-access share (n = 24).

Predictor	Pearson r	95% CI low	95% CI high	p-value	Spearman rho
Median income	-0.549	-0.779	-0.186	0.006	-0.537
Poverty rate	0.691	0.399	0.856	0.000	0.757
No-vehicle households	0.813	0.610	0.916	0.000	0.767

Aggregated across all counties, all Median income, Povery rate, and No-vehicle households all have meaningful Pearson and Spearman corralations with extremely low P-values. At a significance level of $\alpha = 0.01$, we have a statistically significant correlation for all of these predictor variables. Additionally Spearman and Pearson values are very similar, indicating little influence of outliers.

Scatterplots



The assumption of a linear relationship holds through raw observation and rigorous statistical testing, as we view a roughly linear plot for all 3 predictor variables.

Predictors Overlap With Each Other

Table 34: Pearson correlations among county predictors.

Variable	Income	Poverty	No_Vehicle
Income	1.00	-0.32	-0.46
Poverty	-0.32	1.00	0.70
No_Vehicle	-0.46	0.70	1.00

As we see in the table, all predictor variables have meaningful correlation with one another. Naturally a lower median income is correlated with a higher poverty rate. Additionally, a higher poverty

rate seems to significantly contribute to a higher likelihood of homes not having a vehicle.

We can conclude that Income, Poverty rate, and No vehicle all play a part in predicting access to groceries and all play a part in predicting each other.

12. Do more shopping options guarantee better food access?

Coverage of Travel Data

Table 35: Coverage of tract travel data (missing means not recorded, not zero).

Measure	Value
Tracts in access file	134
Tracts with at least one travel alternative	134
Tracts with no travel alternative	0
Residents in tracts with no travel alternative (%)	0
Tracts with no bus alternative (bus measures NA)	4

We will analyze the question of "Do more shopping options guarantee better food access?" with tracts aggregated into county-level data. S_c being study store count, and D_c being number of stores per 10,000 people. These will be used as predictor variables, with the outcome being 'Weighted', defined as population-weighted low-access % from the county-access data set.

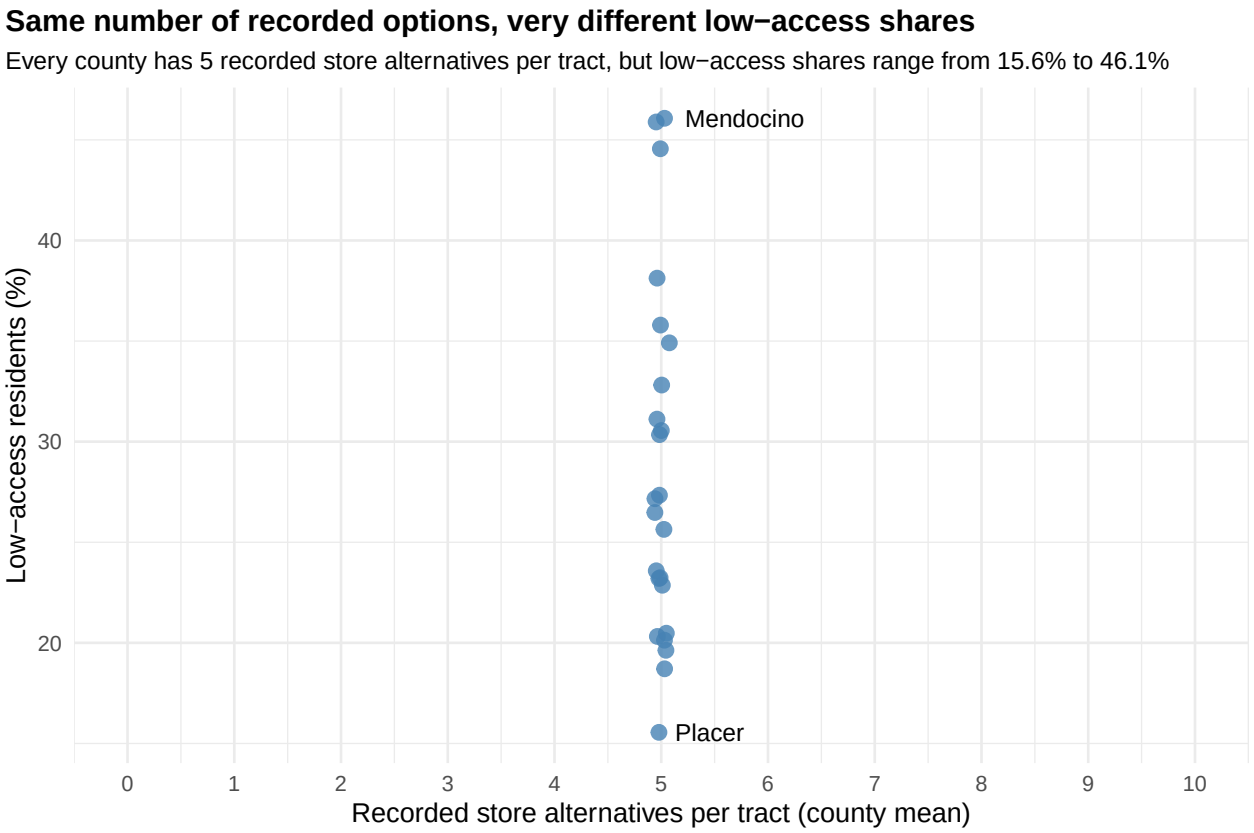
County Level Aggregation

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No counties were excluded.

This aggregation combined counties by tract. All missing data was kept as NA and never imputed. NAs for bus times is a meaningful piece of data, implying no alternative has bus service, there were 4 tracts with this measure. Still after aggregation, there was zero counties with no travel alternative.

There is no gaurentee of better food access when a county has better shopping options



Over all measured tracts, we see the average number of store options is 5, yet the county low-access shares range from 15.5518236 to 46.0731063. We conclude that the number of store options in the recorded dataset cannot guarantee, or even meaningfully predict, the average access to groceries.

NOTE: This is likely a choice made in the simulated data, this constant count may be more revealing of the data set than of any pattern.

13. Can transportation reverse the ranking of the cheapest grocery store?

We start from the 670 recorded tract–store alternatives in the travel file, not from a county join. Each candidate store receives its complete April 15, 2025 `essentials_12` basket cost from Question 4, when one exists. For tract k, eligible store s, and mode m, the travel-adjusted cost is

$$T_{ksm} = B_{s,\text{essentials},2025} + F_{ksm},$$

where F_{ksm} is the recorded round-trip car cost or bus fare. For each tract we find three minimizing stores:

(a) the lowest basket-only cost among all eligible candidates;

(b) the lowest basket + car round-trip cost;

(c) the lowest basket + bus round-trip fare, among candidates with `bus_service_available_sim = "Yes"` and a complete basket.

A candidate is eligible only if its store has all 12 essentials prices on that date. Missing bus fares on alternatives without service are ineligible, not free; a recorded zero fare with service is a valid free trip.

Table 36: Eligibility for the essentials-basket travel scenarios, April 15, 2025.

Measure	Value
Recorded tract-store alternatives	670
Alternatives whose store has a complete essentials basket	496
...of which have bus service	353
...of which have bus service with a zero fare	70
Tracts in the access file	134
Tracts with at least one eligible candidate	134
Tracts with at least one bus-accessible eligible candidate	129

Of the 670 recorded alternatives, 496 lead to a store with a complete essentials basket. 134 of the 134 tracts have at least one such candidate, and 129 have at least one that is also reachable by bus. The remaining tracts are not evaluable for that scenario; they are not assigned a zero cost.

Minimizing stores for each tract

`pick_min()` returns one store per tract. When two stores tie exactly, the lower `store_id` is kept. Tracts with a tied minimum: 0 for basket cost and 0 for car-adjusted cost.

Table 37: Tract-level minimum costs in USD (n = evaluable tracts for each row).

Measure	Tracts	Median	Q1	Q3	Min	Max
(a) Basket only	134	59.23	52.43	60.76	42.91	65.90
(b) Basket + car round trip	134	64.33	59.39	67.91	45.81	75.56
(c) Basket + bus round trip	129	62.63	58.19	66.57	42.91	82.98
Car cost at store chosen in (b)	134	5.45	3.98	7.20	2.42	15.56
Bus fare at store chosen in (c)	129	3.50	2.50	5.00	0.00	7.50

The median tract's cheapest complete essentials basket costs $59.23. Adding a car round trip raises the median minimum to $64.33, and the bus scenario gives $62.63 among the 129 tracts where it is evaluable. The rows use different tract samples, so we won't directly compare the bus one to the vehicle one.

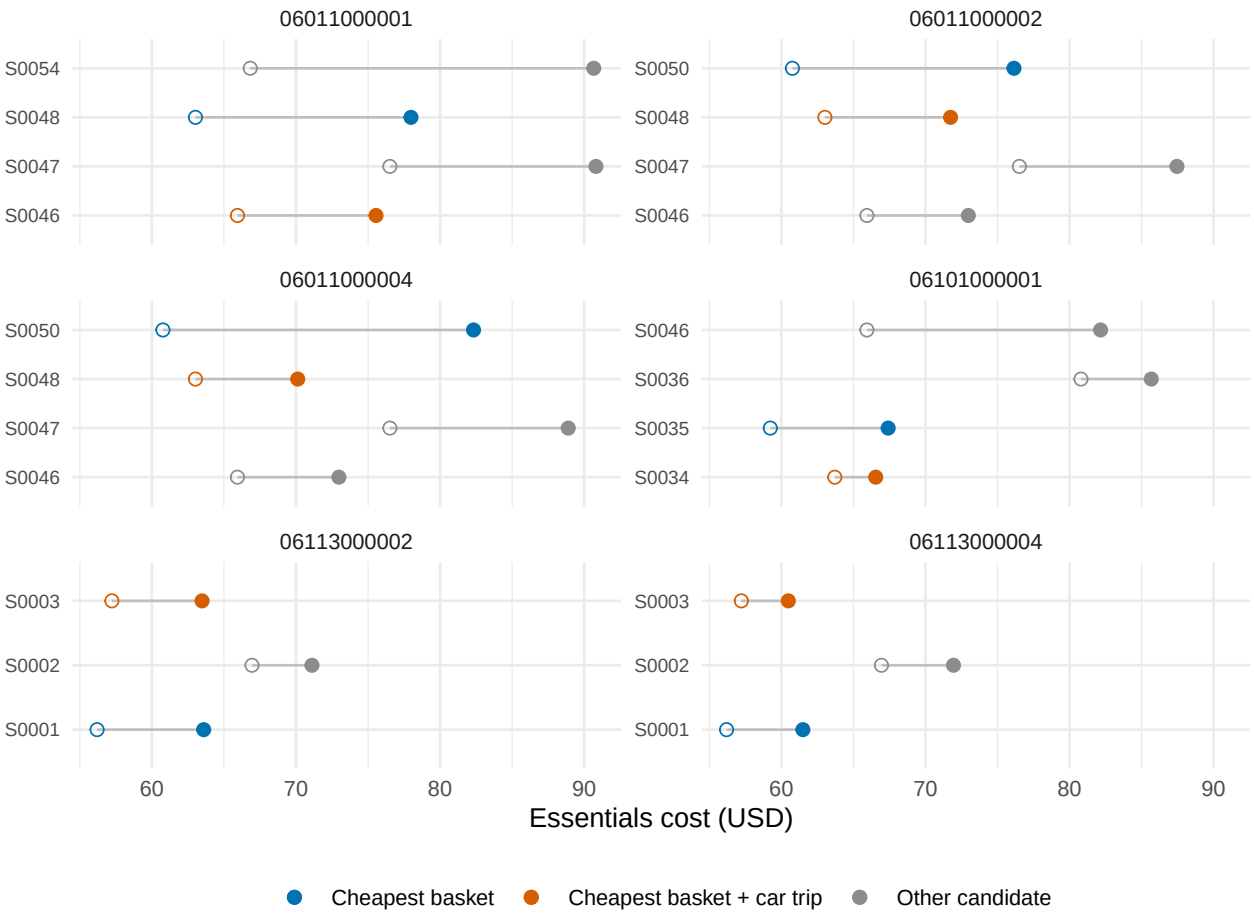
How often does travel change the cheapest store?

Table 38: Changes in the minimizing store; percentages use the 134 tracts with at least one eligible candidate.

Comparison	Tracts	Percent
Car: cheapest store changes after travel	19	14.2
Car: nearest candidate is cheapest after travel	55	41.0
Bus: cheapest store changes after fare (basket-cheapest store on bus)	15	11.2
Bus: basket-cheapest store not reachable by bus	37	27.6
Bus: no bus-accessible eligible store	5	3.7

For households with a vehicle, the cheapest store changes in 19 of 134 evaluable tracts (14.2%). The nearest candidate is also the cheapest after car travel in 55 tracts (41.0%). In the other 79, a farther store is cheaper even after paying for the longer drive, so the nearest store is not always the cheapest trip.

For the bus, the cheapest store changes after the fare in 15 tracts. In another 37 tracts, the store with the cheapest basket has no bus service, so a bus rider cannot reach it at all.



Simulated teaching data. First six of the 19 tracts where the car−optimal store differs; April 15, 2025.

In each panel the blue store has the lowest open circle, but a longer drive pushes its filled circle past the orange store, which ends up the cheapest trip.

Table 39: Simulated 2024 no-vehicle households by tract bus-scenario status.

Bus scenario status	Tracts	No-vehicle households	Share of no-vehicle households (%)
Basket-cheapest store not on bus	37	8141	27.5
Different store than basket-only	15	3614	12.2
No bus-accessible eligible store	5	1138	3.8
Same store as basket-only	77	16754	56.5

The car scenario is hypothetical for households without a vehicle. Weighting tracts by their simulated no-vehicle households shows how many such households live where the bus scenario cannot reach the cheapest basket or any eligible basket.

Interpretation

Transportation can reverse the store ranking. Under the car scenario, the cheapest shelf-price basket is not the cheapest trip in 14.2% of evaluable tracts. Under the bus scenario, the ranking can change because of the fare, and separately because the cheapest basket may not be reachable by bus at all (37 tracts).

A bus alternative with service and a $0 fare is a real free trip and stays in the comparison. An alternative without service has a blank fare and is excluded; treating it as $0 would make unreachable stores look like the cheapest trips. Tracts with no bus-accessible eligible store are reported as not evaluable rather than given a cost.

14. What is the grocery-affordability story for Yolo County?

All values for Yolo (06113) and the comparison counties are simulated. Ranks and medians use only the 24 study counties; rank 1 is the highest value.

This builds one row per county combining store density, owners, income, poverty, no-vehicle share, low-access share, complete essentials basket costs, and the Question 13 travel results. The burden measure assumes 52 identical weekly essentials baskets bought at the county's cheapest complete store, divided by county median household income. It is a rough affordability index: the 12 essentials are not a full grocery budget, the median household is not a low-income household, and travel is not included.

Table 40: Yolo's position among the study counties (rank 1 = highest value).

Measure	Yolo	Study median	Yolo rank	Counties ranked
Stores per 10,000 residents	2.07	2.11	14	24
Distinct owner units	5.00	5.00	7	24
Median household income (USD)	97900.00	78350.00	4	24
Poverty rate (%)	17.60	18.60	16	24
No-vehicle households (%)	10.12	10.89	18	24
Low-access residents (%)	23.58	26.82	15	24
Median complete essentials basket (USD)	57.22	63.57	20	24
52 cheapest baskets / income (%)	2.98	3.36	20	24
Tracts where car trip changes cheapest store (%)	33.33	0.00	3	24
Tracts with no bus-reachable eligible store (%)	0.00	0.00	5	24

Yolo is a high-income county ($97,900, 4th of 24) with slightly below-median poverty (17.6%) and no-vehicle share (10.1%). Its store density (2.07 per 10,000 residents) and number of owners (5) are close to the study median. Its low-access share (23.6%) is below the median of 26.8%.

Groceries are relatively affordable. The median complete essentials basket ($57.22) ranks 20th of 24, so only four counties are cheaper, and 52 of the cheapest baskets take about 3.0% of median income compared with a study median of 3.4%.

The two travel measures need care. In most counties no tract changes its cheapest store after a car trip (the median is 0%), so Yolo's 33.3% (2 of 6 tracts) ranks 3rd. Every Yolo tract can reach an

eligible store by bus. Its rank of 5 on that measure is a tie: 20 counties are at 0%, and `min_rank()` gives all of them the rank after the four counties with a nonzero value.

Table 41: Comparison counties under two selection rules: profile measures.

Set	County	Class	Stores/10k	Income (USD)	Poverty (%)	No vehicle (%)
Yolo	Yolo	Metro	2.07	97900	17.6	10.1
Socioeconomic match	San Mateo	Metro	1.40	92300	18.0	9.7
Socioeconomic match	Sutter	Metro	2.07	92900	18.9	10.8
Socioeconomic match	El Dorado	Metro	1.54	99000	18.7	11.7
Density match	Sutter	Metro	2.07	92900	18.9	10.8
Density match	Glenn	Nonmetro	2.10	59900	20.1	14.0
Density match	Solano	Metro	2.02	67300	15.9	8.7

Table 42: Comparison counties: access, basket, and transportation outcomes.

Set	County	Low access (%)	Complete stores	Median basket (USD)	Burden (%)
Yolo	Yolo	23.6	3	57.22	3.0
Socioeconomic match	San Mateo	20.1	5	69.39	3.4
Socioeconomic match	Sutter	20.5	4	61.47	2.8
Socioeconomic match	El Dorado	32.8	3	63.55	3.2
Density match	Sutter	20.5	4	61.47	2.8
Density match	Glenn	44.6	3	73.49	4.8
Density match	Solano	23.2	3	64.03	4.7

Ranks show where Yolo sits among all 24 counties. The comparison sets show which counties Yolo should be compared with, using two rules:

1) Socioeconomic match: the three metro counties closest to Yolo in standardized income, poverty, and no-vehicle share (San Mateo, Sutter, El Dorado).

2) Density match: the three counties closest to Yolo in stores per 10,000 residents (Sutter, Glenn, Solano).

Among its socioeconomic peers, Yolo's cheapest essentials basket ($56.19) is in the middle: Sutter's is cheaper ($49.16), and San Mateo's ($60.13) and El Dorado's ($61.57) cost more. Its burden of 3.0% is also in the middle. Its low-access share (23.6%) is above San Mateo and Sutter (about 20%) and well below El Dorado (32.8%).

The density rule gives a very different picture. Glenn has nearly the same store density as Yolo, but 44.6% of its residents have low access, its burden is 4.8%, and half of its evaluable tracts have no bus-reachable eligible store. Solano matches on density but has a 4.7% burden. Matching on store counts groups Yolo with counties whose affordability and access are quite different, which agrees with Question 12: more stores per resident does not guarantee better access. The comparison set changes the conclusion, so it has to be stated.

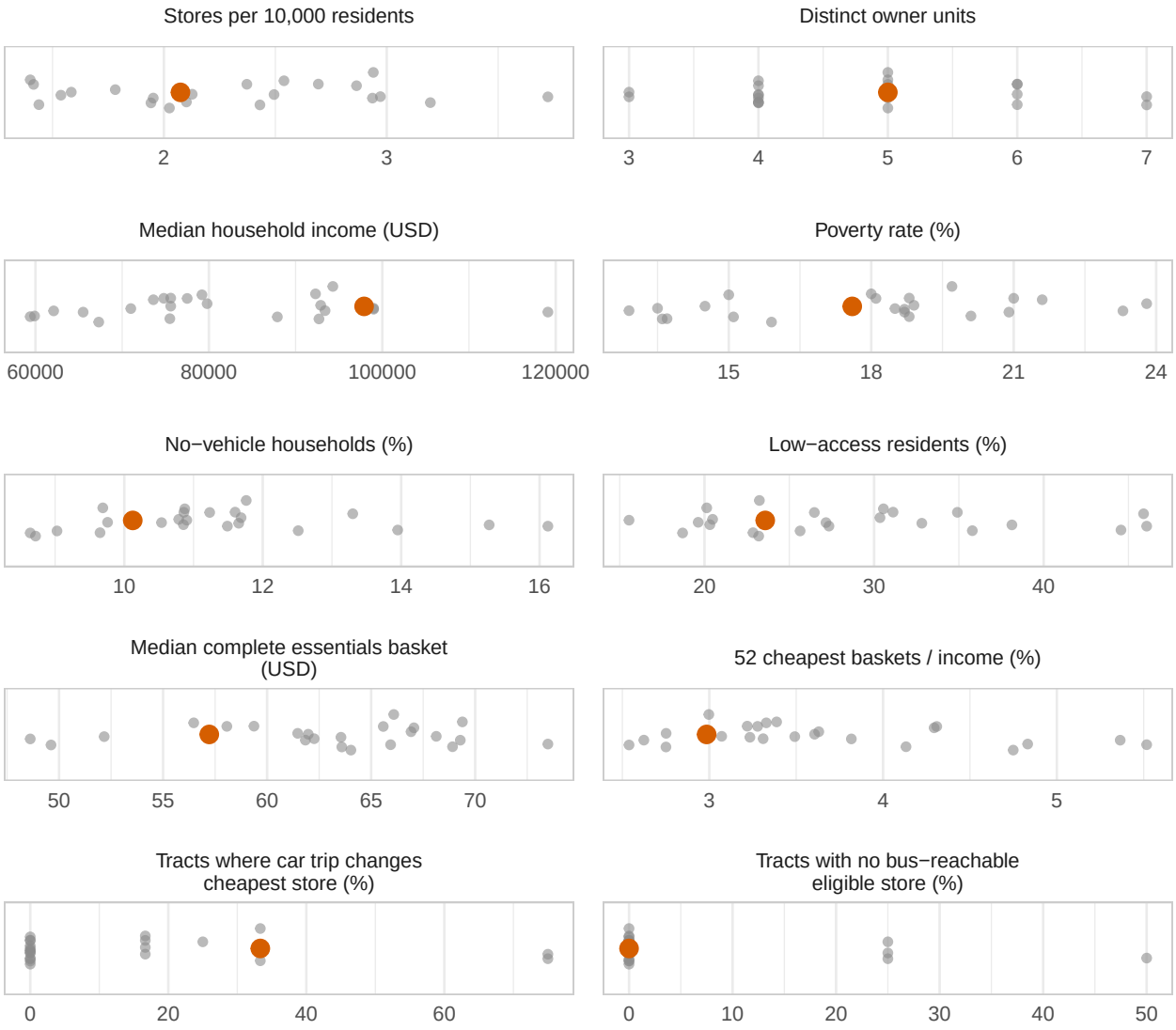


Table 43: Yolo study stores and essentials coverage, April 15, 2025.

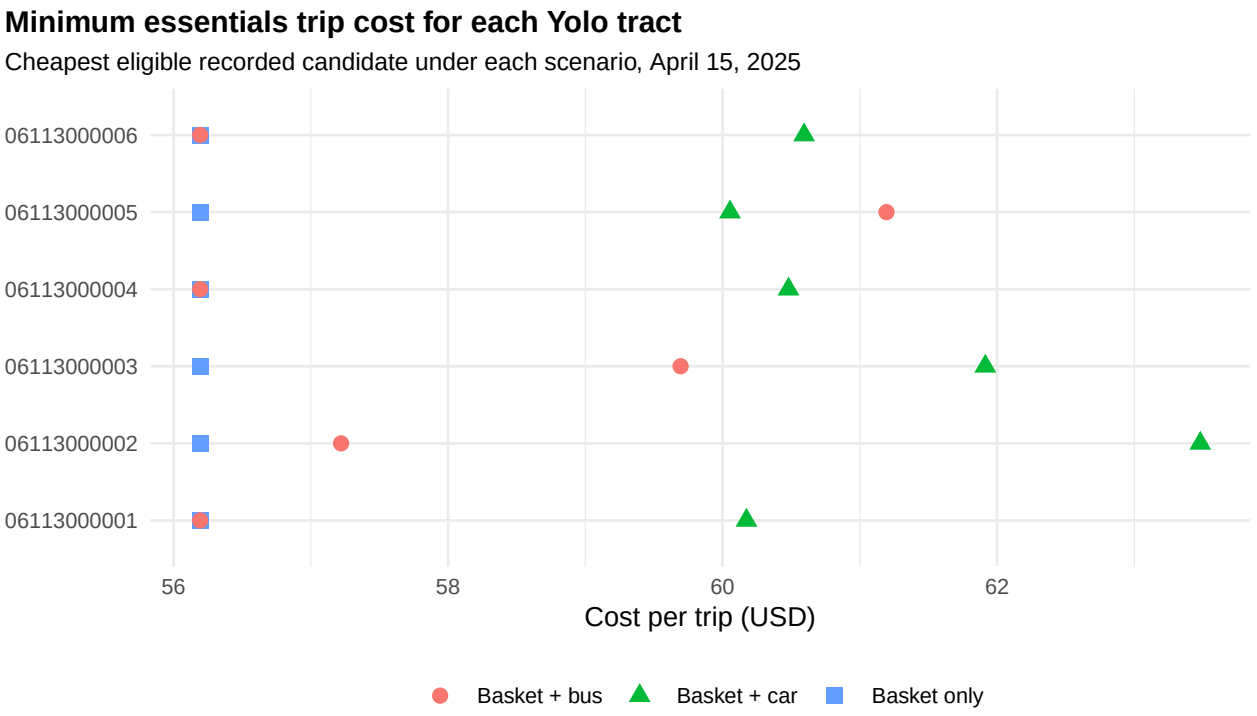
Store	Banner	Type	SNAP	Coverage	Essentials (USD)
S0001	Valley Basket	Supermarket	Yes	12/12	56.19
S0003	Harvest Foods	Supermarket	Yes	12/12	57.22
S0002	GreenGrocer	Supermarket	Yes	12/12	66.95
S0004	Daylight Discount	Discount	Yes	8/12	NA
S0005	SaveMore	Discount	Yes	10/12	NA
S0006	Neighborhood Pantry	Small grocery	Yes	8/12	NA
S0007	Bulk Basket	Warehouse	Yes	7/12	NA

All seven Yolo study stores are SNAP-authorized, but only the three supermarkets have a complete

essentials basket. The two discount stores, the small grocery, and the warehouse store are each missing between 2 and 5 essentials prices, so they cannot be ranked. Daylight Discount had the lower unit price on 21 of 28 shared products in Question 3, yet it is missing 4 of the 12 essentials, so we cannot say whether its basket would be cheapest. The county's basket results describe three supermarkets, not the full store mix.

Table 44: Yolo tracts: access and minimum essentials trip costs (NA = not evaluable).

Tract	Residents	Low access (%)	No-vehicle HH	Basket (USD)	+ car (USD)	+ bus (USD)
06113000001	7052	30.9	286	56.19	60.17	56.19
06113000002	4987	29.8	191	56.19	63.48	57.22
06113000003	8139	13.9	221	56.19	61.91	59.69
06113000004	2990	18.2	127	56.19	60.48	56.19
06113000005	3302	30.4	94	56.19	60.05	61.19
06113000006	7276	22.3	400	56.19	60.59	56.19



Simulated teaching data. Missing points are scenarios that are not evaluable.

Within Yolo, Valley Basket (`S0001`) has the cheapest basket for all six tracts ($56.19). With a car, tracts `06113000002` and `06113000004` switch to Harvest Foods (`S0003`), because a shorter drive outweighs its $1.03 higher basket. By bus, tract `06113000002` also switches to Harvest Foods, where its recorded fare is $0. Three tracts (`...001`, `...004`, `...006`) have a free bus trip to Valley Basket. As a result, the bus trip is cheaper than the car trip in five of the six tracts. Tract-level low-access shares range from 13.9% to 30.9%, and the 1,319 no-vehicle households are concentrated in tracts `...006` (400) and `...001` (286), both of which have a free bus trip to the cheapest store.

Yolo summary

Yolo's grocery affordability is good relative to the study counties, mostly because of high income and one low-priced complete supermarket. Its cheapest essentials basket is affordable at about 3.0% of median income, and adding the median car markup of $4.34 per trip raises this only to about 3.2%. Simulated bus service reaches an eligible store from every tract, often for free. There are two limitations. About a quarter of residents are still classified as low access, and the cheapest store per product (a discount store) cannot be evaluated as a basket because of missing prices. The county-level burden and ranks also hide the experience of low-income households. Since all values are simulated and the county has only 6 tracts and 7 stores, these results describe this scenario, not the real Yolo County.

15. How can a technically successful join produce statistically incorrect results?

Join audits

Table 45: Join keys and intended relationships.

Join	Key	Type	Key unique (right)
Prices to stores	store_id	many-to-one	TRUE
Prices to products	product_id	many-to-one	TRUE
Store counts to county context	county_fips	one-to-one	TRUE
Tract summary to county context	county_fips	one-to-one	TRUE
Travel to tracts	tract_id	many-to-one	TRUE
Travel to stores	store_id	many-to-one	TRUE

Table 46: Row counts before and after each left join, and unmatched keys.

Join	Rows left	Rows right	Rows after	Unmatched left	Unmatched right
Prices to stores	8980	145	8980	0	0
Prices to products	8980	40	8980	0	0
Store counts to county context	24	24	24	0	0
Tract summary to county context	24	24	24	0	0
Travel to tracts	670	134	670	0	0
Travel to stores	670	145	670	0	30

Each join that the analysis relies on uses a key that is unique on the right side, keeps the left row count, and leaves no left rows unmatched. The 30 unmatched stores in the travel-to-stores join are stores that are not a recorded alternative for any tract. They are expected and do not mean the join failed. Passing these checks only shows that a join is mechanically correct. The two operations below also run without errors, but each one changes what the rows represent.

Incorrect operation 1: linking tracts to stores by county only

This joins tracts to stores by county, as many-to-many. Because a county-only join pairs every tract with every store in its county, it creates tract-store pairs that are not recorded alternatives. It also drops recorded alternatives where the store is in a different county from the tract. The anti-join row in the table counts those lost alternatives.

Table 47: Recorded travel links versus a county-only store-tract join.

Measure	Recorded links	County-only join
Tract-store links	670	830.0
Links that are recorded travel alternatives	670	631.0
Recorded alternatives missing (cross-county)	0	39.0
Mean choices per tract (tracts with any link)	5	6.2
Tracts with at least one complete essentials store	134	134.0

The county-only join produces 830 links instead of the 670 recorded ones. Only 631 of them are real alternatives, so 199 are made-up pairs with no travel cost, and 39 real cross-county alternatives are lost. The average tract appears to have 6.19 choices instead of 5. A quick check would not catch this, because both versions give all 134 tracts at least one complete essentials store. Any Question 13 minimum computed from these links would include stores a tract was never recorded as using and leave out cross-county stores it was.

Incorrect operation 2: summing population after a one-to-many join

The travel file has one row per tract-store alternative, so joining tract attributes onto it repeats each tract's population once for every alternative. The join is a valid many-to-one join, but summing population afterward counts each resident several times.

Table 48: Total simulated 2024 tract population, computed two ways.

Method	Rows	Residents
Sum over the tract file (one row per tract)	134	684739
Sum after joining tracts onto travel alternatives	670	3423695

The joined table reports 3,423,695 residents, 5.0 times the true 684,739. Any population-weighted rate computed from it would also weight tracts by how many alternatives they have rather than by residents. Population totals should be computed at the tract level first and joined afterward, as Questions 10 and 13 do.

Both examples show that a join can run without errors and still give wrong statistics. The relationship type, the meaning of the resulting row, and the denominator have to be checked along with the row counts.

16. Formulate an original substantive research question.

Question. Do higher-poverty tracts face a larger travel penalty on the cheapest complete essentials trip, and are they more often unable to reach the cheapest basket by bus?

Chain.

- **Data:** stores (01), products (03), prices (04), basket items (05), county context (06), tracts (08), and travel (09).
- **Cleaning:** unit prices from Q2; complete `essentials_12` baskets only (Q4).
- **Join:** recorded tract–store alternatives only (Q13), with tract poverty from file 08 and county urban/rural class from file 06.
- **Unit of analysis:** the tract.

Table 49: Variables and denominators for Question 16.

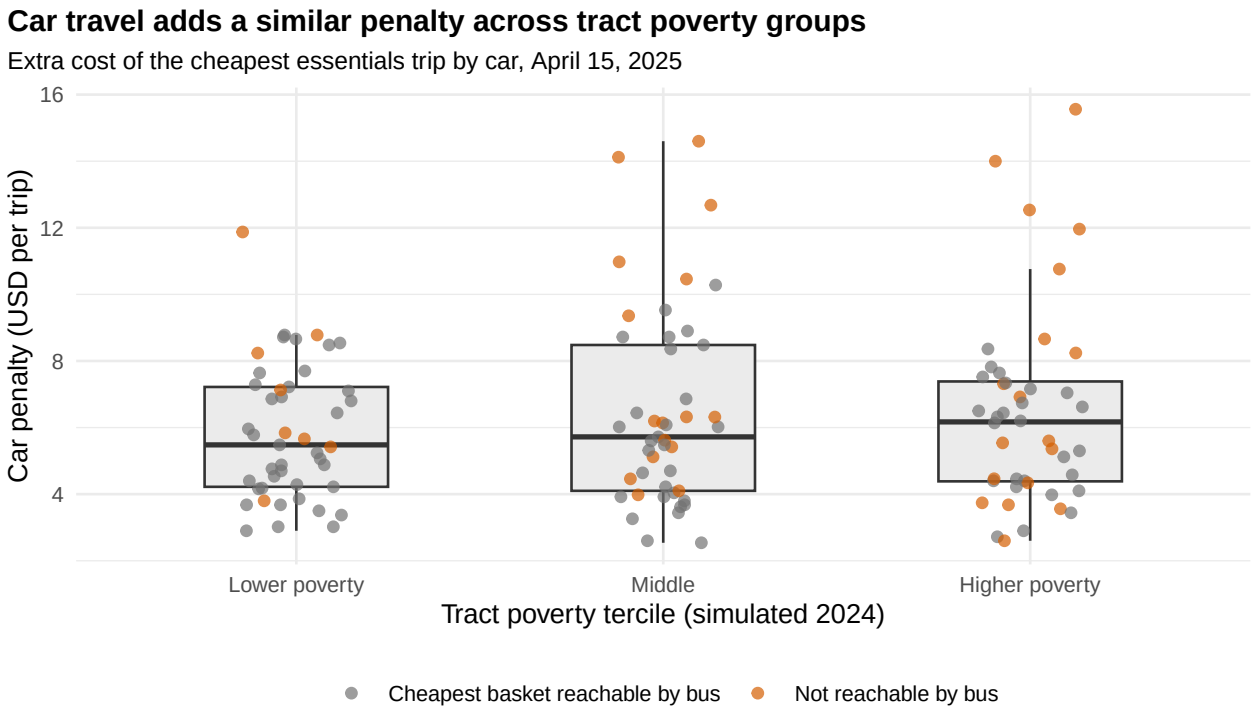
Variable	Definition	Denominator
Poverty group	Tercile of tract poverty rate (file 08)	Evaluable tracts
Car penalty (USD)	Min (basket + car) minus min basket	Evaluable tracts
Car changes store (%)	Cheapest store differs after car trip	Evaluable tracts in group
No bus to cheapest (%)	Cheapest basket not reachable by bus, or no bus-eligible store	Evaluable tracts in group
Eligible candidates	Recorded alternatives with a complete essentials basket	Per tract

Table 50: Evaluable tracts by poverty tercile, essentials bas
April 15, 2025.

Group	Tracts	Median eligible stores	Median poverty (%)	Median car penalty (USD)	IQR p
Lower poverty	45	4	10.3	5.48	
Middle	45	3	17.2	5.72	
Higher poverty	44	4	25.4	6.17	

Each tercile has 44 or 45 evaluable tracts with a median of 3 or 4 eligible stores, so the groups have similar choice sets. The median car penalty rises slightly from \$5.48 (lower poverty) to \$5.72 (middle) and \$6.17 (higher poverty). This \$0.69 difference is small compared with within-group IQRs of \$3 to \$4, and the Spearman correlation between poverty and penalty is only 0.09. The share of tracts where the car trip changes the cheapest store actually falls with poverty (17.8% to 9.1%).

The clearest pattern is in bus access. The share of tracts that cannot reach the cheapest basket by bus rises from 17.8% to 35.6% to 40.9% across the terciles. Higher-poverty tracts do not pay much more per car trip, but they are about twice as likely to be cut off from the cheapest basket by bus.



Simulated teaching data. 134 tracts with at least one eligible candidate; one point per tract.

Table 51: Sensitivity checks for Question 16.

Check	Group	Tracts	Median penalty (USD)	Car changes store (%)	No bus
Metro only (terciles)	Lower poverty	38	5.33	18.4	
Metro only (terciles)	Middle	38	5.37	5.3	
Metro only (terciles)	Higher poverty	38	5.45	7.9	
All tracts (median split)	At/above median	67	6.08	13.4	
All tracts (median split)	Below median	67	5.66	14.9	

The Spearman correlation between tract poverty and car penalty is 0.09 across all 134 evaluable tracts and -0.00 within metro counties (114 tracts).

Robustness. The small car-penalty gradient does not survive the metro-only check. Within metro counties, the median penalty is nearly flat ($5.33, $5.37, $5.45) and the correlation is about zero. The overall gradient comes from the 20 nonmetro tracts, which have a median penalty of about $10.40 and are mostly in the middle and higher terciles (18 of 20). The bus-access gradient does hold. Within metro counties, the no-bus share still rises from 13.2% to 21.1% to 28.9%, and the median split gives 23.9% below the median versus 38.8% at or above it.

Alternative explanation. Poverty is partly a proxy for location. Nonmetro tracts have higher poverty, longer drives, and almost no bus service to the cheapest store (90% of them), so much of the raw difference reflects geography rather than poverty. The metro-only check removes the urban/rural difference but not differences in location within metro counties, such as distance from a transit line.

Limitations. The car scenario is hypothetical for no-vehicle households, who are the group most affected by bus access. Penalties are per trip and are not scaled to household income or shopping

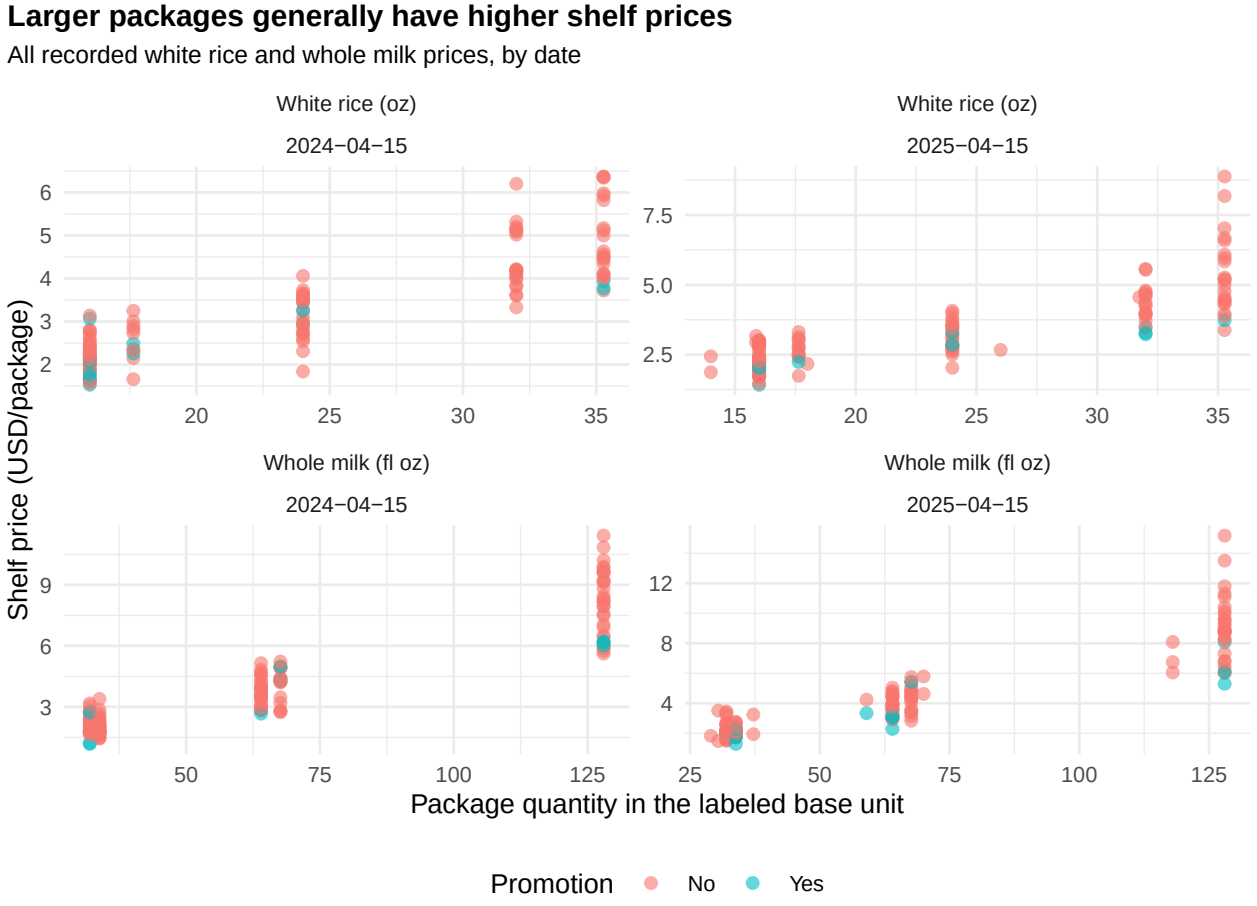
frequency. Bus travel time is not priced. Terciles of about 45 tracts mean that one tract moves a percentage by about 2 points. The data are simulated and cross-sectional, so the results are descriptive associations, not causal effects.

Answer. Higher-poverty tracts do not face a meaningfully larger car penalty once urban/rural location is accounted for. They are, however, more often unable to reach the cheapest complete essentials basket by bus, and this pattern holds within metro counties.

17. Construct a final integrated statistical story.

"Which grocery store is cheapest?" depends on what is compared: a package, a unit price, a complete basket, or a trip from where someone lives. The six figures below build that answer step by step. They draw on stores, products, prices, basket items, county context, tracts, and travel (7 of the 9 files). All data are simulated.

Figure 1: Package size drives shelf price

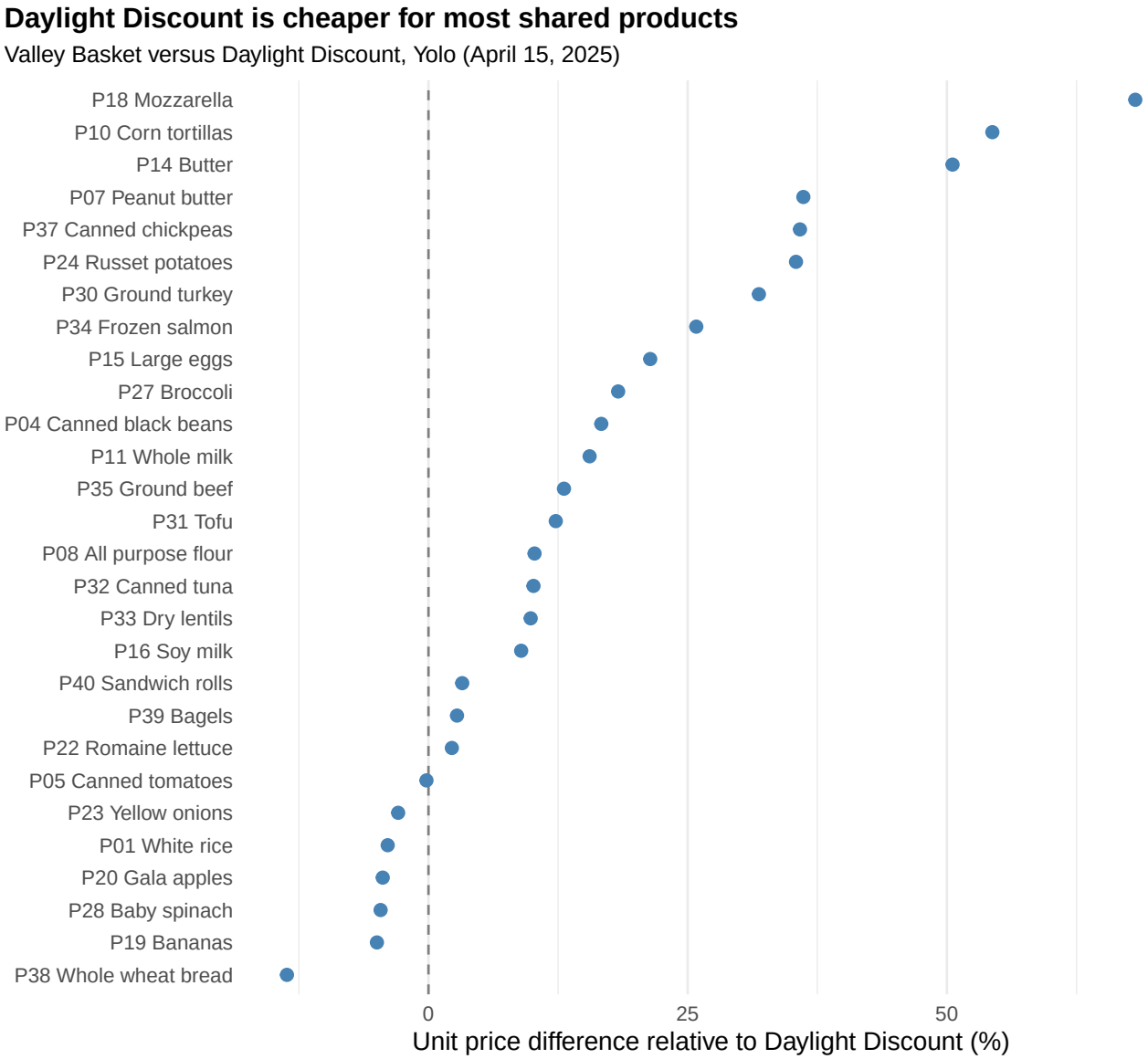


Simulated teaching data. Rice: n = 121 (2024), 126 (2025); milk: n = 128 (2024), 132 (2025).

(a) Does a higher shelf price mean a more expensive product? (b) A scatterplot shows the relationship between two numeric variables, package quantity and price, for every record.

(c) Larger packages cost more, so shelf prices are not comparable across sizes. (d) This is the starting point: the comparison has to be per base unit. (e) The figure shows only two products, and promotions add more variation.

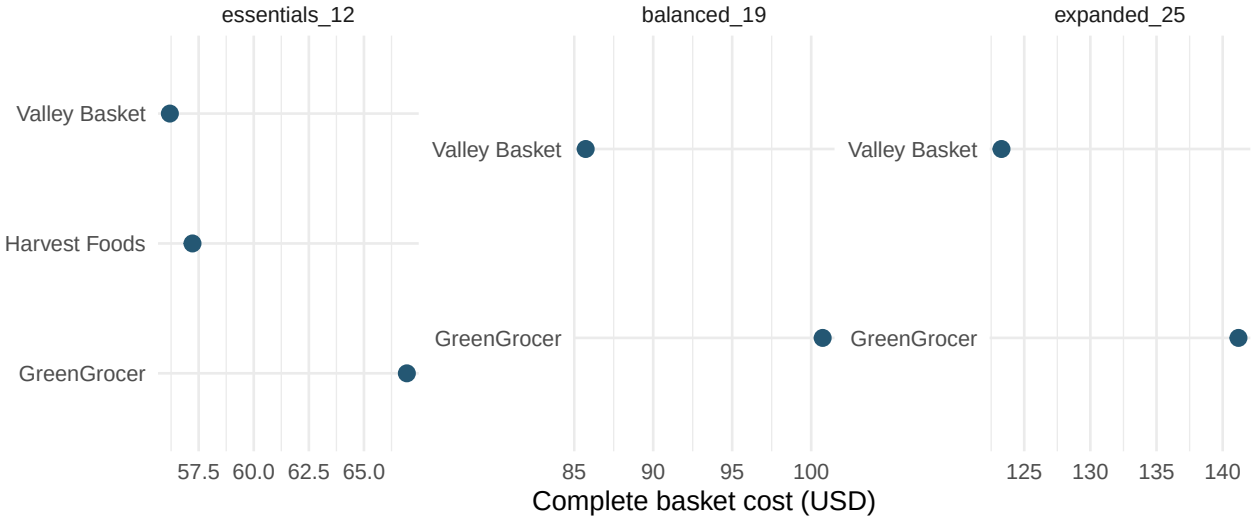
Figure 2: Matched unit prices favor one store, but not for every product



Simulated data. All 28 shared products with promotions included.

(a) Which of two Yolo stores is cheaper per unit for the same products? (b) A dot plot of paired percentage differences compares the stores only on products both of them carry. (c) Daylight Discount is cheaper for 21 of 28 shared products. (d) Once unit prices make the comparison fair, the cheaper store still differs from product to product, and winning most products does not mean winning the basket: Daylight Discount is missing 4 of the 12 essentials, so it cannot be ranked in Figure 3. (e) Only shared products are compared, so the result depends on overlap and on whether promotions are included.

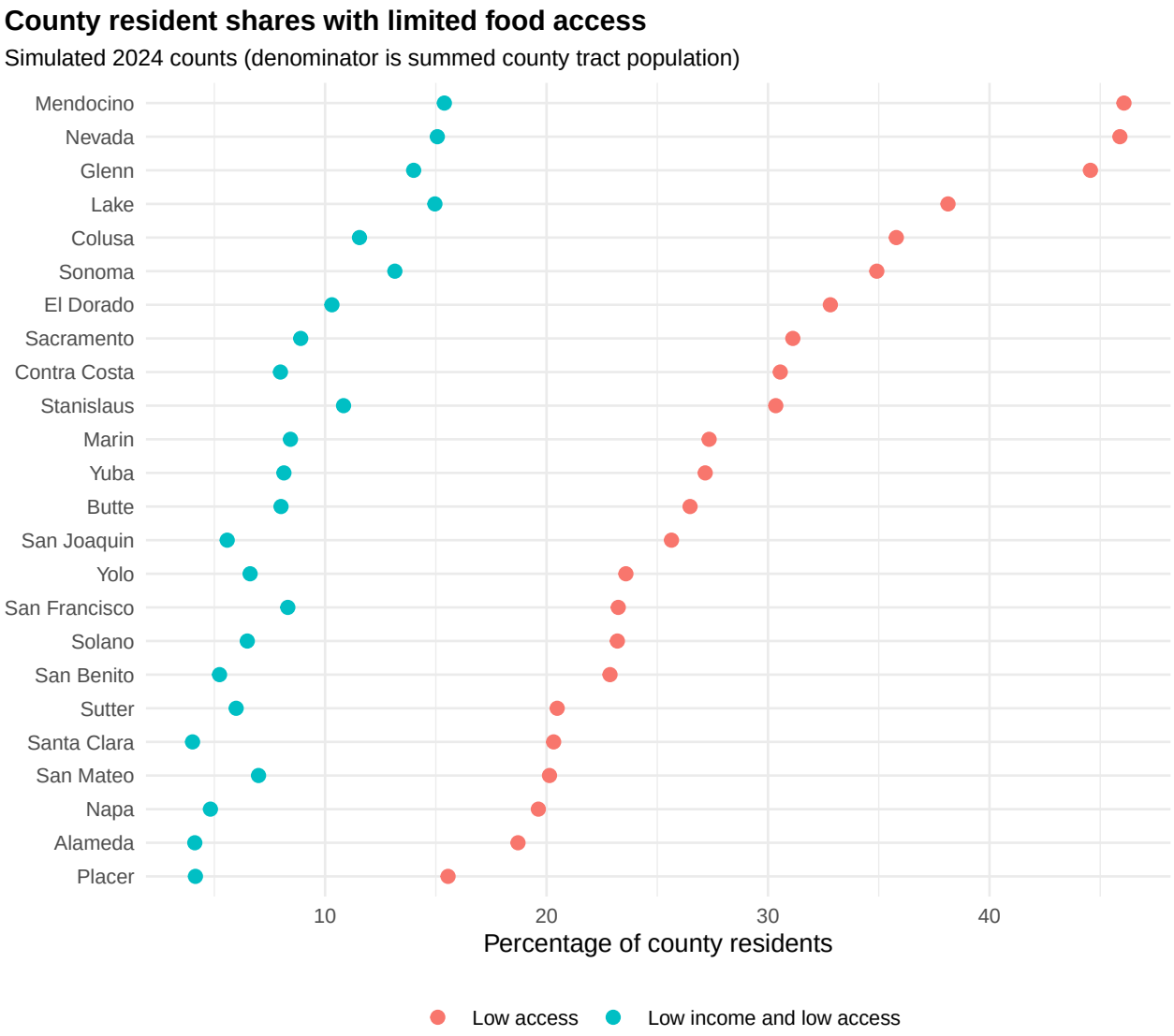
Figure 3: Complete baskets



Simulated teaching data.
Eligible stores: essentials n = 3; balanced n = 2; expanded n = 2.

(a) Which store is cheapest for a fixed list of products in fixed quantities? (b) A dot plot of complete basket costs, faceted by basket, keeps each comparison within one basket. (c) Valley Basket is cheapest for every basket it can be ranked on. The essentials gap is $10.75. (d) Quantity weights and completeness now matter, not just how many products a store wins. (e) Stores missing any price cannot be ranked, so the eligible sample shrinks from 3 stores to 2 for the larger baskets.

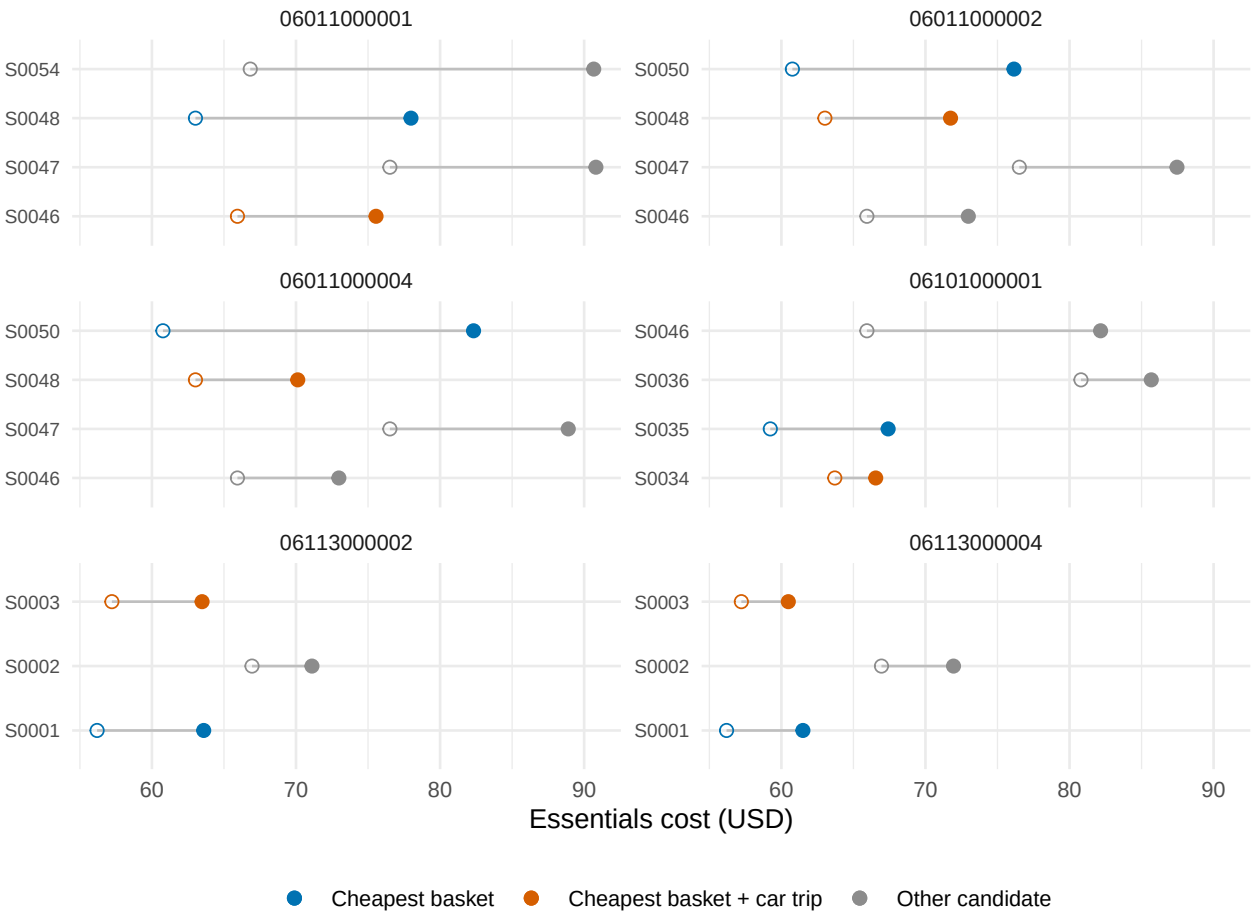
Figure 4: Where people live



Simulated data. 134 constructed tracts across 24 counties.

(a) What share of each county's residents have limited food access? (b) Resident-weighted percentages, shown as dots by county, answer a question about people rather than tracts. (c) County low-access shares range from 15.6% to 46.1%. (d) A cheap basket only helps if residents can get to the store, so this shifts the focus from stores to households. (e) These are county aggregates, so they do not describe any single household (the ecological fallacy).

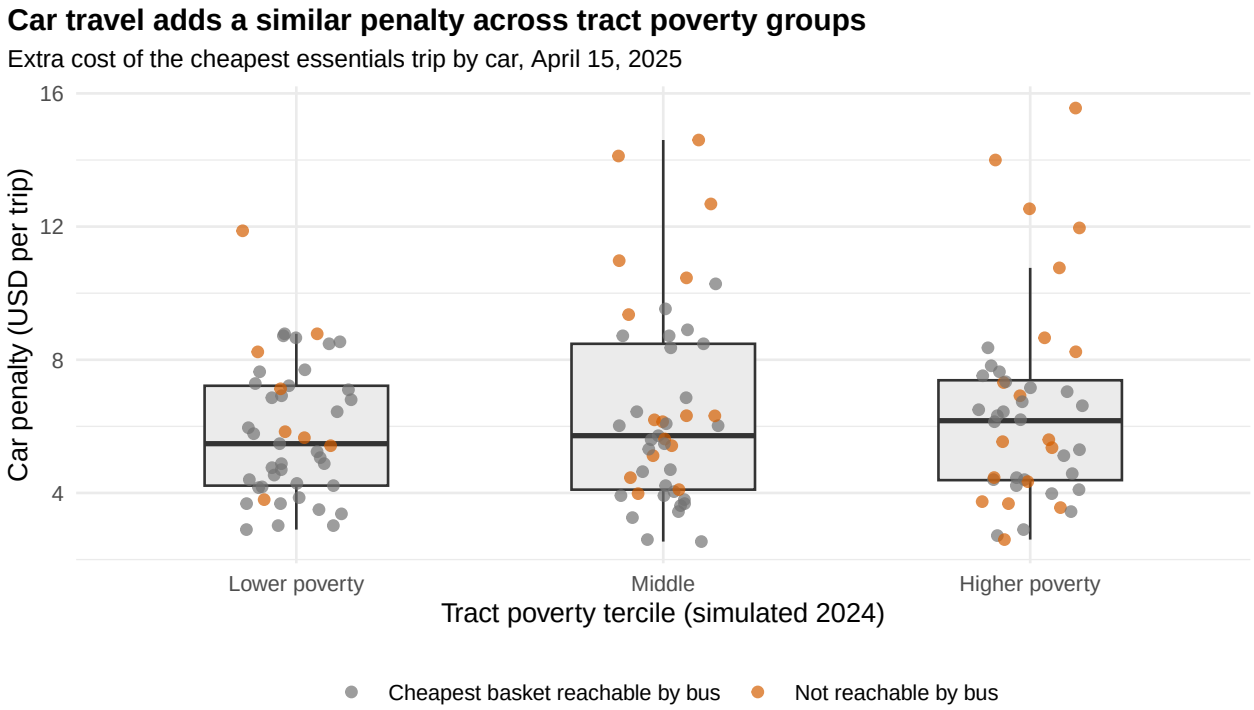
Figure 5: Travel can reverse the ranking



Simulated teaching data. First six of the 19 tracts where the car−optimal store differs; April 15, 2025.

(a) Does adding the trip cost change which store is cheapest? (b) A dot plot per candidate store, for six example tracts, shows basket-only and basket-plus-car cost on the same dollar scale. (c) The car-optimal store differs from the basket-optimal store in 14.2% of evaluable tracts. (d) The answer now depends on where the shopper starts. (e) Travel costs are hypothetical, and bus time is not priced.

Figure 6: Who bears the travel penalty



Simulated teaching data. 134 tracts with at least one eligible candidate; one point per tract.

(a) Is the travel penalty larger in higher-poverty tracts? (b) Boxplots with jittered tract points compare distributions across groups without hiding sample size. (c) The car penalty barely changes with poverty (Spearman 0.09, about zero within metro counties), but the orange points, tracts that cannot reach the cheapest basket by bus, become more common: 17.8% of lower-poverty tracts compared with 40.9% of higher-poverty tracts. (d) The burden of travel shows up as lost bus access to the cheapest store more than as a higher car cost. (e) The car scenario is hypothetical for no-vehicle households, and poverty partly stands in for nonmetro location.

Omitted figures

The Q5–Q9 and Q11–Q12 figures (summary choices, price changes, store types, county density, alternative indicators, and county correlations) describe the store environment rather than the cheapest trip. Several of them rest on only 24 counties, so they were left out to keep the story focused.

Answer

No single shelf-price comparison identifies the cheapest store, because the answer changes with each step:

- **Unit price** removes package-size differences, but it ranks one product at a time.

- **Complete basket cost** combines unit prices with fixed quantities, and it can rank only stores with complete coverage.
- **Transportation-adjusted cost** adds the trip, so the cheapest store differs by tract and by mode of travel.
- **Access and affordability** ask whether a household can reach that store and pay for the trip relative to its resources, which none of the store-level measures capture.

The study cannot conclude anything about real households or retailers. The prices, stores, populations, and travel costs are simulated. Price coverage is incomplete. Measures come from different vintages (2022 indicators, 2024 profiles, 2025 prices). The relationships are cross-sectional and observational, not causal.

Appendix

A. Key definitions and rules

- **Unit price** U_{sjt}: shelf price divided by package quantity, after converting to the product's comparison base unit (Q2).
- **Complete basket cost** $B_{sbt} = \sum_{j \in J_b} q_{bj} U_{sjt}$: computed only when store s has a price for every required product on date t. Missing prices are never treated as zero (Q4).
- **Travel-adjusted cost** $T_{ksm} = B_{s,\text{essentials},2025} + F_{ksm}$: computed only for the recorded tract-store alternatives in file 09, never for county-only pairings (Q13, Q15).
- **Bus eligibility**: an alternative without bus service is excluded, not given a \$0 fare. A recorded \$0 fare with service is a valid free trip (Q13).
- **Weighted shares**: county access percentages sum residents across tracts before dividing, so they describe people rather than tracts (Q10).
- **Ties**: when two stores tie on a minimum, the lower `store_id` is kept; no such ties occur in the 2025 essentials data (Q13).

B. Reproducibility

The report knits from `HW2.Rmd` with the ten CSV files in the same folder. The original files are read once and never modified. All jitter uses seed 141. The code below is every chunk in the order it runs; chunks in Question 17 only reuse earlier figures and are omitted.

C. Code

setup

```
library(tidyverse)
library(kableExtra)
knitr::opts_chunk$set(
  echo = FALSE,
  message = FALSE,
  warning = TRUE,
  fig.width = 6.5,
  fig.height = 4.5,
  out.width = "\\linewidth",
  fig.align = "center"
)

theme_set(
  theme_minimal(base_size = 10) +
    theme(
      plot.title = element_text(size = 11, face = "bold"),
      plot.subtitle = element_text(size = 9),
      plot.caption = element_text(size = 8, hjust = 0),
      plot.title.position = "plot",
```

```
      plot.caption.position = "plot"
      )
  )

# Reproducible placement when using jitter or random sampling later
set.seed(141)

# CSV files are in same folder as this Rmd
data_dir <- "."

report_table <- function(data, caption = NULL,
                         digits = 2, col.names = colnames(data)) {
  knitr::kable(
    data,
    format = "latex",
    booktabs = TRUE,
    longtable = TRUE,
    row.names = FALSE,
    linesep = "",
    digits = digits,
    col.names = col.names,
    caption = caption
  ) |>
    kableExtra::kable_styling(
      latex_options = "repeat_header",
      position = "center",
      font_size = 10
    ) |>
    kableExtra::row_spec(0, bold = TRUE)
}
```

import-data

```
# Preserve ids as character variables at import including leading 0's
# Other numeric fileds are parsed

stores_raw <- read_csv(
  file.path(data_dir, "01_stores.csv"),
  col_types = cols(
    store_id = col_character(),
    banner_id = col_character(),
    ownership_unit_id_sim = col_character(),
    county_fips_raw = col_character(),
    opened_year_sim = col_integer()
  )
)
```

```r
banners_raw <- read_csv(
  file.path(data_dir, "02_banner_ownership.csv"),
  col_types = cols(
    banner_id = col_character()
  )
)

products_raw <- read_csv(
  file.path(data_dir, "03_products.csv"),
  col_types = cols(
    product_id = col_character()
  )
)

prices_raw <- read_csv(
  file.path(data_dir, "04_price_observations.csv"),
  col_types = cols(
    observation_id = col_character(),
    store_id = col_character(),
    product_id = col_character(),
    observation_date = col_date(format = "%Y-%m-%d"),
    package_quantity = col_double(),
    shelf_price_usd = col_double()
  )
)

basket_items_raw <- read_csv(
  file.path(data_dir, "05_basket_items.csv"),
  col_types = cols(
    basket_id = col_character(),
    product_id = col_character(),
    quantity_in_base_units = col_double()
  )
)

county_context_raw <- read_csv(
  file.path(data_dir, "06_county_context.csv"),
  col_types = cols(
    county_fips = col_character(),
    population_2024_sim = col_integer(),
    households_2024_sim = col_integer(),
    median_household_income_2024_usd_sim = col_double(),
    poverty_rate_2024_pct_sim = col_double(),
    households_no_vehicle_2024_sim = col_integer()
  )
)
```

```r
food_environment_raw <- read_csv(
  file.path(data_dir, "07_county_food_environment.csv"),
  col_types = cols(
    county_fips_raw = col_character(),
    population_2022_est_sim = col_integer(),
    food_outlets_2022_est_sim = col_integer(),
    food_outlets_per_10000_2022_sim = col_double(),
    low_access_residents_share_2022_pct_sim = col_double(),
    indicator_reference_year = col_integer()
  )
)

tracts_raw <- read_csv(
  file.path(data_dir, "08_tract_food_access.csv"),
  col_types = cols(
    tract_id = col_character(),
    tract_population_2024_sim = col_integer(),
    households_2024_sim = col_integer(),
    low_access_residents_2024_sim = col_integer(),
    low_income_low_access_residents_2024_sim = col_integer(),
    households_no_vehicle_2024_sim = col_integer(),
    tract_poverty_rate_pct_2024_sim = col_double()
  )
)

travel_raw <- read_csv(
  file.path(data_dir, "09_tract_store_travel.csv"),
  col_types = cols(
    tract_id = col_character(),
    store_id = col_character(),
    driving_distance_oneway_miles_sim = col_double(),
    bus_roundtrip_minutes_sim = col_double(),
    bus_roundtrip_fare_usd_sim = col_double(),
    car_roundtrip_cost_usd_sim = col_double()
  )
)

data_dictionary <- read_csv(
  file.path(data_dir, "10_data_dictionary.csv"),
  col_types = cols(.default = col_character())
)

# Keep original imported tables together for Q1

raw_data <- list(
  stores = stores_raw,
  banners = banners_raw,
```

```
  products = products_raw,
  prices = prices_raw,
  basket_items = basket_items_raw,
  county_context = county_context_raw,
  food_environment = food_environment_raw,
  tracts = tracts_raw,
  travel = travel_raw
)

# stop knitting if an import makes a parsing problem
walk(raw_data, stop_for_problems)
stop_for_problems(data_dictionary)
```

working-data

```
stores <- stores_raw |>
  mutate(
    county_fips = str_pad(county_fips_raw, width = 5, pad = "0")
  )

food_environment <- food_environment_raw |>
  mutate(
    county_fips = str_pad(county_fips_raw, width = 5, pad = "0")
  )

tracts <- tracts_raw |>
  mutate(
    county_fips = substr(tract_id, 1, 5)
  )

banners <- banners_raw
products <- products_raw
prices <- prices_raw
basket_items <- basket_items_raw
county_context <- county_context_raw
travel <- travel_raw

# Check
stopifnot(
  all(str_detect(county_context$county_fips, "^[0-9]{5}$")),
  all(str_detect(stores$county_fips, "^[0-9]{5}$")),
  all(str_detect(food_environment$county_fips, "^[0-9]{5}$")),
  all(str_detect(tracts$tract_id, "^[0-9]{11}$")),
  all(str_detect(travel$tract_id, "^[0-9]{11}$")),
  all(stores$county_fips %in% county_context$county_fips),
  all(food_environment$county_fips %in% county_context$county_fips),
  all(tracts$county_fips %in% county_context$county_fips)
```

```
)
```

q1-inventory

```
# Count rows and columns in original tables
inventory <- tibble(
  Dataset = c(
    "01 Stores",
    "02 Banners",
    "03 Products",
    "04 Prices",
    "05 Basket items",
    "06 County context",
    "07 Food environment",
    "08 Tract access",
    "09 Travel"
  ),
  Rows = map_int(raw_data, nrow),
  Columns = map_int(raw_data, ncol)
)

report_table(
  inventory,
  caption = "Dimensions of the nine original substantive datasets."
)
```

q1-key-audit

```
# List proposed key for each table in raw_data
key_columns <- list(
  stores = "store_id",
  banners = "banner_id",
  products = "product_id",
  prices = c("store_id", "product_id", "observation_date"),
  basket_items = c("basket_id", "product_id"),
  county_context = "county_fips",
  food_environment = "county_fips_raw",
  tracts = "tract_id",
  travel = c("tract_id", "store_id")
)

# to apply same audit to every table
audit_key <- function(data, key) {
  key_values <- select(data, all_of(key))

  tibble(
    Rows = nrow(data),
```

```
    Distinct = nrow(distinct(key_values)),
    Repeated = sum(duplicated(key_values)),
    Missing = sum(!complete.cases(key_values))
  )
}

key_audit <- map2_dfr(raw_data, key_columns, audit_key) |>
  mutate(Dataset = inventory$Dataset, .before = 1)

report_table(
  key_audit,
  caption = "Audit of the proposed keys listed above."
)
```

q1-missingness

```
# count missing values separately for every column of every dataset
missingness <- imap_dfr(raw_data, function(data, dataset_name) {
  tibble(
    Dataset = dataset_name,
    Field = names(data),
    Rows = nrow(data),
    Missing = map_int(data, function(x) sum(is.na(x))),
    Percent = 100 * Missing / Rows
  )
})

# display important fileds

important_fields <- tribble(
  ~Dataset, ~Field, ~Measure,
  "prices", "shelf_price_usd", "Shelf price",
  "prices", "package_quantity", "Package quantity",
  "county_context", "population_2024_sim", "County population",
  "food_environment", "low_access_residents_share_2022_pct_sim", "2022 low-access share",
  "tracts", "low_access_residents_2024_sim", "Tract low-access count",
  "travel", "bus_roundtrip_minutes_sim", "Bus round-trip time",
  "travel", "bus_roundtrip_fare_usd_sim", "Bus round-trip fare"
)

missingness_display <- important_fields |>
  left_join(missingness, by = c("Dataset", "Field")) |>
  select(Measure, Rows, Missing, Percent)

report_table(
  missingness_display,
  digits = 1,
```

```
  caption = "Missingness in selected fields (percentages use rows in each file)."
)
```

q1-bus-check

```
bus_check <- travel |>
  group_by(bus_service_available_sim) |>
  summarise(
    Alternatives = n(),
    Missing_time = sum(is.na(bus_roundtrip_minutes_sim)),
    Missing_fare = sum(is.na(bus_roundtrip_fare_usd_sim)),
    Zero_fare = sum(bus_roundtrip_fare_usd_sim == 0, na.rm = TRUE),
    .groups = "drop"
  )

report_table(
  bus_check,
  col.names = c(
    "Bus service", "Alternatives", "Missing time",
    "Missing fare", "Zero fare"
  ),
  caption = "Bus missingness and recorded zero fares by service availability."
)
```

q1-relationships

```
relationships <- tribble(
  ~From, ~To, ~Key, ~Relationship,
  "Prices", "Stores", "store_id", "Many-to-one",
  "Prices", "Products", "product_id", "Many-to-one",
  "Stores", "Banners", "banner_id", "Many-to-one",
  "Stores", "County context", "county_fips", "Many-to-one",
  "Basket items", "Products", "product_id", "Many-to-one",
  "Food environment", "County context", "county_fips", "One-to-one",
  "Tracts", "County context", "county_fips", "Many-to-one",
  "Travel", "Tracts", "tract_id", "Many-to-one",
  "Travel", "Stores", "store_id", "Many-to-one"
)

report_table(
  relationships,
  caption = "Intended relationships between datasets."
)
```

q1-invalid-join-size

```
# count the combos a county only store tract join would create
# without constructing the expanded table
stores_by_county <- stores |>
  count(county_fips, name = "n_stores")

tracts_by_county <- tracts |>
  count(county_fips, name = "n_tracts")

county_combinations <- stores_by_county |>
  inner_join(tracts_by_county, by = "county_fips") |>
  mutate(combinations = n_stores * n_tracts)

join_comparison <- tibble(
  Method = c(
    "All within-county store-tract combinations",
    "Recorded tract-store travel alternatives"
  ),
  Rows = c(
    sum(county_combinations$combinations),
    nrow(travel)
  )
)

report_table(
  join_comparison,
  caption = "Different row counts reflect different definitions of a link."
)
```

q2-standardize

```
# each package unit maps to compatible base unit and conversion factor
conversions <- tribble(
  ~package_unit, ~converted_unit, ~factor,
  "oz", "oz", 1,
  "lb", "oz", 16,
  "g", "oz", 0.03527396195,
  "kg", "oz", 35.27396195,
  "fl_oz", "fl_oz", 1,
  "liter", "fl_oz", 33.8140227,
  "count", "count", 1,
  "dozen", "count", 12
)

prices_units <- prices |>
  left_join(
    products,
    by = "product_id",
```

```
    relationship = "many-to-one"
  ) |>
  left_join(
    conversions,
    by = "package_unit",
    relationship = "many-to-one"
  ) |>
  mutate(
    quantity_base = package_quantity * factor,
    valid = !is.na(comparison_base_unit) &
      !is.na(converted_unit) &
      !is.na(quantity_base) &
      !is.na(shelf_price_usd) &
      converted_unit == comparison_base_unit &
      quantity_base > 0 &
      shelf_price_usd > 0,
    unit_price = if_else(
      valid, shelf_price_usd / quantity_base, NA_real_
    )
  )

# confirm joins preserve rows and conversions valid
stopifnot(
  nrow(prices_units) == nrow(prices),
  all(prices_units$valid)
)
```

q2-date-summary

```
date_summary <- prices_units |>
  group_by(observation_date) |>
  summarise(
    Observations = n(),
    Products = n_distinct(product_id),
    Stores = n_distinct(store_id),
    Mean = mean(shelf_price_usd),
    Median = median(shelf_price_usd),
    SD = sd(shelf_price_usd),
    Q1 = quantile(shelf_price_usd, 0.25),
    Q3 = quantile(shelf_price_usd, 0.75),
    IQR = IQR(shelf_price_usd),
    P90 = quantile(shelf_price_usd, 0.90),
    Promotion_pct = 100 * mean(promotion_flag == "Yes"),
    .groups = "drop"
  )

# split summaries into two tables to fit in the pdf
```

```r
report_table(
  date_summary |>
    select(observation_date, Observations, Products, Stores, Promotion_pct),
  digits = 2,
  col.names = c("Date", "Observations", "Products", "Stores", "Promotion (%)"),
  caption = "Coverage and promotional prevalence by date."
)

report_table(
  date_summary |>
    select(observation_date, Mean, Median, SD, Q1, Q3, IQR, P90),
  digits = 2,
  col.names = c("Date", "Mean", "Median", "SD", "Q1", "Q3", "IQR", "P90"),
  caption = "Shelf price summaries, in dollars per package."
)
```

q2-product-summary

```r
selected_prices <- prices_units |>
  filter(product_id %in% c("P01", "P11"))

product_summary <- selected_prices |>
  group_by(generic_product, observation_date, comparison_base_unit) |>
  summarise(
    n = n(),
    Mean = mean(unit_price),
    Median = median(unit_price),
    IQR = IQR(unit_price),
    .groups = "drop"
  )

report_table(
  product_summary,
  digits = 3,
  col.names = c(
    "Product", "Date", "Base unit", "n",
    "Mean", "Median", "IQR"
  ),
  caption = "Product specific unit prices (USD/base unit)."
)
```

q2-quantity-plo

```r
selected_prices |>
  mutate(
    Product = case_when(
      product_id == "P01" ~ "White rice (oz)",
```

```
      product_id == "P11" ~ "Whole milk (fl oz)"
    )
  ) |>
  ggplot(aes(
    x = quantity_base,
    y = shelf_price_usd,
    color = promotion_flag
  )) +
  geom_point(alpha = 0.6, size = 1.8) +
  facet_wrap(
    vars(Product, observation_date),
    ncol = 2,
    scales = "free"
  ) +
  labs(
    title = "Larger packages generally have higher shelf prices",
    subtitle = "All recorded white rice and whole milk prices, by date",
    x = "Package quantity in the labeled base unit",
    y = "Shelf price (USD/package)",
    color = "Promotion",
    caption = paste(
      "Simulated teaching data.",
      "Rice: n = 121 (2024), 126 (2025); milk: n = 128 (2024), 132 (2025)."
    )
  ) +
  theme(legend.position = "bottom")
```

q2-promotions

```
promotion_summary <- prices_units |>
  group_by(category, observation_date) |>
  summarise(
    n = n(),
    Promotion_pct = 100 * mean(promotion_flag == "Yes"),
    .groups = "drop"
  ) |>
  mutate(Year = format(observation_date, "%Y")) |>
  select(category, Year, n, Promotion_pct) |>
  pivot_wider(
    names_from = Year,
    values_from = c(n, Promotion_pct)
  )

report_table(
  promotion_summary,
  digits = 1,
  col.names = c("Category", "n 2024", "n 2025", "Promo 2024 (%)", "Promo 2025 (%)"),
```

```
  caption = "Promotion percentages use recorded prices within each category and date."
)
```

q2-price-reversal

```
price_example <- prices_units |>
  filter(
    product_id == "P01",
    observation_date == as.Date("2024-04-15"),
    store_id %in% c("S0001", "S0011")
  ) |>
  arrange(store_id) |>
  select(store_id, quantity_base, shelf_price_usd, unit_price)

report_table(
  price_example,
  digits = c(0, 0, 2, 4),
  col.names = c("Store", "Quantity (oz)", "Shelf price (USD)", "Unit price (USD/oz)"),
  caption = "White rice on April 15, 2024 (both records are nonpromotional)."
)
```

q3-matching

```
a_prices <- prices_units |>
  filter(
    observation_date == as.Date("2025-04-15"),
    store_id == "S0001"
  ) |>
  select(
    product_id, generic_product, comparison_base_unit,
    U_a = unit_price, promo_a = promotion_flag
  )

b_prices <- prices_units |>
  filter(
    observation_date == as.Date("2025-04-15"),
    store_id == "S0004"
  ) |>
  select(
    product_id,
    U_b = unit_price, promo_b = promotion_flag
  )

matched <- inner_join(
  a_prices, b_prices,
  by = "product_id",
  relationship = "one-to-one"
```

```
) |>
  mutate(
    Delta = U_a - U_b,
    R = 100 * Delta / U_b
  )

# remove promotional records from each store before looking at overlap
a_regular <- a_prices |> filter(promo_a == "No")
b_regular <- b_prices |> filter(promo_b == "No")

matched_regular <- matched |>
  filter(promo_a == "No", promo_b == "No")
```

q3-overlap

```
overlap_summary <- function(a, b) {
  shared <- length(intersect(a$product_id, b$product_id))
  combined <- length(union(a$product_id, b$product_id))

  c(
    nrow(a),
    nrow(b),
    shared,
    nrow(a) - shared,
    nrow(b) - shared,
    100 * shared / nrow(a),
    100 * shared / nrow(b),
    100 * shared / combined
  )
}

overlap <- tibble(
  Measure = c(
    "Products at a",
    "Products at b",
    "Shared products",
    "Only at a",
    "Only at b",
    "Shared / products at a (%)",
    "Shared / products at b (%)",
    "Shared / union (%)"
  ),
  All = overlap_summary(a_prices, b_prices),
  No_promotions = overlap_summary(a_regular, b_regular)
)

report_table(
```

```
  overlap,
  digits = 1,
  col.names = c("Measure", "All records", "No promotions"),
  caption = "Product overlap on April 15, 2025."
)
```

q3-differences

```
difference_summary <- function(data) {
  c(
    nrow(data),
    mean(data$R),
    median(data$R),
    IQR(data$R),
    100 * mean(data$Delta < 0),
    100 * mean(data$Delta > 0),
    100 * mean(data$Delta == 0)
  )
}

price_comparison <- tibble(
  Measure = c(
    "Matched products",
    "Mean R (%)",
    "Median R (%)",
    "IQR of R (percentage points)",
    "Store a cheaper (%)",
    "Store b cheaper (%)",
    "Equal prices (%)"
  ),
  All = difference_summary(matched),
  No_promotions = difference_summary(matched_regular)
)

report_table(
  price_comparison,
  digits = 2,
  col.names = c("Measure", "All records", "No promotions"),
  caption = "Matched percentage differences (store b is baseline)."
)

rice_comparison <- matched |>
  filter(product_id == "P01")
```

q3-plot

```
matched |>
  mutate(
    Product = paste(product_id, generic_product),
    Product = reorder(Product, R)
  ) |>
  ggplot(aes(x = R, y = Product)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(size = 2, color = "steelblue") +
  labs(
    title = "Daylight Discount is cheaper for most shared products",
    subtitle = "Valley Basket versus Daylight Discount, Yolo (April 15, 2025)",
    x = "Unit price difference relative to Daylight Discount (%)",
    y = NULL,
    caption = "Simulated data. All 28 shared products with promotions included."
  ) +
  theme(
    axis.text.y = element_text(size = 8),
    panel.grid.major.y = element_blank()
  )
```

q4-basket-costs

```
basket_definitions <- basket_items |>
  left_join(
    products |>
      select(product_id, comparison_base_unit),
    by = "product_id",
    relationship = "many-to-one"
  )

stopifnot(
  !anyDuplicated(basket_items[c("basket_id", "product_id")]),
  all(!is.na(basket_definitions$comparison_base_unit)),
  all(basket_definitions$base_unit ==
        basket_definitions$comparison_base_unit),
  all(!is.na(basket_definitions$quantity_in_base_units)),
  all(basket_definitions$quantity_in_base_units > 0)
)

prices_2025 <- prices_units |>
  filter(observation_date == as.Date("2025-04-15")) |>
  select(store_id, product_id, unit_price)

# make every required store basket product combo
basket_lines_2025 <- stores |>
  select(store_id) |>
  cross_join(basket_definitions) |>
```

```
  left_join(
    prices_2025,
    by = c("store_id", "product_id"),
    relationship = "many-to-one"
  )

basket_costs_2025 <- basket_lines_2025 |>
  group_by(store_id, basket_id) |>
  summarise(
    required = n(),
    observed = sum(!is.na(unit_price)),
    coverage = observed / required,
    basket_cost = if (all(!is.na(unit_price))) {
      sum(quantity_in_base_units * unit_price)
    } else {
      NA_real_
    },
    .groups = "drop"
  ) |>
  left_join(
    stores |> select(store_id, store_label, county_fips),
    by = "store_id",
    relationship = "many-to-one"
  )

basket_order <- c("essentials_12", "balanced_19", "expanded_25")

yolo_baskets <- basket_costs_2025 |>
  filter(county_fips == "06113") |>
  mutate(basket_id = factor(basket_id, levels = basket_order))
```

q4-coverage

```
coverage_table <- yolo_baskets |>
  mutate(
    Coverage = sprintf(
      "%d/%d (%.1f%%)", observed, required, 100 * coverage
    )
  ) |>
  select(store_id, basket_id, Coverage) |>
  pivot_wider(names_from = basket_id, values_from = Coverage) |>
  select(store_id, all_of(basket_order)) |>
  arrange(store_id)

report_table(
  coverage_table,
  col.names = c("Store", "Essentials", "Balanced", "Expanded"),
```

```
  caption = "Observed/required products and coverage percentage, Yolo, April 15, 2025."
)
```

q4-rankings

```
eligible_baskets <- yolo_baskets |>
  filter(coverage == 1, !is.na(basket_cost)) |>
  group_by(basket_id) |>
  mutate(rank = min_rank(basket_cost)) |>
  ungroup()

rank_table <- eligible_baskets |>
  arrange(basket_id, rank) |>
  select(basket_id, store_id, basket_cost, rank)

report_table(
  rank_table,
  digits = c(0, 0, 2, 0),
  col.names = c("Basket", "Store", "Cost (USD)", "Rank"),
  caption = "Rankings among stores with complete recorded baskets."
)

basket_summary <- eligible_baskets |>
  group_by(basket_id) |>
  summarise(
    Complete_stores = n(),
    Minimum = min(basket_cost),
    Maximum = max(basket_cost),
    Gap = Maximum - Minimum,
    Gap_pct = 100 * Gap / Minimum,
    .groups = "drop"
  )

report_table(
  basket_summary,
  digits = 2,
  col.names = c(
    "Basket", "Stores", "Min (USD)", "Max (USD)",
    "Gap (USD)", "Gap (%)"
  ),
  caption = "Within-basket ranges; percentage gaps use the cheapest eligible cost."
)
```

q4-plot

```
eligible_baskets |>
  mutate(
    Store = factor(
      store_id,
      levels = c("S0002", "S0003", "S0001"),
      labels = c("GreenGrocer", "Harvest Foods", "Valley Basket")
    )
  ) |>
  ggplot(aes(x = basket_cost, y = Store)) +
  geom_point(size = 2.5, color = "#245773") +
  facet_wrap(
    vars(basket_id),
    nrow = 1,
    scales = "free"
  ) +
  labs(
    title = "Complete basket costs in Yolo County",
    subtitle = "April 15, 2025 (only stores with all required prices)",
    x = "Complete basket cost (USD)",
    y = NULL,
    caption = paste(
      "Simulated teaching data.",
      "Eligible stores: essentials n = 3; balanced n = 2; expanded n = 2.",
      sep = "\n"
    )
  )
```

q5-category

```
category_comparison <- matched |>
  group_by(category = products$category[
    match(product_id, products$product_id)
  ]) |>
  summarise(
    Products = n(),
    Median_R = median(R),
    .groups = "drop"
  )

report_table(
  category_comparison,
  digits = 2,
  col.names = c("Category", "Shared products", "Median R (%)"),
  caption = "Matched category comparisons (Daylight Discount is baseline)."
)
```

q5-county-summary

```
yolo_essentials <- eligible_baskets |>
  filter(basket_id == "essentials_12")

essentials_summary <- yolo_essentials |>
  summarise(
    Stores = n(),
    Minimum = min(basket_cost),
    Median = median(basket_cost),
    Mean = mean(basket_cost),
    Maximum = max(basket_cost),
    Range = Maximum - Minimum
  )

report_table(
  essentials_summary,
  digits = 2,
  caption = paste(
    "Yolo essentials-basket costs on April 15, 2025;",
    "all monetary summaries are USD."
  )
)
```

q6-match

```
two_waves <- prices_units |>
  filter(observation_date %in% as.Date(c("2024-04-15", "2025-04-15"))) |>
  mutate(year = format(observation_date, "%Y")) |>
  select(
    store_id, product_id, year,
    unit_price, shelf_price_usd, quantity_base, promotion_flag
  )

stopifnot(
  !anyDuplicated(two_waves[c("store_id", "product_id", "year")])
)

expected_matches <- two_waves |>
  count(store_id, product_id) |>
  filter(n == 2) |>
  nrow()

prices_wide <- two_waves |>
  pivot_wider(
    names_from = year,
    values_from = c(
      unit_price, shelf_price_usd, quantity_base, promotion_flag
    )
```

```
  )

stopifnot(
  nrow(prices_wide) ==
    nrow(distinct(two_waves, store_id, product_id))
)

matched_changes <- prices_wide |>
  filter(
    !is.na(unit_price_2024),
    !is.na(unit_price_2025)
  ) |>
  mutate(
    Delta = unit_price_2025 - unit_price_2024,
    R = 100 * Delta / unit_price_2024,
    Raw_R = 100 * (shelf_price_usd_2025 - shelf_price_usd_2024) /
      shelf_price_usd_2024
  ) |>
  left_join(
    products |> select(product_id, category),
    by = "product_id",
    relationship = "many-to-one"
  )

stopifnot(
  nrow(matched_changes) == expected_matches,
  !anyDuplicated(matched_changes[c("store_id", "product_id")])
)

coverage_summary <- tibble(
  Measure = c(
    "Potential pairs: 145 stores x 40 products",
    "Pairs observed in at least one wave",
    "Pairs observed in both waves",
    "2024 only pairs",
    "2025 only pairs",
    "Matched / potential pairs (%)",
    "Matched / observed union (%)"
  ),
  Value = c(
    nrow(stores) * nrow(products),
    nrow(prices_wide),
    nrow(matched_changes),
    sum(!is.na(prices_wide$unit_price_2024) &
          is.na(prices_wide$unit_price_2025)),
    sum(is.na(prices_wide$unit_price_2024) &
          !is.na(prices_wide$unit_price_2025)),
```

```
    100 * nrow(matched_changes) / (nrow(stores) * nrow(products)),
    100 * nrow(matched_changes) / nrow(prices_wide)
  )
)

report_table(
  coverage_summary,
  digits = 2,
  caption = "Coverage of store product pairs across the two price waves."
)
```

q6-summary

```
regular_changes <- matched_changes |>
  filter(
    promotion_flag_2024 == "No",
    promotion_flag_2025 == "No"
  )

summarize_changes <- function(data) {
  # Small tolerance prevents floating-point noise from creating changes.
  increasing <- data$R > 1e-8
  decreasing <- data$R < -1e-8
  unchanged <- abs(data$R) <= 1e-8

  c(
    nrow(data),
    mean(data$R),
    median(data$R),
    sd(data$R),
    quantile(data$R, 0.25),
    quantile(data$R, 0.75),
    IQR(data$R),
    sum(increasing),
    100 * mean(increasing),
    sum(decreasing),
    100 * mean(decreasing),
    sum(unchanged),
    100 * mean(unchanged)
  )
}

change_summary <- tibble(
  Measure = c(
    "Matched pairs",
    "Mean change (%)",
    "Median change (%)",
```

```
    "SD (percentage points)",
    "Q1 (%)",
    "Q3 (%)",
    "IQR (percentage points)",
    "Increasing: count",
    "Increasing (%)",
    "Decreasing: count",
    "Decreasing (%)",
    "Unchanged: count",
    "Unchanged (%)"
  ),
  All = summarize_changes(matched_changes),
  No_promotions = summarize_changes(regular_changes)
)

report_table(
  change_summary,
  digits = 2,
  col.names = c("Measure", "All matched pairs", "No promotions"),
  caption = "Unit-price changes from April 15, 2024 to April 15, 2025."
)
```

q6-categories

```
category_changes <- matched_changes |>
  group_by(category) |>
  summarise(
    Pairs = n(),
    Mean = mean(R),
    Median = median(R),
    .groups = "drop"
  )

report_table(
  category_changes,
  digits = 2,
  col.names = c("Category", "Matched pairs", "Mean (%)", "Median (%)"),
  caption = "Matched percentage changes by product category."
)
```

q6-plot

```
ggplot(matched_changes, aes(x = category, y = R)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_boxplot(fill = "steelblue", outlier.alpha = 0.35) +
  labs(
    title = "Matched unit price changes vary within and across categories",
```

```
    subtitle = "April 15, 2024 to April 15, 2025 (promotions included)",
    x = "Product category",
    y = "Unit price change relative to 2024 (%)",
    caption = paste(
      "Simulated data. n = 3,638 matched store product pairs.",
      "Category sample sizes are reported in the table."
    )
  )
```

q6-package-example

```
package_example <- matched_changes |>
  filter(store_id == "S0009", product_id == "P01") |>
  transmute(
    `2024 oz` = quantity_base_2024,
    `2025 oz` = quantity_base_2025,
    `2024 USD` = shelf_price_usd_2024,
    `2025 USD` = shelf_price_usd_2025,
    `Shelf change (%)` = Raw_R,
    `Unit change (%)` = R
  )

report_table(
  package_example,
  digits = 2,
  caption = "White rice at S0009: package-price and unit-price changes."
)
```

q7-prepare

```
stores_owners <- stores |>
  left_join(
    banners,
    by = "banner_id",
    relationship = "many-to-one"
  )

stopifnot(
  nrow(stores_owners) == nrow(stores),
  !anyNA(stores_owners$banner_name),
  !anyNA(stores_owners$store_type),
  !anyNA(stores_owners$ownership_unit_id_sim),
  !anyNA(stores_owners$snap_authorized_sim),
  all(stores_owners$snap_authorized_sim %in% c("Yes", "No"))
)

county_ownership <- stores_owners |>
```

```
  group_by(county_fips, county_label) |>
  summarise(
    Stores = n(),
    Banners = n_distinct(banner_id),
    Owners = n_distinct(ownership_unit_id_sim),
    Owners_per_store = Owners / Stores,
    Largest_banner_pct = 100 * max(table(banner_id)) / Stores,
    Largest_owner_pct = 100 * max(table(ownership_unit_id_sim)) / Stores,
    .groups = "drop"
  )
```

q7-county-table

```
report_table(
  county_ownership |>
    arrange(county_label) |>
    select(-county_fips),
  digits = c(0, 0, 0, 0, 2, 1, 1),
  col.names = c(
    "County", "Stores", "Banners", "Owners",
    "Owners/store", "Top banner (%)", "Top owner (%)"
  ),
  caption = "Undated simulated store roster (shares use stores within each county)."
)
```

q7-equal-count

```
yolo_store_count <- county_ownership |>
  filter(county_fips == "06113") |>
  pull(Stores)

comparison_counties <- county_ownership |>
  filter(Stores == yolo_store_count) |>
  arrange(county_label)

report_table(
  comparison_counties |>
    select(county_label, Stores, Banners, Owners),
  col.names = c("County", "Stores", "Banners", "Owners"),
  caption = "Reproducible comparison (all counties with Yolo's store count)."
)
```

q7-store-types

```
store_type_shares <- stores_owners |>
  count(county_fips, county_label, store_type, name = "Stores") |>
```

```
  group_by(county_fips, county_label) |>
  mutate(Share = Stores / sum(Stores)) |>
  ungroup()

ggplot(
  store_type_shares,
  aes(x = Share, y = county_label, fill = store_type)
) +
  geom_col() +
  scale_x_continuous(labels = scales::label_percent()) +
  labs(
    title = "Store type composition varies across counties",
    subtitle = "Undated study roster (denominator is each county's store count)",
    x = "Share of county study stores",
    y = NULL,
    fill = "Store type",
    caption = "Simulated data. 145 stores in 24 counties."
  ) +
  theme(legend.position = "bottom") +
  guides(fill = guide_legend(nrow = 2))
```

q7-snap

```
snap_summary <- stores_owners |>
  group_by(county_label) |>
  summarise(
    Stores = n(),
    Yes = sum(snap_authorized_sim == "Yes"),
    No = sum(snap_authorized_sim == "No"),
    Yes_pct = 100 * Yes / Stores,
    No_pct = 100 * No / Stores,
    .groups = "drop"
  )

report_table(
  snap_summary,
  digits = 1,
  col.names = c("County", "Stores", "Yes", "No", "Yes (%)", "No (%)"),
  caption = "Simulated SNAP status (percentages use all study stores in each county)."
)
```

q8-county-data

```
county_store_counts <- stores |>
  count(county_fips, name = "S_c")

stopifnot(
```

```
  !anyNA(county_context$county_fips),
  !anyDuplicated(county_context$county_fips),
  !anyNA(stores$county_fips),
  all(stores$county_fips %in% county_context$county_fips)
)

county_stores <- county_context |>
  left_join(
    county_store_counts,
    by = "county_fips",
    relationship = "one-to-one"
  ) |>
  mutate(
    S_c = coalesce(S_c, 0L),
    N_c = population_2024_sim,
    D_c = if_else(
      !is.na(N_c) & N_c > 0,
      10000 * S_c / N_c,
      NA_real_
    )
  )


stopifnot(
  nrow(county_stores) == nrow(county_context),
  !anyNA(county_stores$D_c),
  !anyNA(county_stores$urban_rural_sim)
)

county_stores <- county_stores |>
  mutate(
    count_rank = min_rank(desc(S_c)),
    density_rank = min_rank(desc(D_c)),
    rank_change = count_rank - density_rank
  )
```

q8-summary

```
county_summary <- county_stores |>
  select(S_c, N_c, D_c) |>
  pivot_longer(everything(), names_to = "Measure", values_to = "Value") |>
  group_by(Measure) |>
  summarise(
    Mean = mean(Value),
    Median = median(Value),
    SD = sd(Value),
    Q1 = quantile(Value, 0.25),
```

```
    Q3 = quantile(Value, 0.75),
    IQR = IQR(Value),
    Min = min(Value),
    Max = max(Value),
    .groups = "drop"
  )

county_summary_display <- county_summary |>
  pivot_longer(
    cols = -Measure,
    names_to = "Statistic",
    values_to = "Value"
  ) |>
  mutate(
    Statistic = factor(
      Statistic,
      levels = c("Mean", "Median", "SD", "Q1", "Q3", "IQR", "Min", "Max")
    )
  ) |>
  pivot_wider(names_from = Measure, values_from = Value) |>
  arrange(Statistic) |>
  select(Statistic, S_c, N_c, D_c)

report_table(
  county_summary_display,
  digits = 2,
  col.names = c(
    "Statistic", "Stores", "2024 residents", "Stores per 10,000"
  ),
  caption = "County level summaries across 24 simulated counties."
)
```

q8-rankings

```
report_table(
  county_stores |>
    arrange(density_rank) |>
    select(county_name, S_c, N_c, D_c, count_rank, density_rank),
  digits = c(0, 0, 0, 2, 0, 0),
  col.names = c(
    "County", "Stores", "2024 population", "Density",
    "Count rank", "Density rank"
  ),
  caption = "County rankings by stores count and stores per 10,000 residents."
)
```

q8-rank-changes

```
largest_changes <- county_stores |>
  arrange(desc(abs(rank_change)), county_name) |>
  slice_head(n = 5)

report_table(
  largest_changes |>
    select(county_name, count_rank, density_rank, rank_change),
  col.names = c("County", "Count rank", "Density rank", "Change"),
  caption = "Five largest absolute rank changes (ties in selected are reslved alphabetically).
)
```

q8-groups

```
group_summary <- county_stores |>
  group_by(urban_rural_sim) |>
  summarise(
    Counties = n(),
    Mean_count = mean(S_c),
    Mean_density = mean(D_c),
    Median_density = median(D_c),
    IQR_density = IQR(D_c),
    .groups = "drop"
  )

report_table(
  group_summary,
  digits = 2,
  col.names = c(
    "Group", "Counties", "Mean stores",
    "Mean density", "Median density", "Density IQR"
  ),
  caption = "County level summaries by simulated metro classification"
)
```

q8-plot

```
ggplot(
  county_stores,
  aes(
    x = D_c,
    y = reorder(county_name, D_c),
    color = urban_rural_sim
  )
) +
  geom_point(size = 2.3) +
  labs(
    title = "Population adjustment changes the comparison of store availability",
```

```
    subtitle = "Undated study store counts divided by simulated 2024 population",
    x = "Study stores per 10,000 residents",
    y = NULL,
    color = "County group",
    caption = "Simulated data. All 24 counties, 145 study stores."
  ) +
  theme(legend.position = "bottom")
```

q9-compare

```
# normalize alternative file's fips
environment_2022 <- food_environment_raw |>
  mutate(
    county_fips = str_pad(county_fips_raw, width = 5, pad = "0")
  )

stopifnot(
  !anyNA(environment_2022$county_fips),
  !anyDuplicated(environment_2022$county_fips),
  setequal(county_stores$county_fips, environment_2022$county_fips)
)

county_comparison <- county_stores |>
  left_join(
    environment_2022 |>
      select(
        county_fips,
        population_2022_est_sim,
        food_outlets_2022_est_sim,
        food_outlets_per_10000_2022_sim
      ),
    by = "county_fips",
    relationship = "one-to-one"
  ) |>
  mutate(
    outlets_2022 = food_outlets_2022_est_sim,
    density_2022 = food_outlets_per_10000_2022_sim,
    count_difference = outlets_2022 - S_c,
    count_pct = if_else(
      !is.na(S_c) & S_c > 0,
      100 * count_difference / S_c,
      NA_real_
    ),
    density_difference = density_2022 - D_c,
    density_pct = if_else(
      !is.na(D_c) & D_c > 0,
      100 * density_difference / D_c,
```

```
      NA_real_
    ),
    alternative_rank = min_rank(desc(density_2022)),
    rank_change = density_rank - alternative_rank
  )

# stop if not required measurements are missing

stopifnot(
  nrow(county_comparison) == nrow(county_stores),
  !anyNA(county_comparison[c(
    "S_c", "D_c", "outlets_2022",
    "density_2022", "population_2022_est_sim"
  )]),
  all(county_comparison$population_2022_est_sim > 0),
  all(abs(
    county_comparison$density_2022 -
      10000 * county_comparison$outlets_2022 /
      county_comparison$population_2022_est_sim
  ) <= 0.0005)
)

report_table(
  county_comparison |>
    arrange(county_name) |>
    select(
      county_name, S_c, outlets_2022,
      count_difference, count_pct,
      density_difference, density_pct
    ),
  digits = c(0, 0, 0, 0, 1, 3, 1),
  col.names = c(
    "County", "Study n", "2022 n",
    "Count diff.", "Count (%)", "Rate diff.", "Rate (%)"
  ),
  caption = "Alternative minus study measures (percentage differences use study baseline)."
)
```

q9-association

```
count_rho <- cor(
  county_comparison$S_c,
  county_comparison$outlets_2022,
  method = "spearman"
)

density_rho <- cor(
```

```
  county_comparison$D_c,
  county_comparison$density_2022,
  method = "spearman"
)

rank_disagreements <- county_comparison |>
  arrange(desc(abs(rank_change)), county_name) |>
  slice_head(n = 5)

report_table(
  rank_disagreements |>
    select(
      county_name, D_c, density_2022,
      density_rank, alternative_rank, rank_change
    ),
  digits = c(0, 3, 3, 0, 0, 0),
  col.names = c(
    "County", "Study rate", "2022 rate",
    "Study rank", "2022 rank", "Change"
  ),
  caption = "Five largest density rank disagreements (selection ties resolved alphabetically).
)
```

q9-plt

```
ggplot(
  county_comparison,
  aes(x = D_c, y = density_2022, color = urban_rural_sim)
) +
  geom_abline(
    intercept = 0, slope = 1,
    linetype = "dashed", color = "gray50"
  ) +
  geom_point(size = 2.5) +
  coord_equal(
    xlim = c(0, 6),
    ylim = c(0, 6),
    expand = FALSE
  ) +
  labs(
    title = "County outlet densities differ between measures",
    subtitle = "Study roster versus alternative 2022 indicator",
    x = "Undated study stores per 10,000 residents\n(2024 population denominator)",
    y = "2022 outlets per 10,000 residents\n(2022 population denominator)",
    color = "County group",
    caption = "Simulated data. n = 24 matched counties (dashed line indicates equality)."
  ) +
```

```
  theme(legend.position = "bottom")
```

q10-aggregate

```
tract_access <- tracts_raw |>
  mutate(county_fips = substr(tract_id, 1, 5))

required_counts <- c(
  "tract_population_2024_sim",
  "households_2024_sim",
  "low_access_residents_2024_sim",
  "low_income_low_access_residents_2024_sim"
)

# stop if counts are missing
stopifnot(
  !anyNA(tract_access$tract_id),
  !anyDuplicated(tract_access$tract_id),
  all(str_detect(tract_access$tract_id, "^[0-9]{11}$")),
  all(tract_access$county_fips %in% county_context$county_fips),
  !anyNA(tract_access[required_counts]),
  all(as.matrix(tract_access[required_counts]) >= 0),
  all(tract_access$tract_population_2024_sim > 0),
  all(tract_access$low_access_residents_2024_sim <=
        tract_access$tract_population_2024_sim),
  all(tract_access$low_income_low_access_residents_2024_sim <=
        tract_access$low_access_residents_2024_sim)
)

county_access <- tract_access |>
  group_by(county_fips) |>
  summarise(
    Tracts = n(),
    Population = sum(tract_population_2024_sim),
    Households = sum(households_2024_sim),
    Low_access = sum(low_access_residents_2024_sim),
    Low_income_low_access = sum(low_income_low_access_residents_2024_sim),
    Weighted = 100 * Low_access / Population,
    Unweighted = mean(
      100 * low_access_residents_2024_sim / tract_population_2024_sim
    ),
    Absolute_gap = abs(Weighted - Unweighted),
    Low_income_low_access_pct = 100 * Low_income_low_access / Population,
    .groups = "drop"
  ) |>
  left_join(
    county_context |>
```

```
      select(
        county_fips, county_name,
        population_2024_sim, households_2024_sim
      ),
    by = "county_fips",
    relationship = "one-to-one"
  ) |>
  mutate(
    Population_difference = Population - population_2024_sim,
    Household_difference = Households - households_2024_sim
  )

stopifnot(
  setequal(county_access$county_fips, county_context$county_fips),
  all(county_access$Population_difference == 0),
  all(county_access$Household_difference == 0)
)

report_table(
  county_access |>
    arrange(county_name) |>
    select(
      county_name, Tracts, Population, Households,
      Low_access, Low_income_low_access
    ),
  digits = 0,
  col.names = c(
    "County", "Tracts", "Residents", "Households",
    "Low access", "Low income + low access"
  ),
  caption = "Summed simulated 2024 tract counts by county"
)
```

q10-percentages

```
report_table(
  county_access |>
    arrange(county_name) |>
    select(
      county_name, Weighted, Unweighted,
      Absolute_gap, Low_income_low_access_pct
    ),
  digits = 2,
  col.names = c(
    "County", "Weighted (%)", "Unweighted (%)",
    "Gap (pp)", "Low income + low access (%)"
  ),
```

```
  caption = "Low-access summaries (absolute gaps are percentage points)."
)
```

q10-long

```
access_long <- tract_access |>
  select(
    tract_id, county_fips, tract_population_2024_sim,
    low_access_residents_2024_sim,
    low_income_low_access_residents_2024_sim
  ) |>
  pivot_longer(
    cols = c(
      low_access_residents_2024_sim,
      low_income_low_access_residents_2024_sim
    ),
    names_to = "indicator",
    values_to = "count"
  )

rows_per_tract <- access_long |>
  count(tract_id)

stopifnot(
  nrow(access_long) == 2 * nrow(tract_access),
  all(rows_per_tract$n == 2),
  !anyDuplicated(access_long[c("tract_id", "indicator")])
)

# within each indicator each tract's population appears once
county_access_long <- access_long |>
  group_by(county_fips, indicator) |>
  summarise(
    Residents = sum(count),
    Population = sum(tract_population_2024_sim),
    Share = 100 * Residents / Population,
    .groups = "drop"
  ) |>
  left_join(
    county_context |> select(county_fips, county_name),
    by = "county_fips",
    relationship = "many-to-one"
  ) |>
  mutate(
    Indicator = recode(
      indicator,
      low_access_residents_2024_sim = "Low access",
```

```
      low_income_low_access_residents_2024_sim = "Low income and low access"
    )
  )
```

q10-plot

```
county_order <- county_access |>
  arrange(Weighted) |>
  pull(county_name)

county_access_long |>
  mutate(county_name = factor(county_name, levels = county_order)) |>
  ggplot(aes(x = Share, y = county_name, color = Indicator)) +
  geom_point(size = 2.2) +
  labs(
    title = "County resident shares with limited food access",
    subtitle = "Simulated 2024 counts (denominator is summed county tract population)",
    x = "Percentage of county residents",
    y = NULL,
    color = NULL,
    caption = "Simulated data. 134 constructed tracts across 24 counties."
  ) +
  theme(legend.position = "bottom")
```

q11-aggregate

```
county_q11 <- county_context |>
  transmute(
    county_fips, county_name,
    Income = median_household_income_2024_usd_sim,
    Poverty = poverty_rate_2024_pct_sim,
    No_Vehicle = 100 * households_no_vehicle_2024_sim / households_2024_sim,
  ) |>
  left_join(
    county_access |> select(county_fips, Weighted, Low_income_low_access_pct),
    by = "county_fips",
    relationship = "one-to-one"
  )

stopifnot(
  nrow(county_q11) == 24,
  !anyNA(county_q11),
  all(county_q11$No_Vehicle >= 0 & county_q11$No_Vehicle <= 100),
  all(county_q11$Poverty >= 0 & county_q11$Poverty <= 100),
  all(county_q11$Income >0)
)
```

q11-summary

```
q11_summary <- county_q11 |>
  select(Income, Poverty, No_Vehicle, Weighted) |>
  pivot_longer(everything(), names_to = "Measure", values_to = "Value") |>
  group_by(Measure) |>
  summarise(
    Mean = mean(Value), Median = median(Value), SD = sd(Value),
    Q1 = quantile(Value, 0.25), Q3 = quantile(Value, 0.75),
    Min = min(Value), Max = max(Value), .groups = "drop"
  ) |>
  mutate(Measure = recode(Measure,
                          Income = "Meadian Income (USD)",
                          Poverty = "Poverty Rate (%)",
                          No_Vehicle = "No-vehicle households (%)",
                          Weighted = "Low-access residents (%)"
                          ))
report_table(q11_summary, digits = 2, caption = "County-level summarries across 24 counties.")
```

q11-cor

```
predictors <- c("Income", "Poverty", "No_Vehicle")

cor_table <- map_dfr(predictors, function(x) {
  ct <- cor.test(county_q11[[x]], county_q11$Weighted)
  tibble(
    Predictor = x,
    Pearson = unname(ct$estimate),
    CI_low = ct$conf.int[1],
    CI_high = ct$conf.int[2],
    p_value = ct$p.value,
    Spearman = cor(county_q11[[x]], county_q11$Weighted, method = "spearman")
  )
}) |>
  mutate(Predictor = recode(Predictor,
    Income = "Median income", Poverty = "Poverty rate",
    No_Vehicle = "No-vehicle households"))

report_table(cor_table, digits = 3,
  col.names = c("Predictor", "Pearson r", "95% CI low", "95% CI high",
                "p-value", "Spearman rho"),
  caption = "Association with county low-access share (n = 24).")
```

q-11 plots

```
county_q11 |>
  pivot_longer(c(Income, Poverty, No_Vehicle),
               names_to = "Predictor", values_to = "X") |>
  mutate(Predictor = recode(Predictor,
    Income = "Median income (USD)",
    Poverty = "Poverty rate (%)",
    No_Vehicle = "No-vehicle households (%)")) |>
  ggplot(aes(x = X, y = Weighted)) +
  geom_point(alpha = 0.8, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "gray30", linewidth = 0.6) +
  facet_wrap(vars(Predictor), scales = "free_x") +
  labs(
    title = "County low-access share versus income, poverty, and vehicle access",
    x = NULL, y = "Low-access residents (%)",
    caption = "Simulated data. n = 24 counties; lines are least-squares fits."
  )
```

q-11 overlap

```
predictor_cor <- cor(county_q11[predictors]) |>
  as.data.frame() |>
  rownames_to_column("Variable")

report_table(predictor_cor, digits = 2,
  caption = "Pearson correlations among county predictors.")
```

q12-tract

```
county_q12 <- county_comparison |>
  select(county_fips, county_name, urban_rural_sim, population_2024_sim,
         S_c, D_c, density_2022) |>
  left_join(
    county_access |> select(county_fips, Weighted, Low_income_low_access_pct),
    by = "county_fips", relationship = "one-to-one"
  ) |>
  left_join(
    county_q11 |> select(county_fips, Poverty, No_Vehicle),
    by = "county_fips", relationship = "one-to-one"
  )

stopifnot(nrow(county_q12) == 24, !anyNA(county_q12))

tract_options <- travel |>
  group_by(tract_id) |>
  summarise(
    Alternatives = n(),
    Nearest_miles = min(driving_distance_oneway_miles_sim),
```

```
    # Define bus availability from the data itself: time/fare are NA without service
    Bus_alternatives = sum(!is.na(bus_roundtrip_minutes_sim)),
    Best_bus_minutes = if (all(is.na(bus_roundtrip_minutes_sim))) {
      NA_real_
    } else {
      min(bus_roundtrip_minutes_sim, na.rm = TRUE)
    },
    .groups = "drop"
  )

tract_q12 <- tract_access |>
  left_join(tract_options, by = "tract_id", relationship = "one-to-one") |>
  mutate(
    Low_access_pct = 100 * low_access_residents_2024_sim /
      tract_population_2024_sim,
    Has_travel_data = !is.na(Alternatives)
  )

# Orphan check: travel rows whose tract is not in the tract file
stopifnot(all(travel$tract_id %in% tract_access$tract_id))

missing_audit <- tibble(
  Measure = c(
    "Tracts in access file",
    "Tracts with at least one travel alternative",
    "Tracts with no travel alternative",
    "Residents in tracts with no travel alternative (%)",
    "Tracts with no bus alternative (bus measures NA)"
  ),
  Value = c(
    nrow(tract_q12),
    sum(tract_q12$Has_travel_data),
    sum(!tract_q12$Has_travel_data),
    100 * sum(tract_q12$tract_population_2024_sim[!tract_q12$Has_travel_data]) /
      sum(tract_q12$tract_population_2024_sim),
    sum(tract_q12$Has_travel_data & tract_q12$Bus_alternatives == 0)
  )
)

report_table(
  missing_audit, digits = 1,
  caption = "Coverage of tract travel data (missing means not recorded, not zero)."
)
```

q12-aggregate

```
county_options <- tract_q12 |>
  group_by(county_fips) |>
  summarise(
    Tracts = n(),
    Tracts_with_data = sum(Has_travel_data),
    Pop_with_data_pct = 100 *
      sum(tract_population_2024_sim[Has_travel_data]) /
      sum(tract_population_2024_sim),
    # na.rm = TRUE drops only tracts with no travel rows, and coverage is reported above
    Mean_alternatives = if (any(Has_travel_data)) {
      mean(Alternatives, na.rm = TRUE)
    } else NA_real_,
    Weighted_nearest_miles = if (any(Has_travel_data)) {
      weighted.mean(Nearest_miles, tract_population_2024_sim, na.rm = TRUE)
    } else NA_real_,
    .groups = "drop"
  )

# Counties with no usable travel data stay NA (not 0) and are excluded from the models
county_q12 <- county_q12 |>
  left_join(county_options, by = "county_fips", relationship = "one-to-one")

excluded_counties <- county_q12 |>
  filter(is.na(Mean_alternatives)) |>
  pull(county_name)

excluded_counties
```

q12 plot

```
alts_per_county <- travel |>
  count(tract_id, name = "Alternatives") |>
  left_join(
    tract_access |> select(tract_id, county_fips),
    by = "tract_id", relationship = "one-to-one"
  ) |>
  group_by(county_fips) |>
  summarise(Mean_alternatives = mean(Alternatives), .groups = "drop")

plot_q12 <- county_access |>
  select(county_fips, county_name, Weighted) |>
  left_join(alts_per_county, by = "county_fips", relationship = "one-to-one")

stopifnot(nrow(plot_q12) == 24, !anyNA(plot_q12))

extremes <- plot_q12 |>
```

```
  filter(Weighted == min(Weighted) | Weighted == max(Weighted))

ggplot(plot_q12, aes(x = Mean_alternatives, y = Weighted)) +
  geom_point(
    position = position_jitter(width = 0.08, height = 0, seed = 141),
    size = 2.3, color = "steelblue", alpha = 0.8
  ) +
  geom_text(
    data = extremes, aes(label = county_name),
    hjust = -0.2, size = 3
  ) +
  scale_x_continuous(limits = c(0, 10), breaks = 0:10) +
  labs(
    title = "Same number of recorded options, very different low-access shares",
    subtitle = sprintf(
      "Every county has %.0f recorded store alternatives per tract, but low-access shares rang
      mean(plot_q12$Mean_alternatives), min(plot_q12$Weighted), max(plot_q12$Weighted)
    ),
    x = "Recorded store alternatives per tract (county mean)",
    y = "Low-access residents (%)",
    caption = "Simulated data. n = 24 counties; points are jittered horizontally for visibilit
  )
```

q13-eligible

```
# Complete essentials-basket costs from Question 4 (one row per store)
ess_costs <- basket_costs_2025 |>
  filter(basket_id == "essentials_12", coverage == 1, !is.na(basket_cost)) |>
  select(store_id, basket_cost)

stopifnot(!anyDuplicated(ess_costs$store_id))

# Bus fields must be present exactly when service is available
stopifnot(
  all(travel$bus_service_available_sim %in% c("Yes", "No")),
  all(!is.na(travel$bus_roundtrip_fare_usd_sim[travel$bus_service_available_sim == "Yes"])),
  all(is.na(travel$bus_roundtrip_fare_usd_sim[travel$bus_service_available_sim == "No"])),
  !anyNA(travel$car_roundtrip_cost_usd_sim)
)

# Recorded alternatives whose store has a complete essentials basket
cand <- travel |>
  inner_join(ess_costs, by = "store_id", relationship = "many-to-one") |>
  mutate(
    bus_ok    = bus_service_available_sim == "Yes",
    car_cost  = car_roundtrip_cost_usd_sim,
    bus_fare  = if_else(bus_ok, bus_roundtrip_fare_usd_sim, NA_real_),
```

```
    car_total = basket_cost + car_cost,
    bus_total = if_else(bus_ok, basket_cost + bus_fare, NA_real_)
  ) |>
  arrange(tract_id, store_id)   # fixes tie-breaking order below

# The inner join can only drop rows (stores without a complete basket)
stopifnot(
  nrow(cand) == sum(travel$store_id %in% ess_costs$store_id),
  !anyDuplicated(cand[c("tract_id", "store_id")])
)

eligibility <- tibble(
  Measure = c(
    "Recorded tract-store alternatives",
    "Alternatives whose store has a complete essentials basket",
    "  ...of which have bus service",
    "  ...of which have bus service with a zero fare",
    "Tracts in the access file",
    "Tracts with at least one eligible candidate",
    "Tracts with at least one bus-accessible eligible candidate"
  ),
  Value = c(
    nrow(travel),
    nrow(cand),
    sum(cand$bus_ok),
    sum(cand$bus_ok & cand$bus_fare == 0),
    nrow(tract_access),
    n_distinct(cand$tract_id),
    n_distinct(cand$tract_id[cand$bus_ok])
  )
)

report_table(
  eligibility, digits = 0,
  caption = "Eligibility for the essentials-basket travel scenarios, April 15, 2025."
)
```

q13-best

```
# Choose the single cheapest candidate per tract on a given cost column.
# Rows are sorted by store_id, so ties go to the lower store_id.
pick_min <- function(data, cost) {
  data |>
    filter(!is.na({{ cost }})) |>
    group_by(tract_id) |>
    slice_min({{ cost }}, n = 1, with_ties = FALSE) |>
    ungroup()
```

```
}

best_basket <- pick_min(cand, basket_cost) |>
  select(tract_id,
         store_basket = store_id, min_basket = basket_cost,
         basket_store_bus_ok = bus_ok,
         basket_store_car_total = car_total,
         basket_store_bus_total = bus_total)

best_car <- pick_min(cand, car_total) |>
  select(tract_id, store_car = store_id, min_car_total = car_total,
         car_cost_chosen = car_cost)

best_bus <- pick_min(filter(cand, bus_ok), bus_total) |>
  select(tract_id, store_bus = store_id, min_bus_total = bus_total,
         bus_fare_chosen = bus_fare,
         bus_minutes_chosen = bus_roundtrip_minutes_sim)

nearest <- pick_min(cand, driving_distance_oneway_miles_sim) |>
  select(tract_id, store_nearest = store_id)

# Count tracts where the minimum is tied (with_ties = FALSE hides these)
tie_counts <- c(
  Basket = sum(count(filter(cand, basket_cost ==
                 ave(basket_cost, tract_id, FUN = min)), tract_id)$n > 1),
  Car = sum(count(filter(cand, car_total ==
                 ave(car_total, tract_id, FUN = min)), tract_id)$n > 1)
)

tract_q13 <- tract_access |>
  select(tract_id, county_fips, households_2024_sim,
         households_no_vehicle_2024_sim) |>
  left_join(best_basket, by = "tract_id", relationship = "one-to-one") |>
  left_join(best_car,    by = "tract_id", relationship = "one-to-one") |>
  left_join(best_bus,    by = "tract_id", relationship = "one-to-one") |>
  left_join(nearest,     by = "tract_id", relationship = "one-to-one") |>
  mutate(
    evaluable  = !is.na(store_basket),
    car_switch = store_car != store_basket,
    # Extra cost of choosing on shelf price alone once the car trip is paid
    car_extra  = basket_store_car_total - min_car_total,
    nearest_is_car_best = store_nearest == store_car,
    bus_status = case_when(
      is.na(store_basket)       ~ "No eligible store",
      is.na(store_bus)          ~ "No bus-accessible eligible store",
      !basket_store_bus_ok      ~ "Basket-cheapest store not on bus",
      store_bus == store_basket ~ "Same store as basket-only",
```

```
      TRUE                        ~ "Different store than basket-only"
    ),
    bus_extra = if_else(
      bus_status == "Different store than basket-only",
      basket_store_bus_total - min_bus_total, NA_real_
    )
  )

stopifnot(nrow(tract_q13) == nrow(tract_access))
```

q13-min-costs

```
cost_summary <- function(x, label) {
  x <- x[!is.na(x)]
  tibble(Measure = label, Tracts = length(x),
         Median = median(x), Q1 = quantile(x, 0.25),
         Q3 = quantile(x, 0.75), Min = min(x), Max = max(x))
}

min_costs <- bind_rows(
  cost_summary(tract_q13$min_basket,      "(a) Basket only"),
  cost_summary(tract_q13$min_car_total,   "(b) Basket + car round trip"),
  cost_summary(tract_q13$min_bus_total,   "(c) Basket + bus round trip"),
  cost_summary(tract_q13$car_cost_chosen, "Car cost at store chosen in (b)"),
  cost_summary(tract_q13$bus_fare_chosen, "Bus fare at store chosen in (c)")
)

report_table(
  min_costs, digits = 2,
  caption = "Tract-level minimum costs in USD (n = evaluable tracts for each row)."
)
```

q13-switches

```
ev <- filter(tract_q13, evaluable)

switch_table <- tibble(
  Comparison = c(
    "Car: cheapest store changes after travel",
    "Car: nearest candidate is cheapest after travel",
    "Bus: cheapest store changes after fare (basket-cheapest store on bus)",
    "Bus: basket-cheapest store not reachable by bus",
    "Bus: no bus-accessible eligible store"
  ),
  Tracts = c(
    sum(ev$car_switch),
    sum(ev$nearest_is_car_best),
```

```
    sum(ev$bus_status == "Different store than basket-only"),
    sum(ev$bus_status == "Basket-cheapest store not on bus"),
    sum(ev$bus_status == "No bus-accessible eligible store")
  )
) |>
  mutate(Percent = 100 * Tracts / nrow(ev))

report_table(
  switch_table, digits = c(0, 0, 1),
  caption = paste0("Changes in the minimizing store; percentages use the ",
                   nrow(ev), " tracts with at least one eligible candidate.")
)
```

q13-plot

```
# Six tracts where the car trip changes the cheapest store
switch_tracts <- ev |>
  filter(car_switch) |>
  slice_head(n = 6) |>
  pull(tract_id)

cand |>
  filter(tract_id %in% switch_tracts) |>
  left_join(tract_q13 |> select(tract_id, store_basket, store_car),
            by = "tract_id", relationship = "many-to-one") |>
  mutate(Role = case_when(
    store_id == store_basket ~ "Cheapest basket",
    store_id == store_car    ~ "Cheapest basket + car trip",
    TRUE                     ~ "Other candidate"
  )) |>
  ggplot(aes(y = store_id, color = Role)) +
  geom_segment(aes(x = basket_cost, xend = car_total, yend = store_id),
               color = "gray75") +
  geom_point(aes(x = basket_cost), shape = 1, size = 2) +
  geom_point(aes(x = car_total), size = 2) +
  facet_wrap(vars(tract_id), scales = "free_y", ncol = 2) +
  scale_color_manual(values = c(`Cheapest basket` = "#0072B2",
                                `Cheapest basket + car trip` = "#D55E00",
                                `Other candidate` = "gray55")) +
  labs(
    title = "Adding the car trip can change which store is cheapest",
    subtitle = "Open circle = basket only; filled circle = basket + car round trip",
    x = "Essentials cost (USD)", y = NULL, color = NULL,
    caption = paste0("Simulated teaching data. First six of the ", sum(ev$car_switch),
                     " tracts where the car-optimal store differs; April 15, 2025.")
  ) +
  theme(legend.position = "bottom", axis.text.y = element_text(size = 7))
```

q13-novehicle

```
no_vehicle_at_risk <- tract_q13 |>
  group_by(bus_status) |>
  summarise(
    Tracts = n(),
    No_vehicle_households = sum(households_no_vehicle_2024_sim),
    .groups = "drop"
  ) |>
  mutate(Share_pct = 100 * No_vehicle_households /
           sum(No_vehicle_households))

report_table(
  no_vehicle_at_risk, digits = c(0, 0, 0, 1),
  col.names = c("Bus scenario status", "Tracts",
                "No-vehicle households", "Share of no-vehicle households (%)"),
  caption = "Simulated 2024 no-vehicle households by tract bus-scenario status."
)
```

q14-profile

```
yolo_fips <- "06113"

# County essentials-basket summary (complete baskets only)
county_basket <- basket_costs_2025 |>
  filter(basket_id == "essentials_12") |>
  group_by(county_fips) |>
  summarise(
    Complete_essentials = sum(coverage == 1),
    Min_essentials = if (any(coverage == 1)) min(basket_cost[coverage == 1]) else NA_real_,
    Median_essentials = if (any(coverage == 1)) median(basket_cost[coverage == 1]) else NA_rea
    .groups = "drop"
  )

# County transportation summary from the Question 13 tract table
county_travel <- tract_q13 |>
  group_by(county_fips) |>
  summarise(
    Evaluable_tracts = sum(evaluable),
    Car_switch_pct = if (any(evaluable)) 100 * mean(car_switch[evaluable]) else NA_real_,
    Car_markup = median(min_car_total - min_basket, na.rm = TRUE),
    No_bus_option_pct = 100 * mean(bus_status %in% c(
      "No eligible store", "No bus-accessible eligible store")),
    .groups = "drop"
  )

county_profile <- county_stores |>
  select(county_fips, county_name, urban_rural_sim, S_c, N_c, D_c) |>
```

```
  left_join(county_ownership |> select(county_fips, Banners, Owners),
            by = "county_fips", relationship = "one-to-one") |>
  left_join(county_q11 |> select(county_fips, Income, Poverty, No_Vehicle),
            by = "county_fips", relationship = "one-to-one") |>
  left_join(county_access |> select(county_fips, Weighted, Low_income_low_access_pct),
            by = "county_fips", relationship = "one-to-one") |>
  left_join(county_basket, by = "county_fips", relationship = "one-to-one") |>
  left_join(county_travel, by = "county_fips", relationship = "one-to-one") |>
  mutate(
    Complete_essentials = coalesce(Complete_essentials, 0L),
    # 52 weekly cheapest complete essentials baskets / median household income
    Burden_pct = 100 * 52 * Min_essentials / Income
  )

stopifnot(nrow(county_profile) == 24, yolo_fips %in% county_profile$county_fips)
```

q14-rank

```
rank_measures <- c(
  D_c = "Stores per 10,000 residents",
  Owners = "Distinct owner units",
  Income = "Median household income (USD)",
  Poverty = "Poverty rate (%)",
  No_Vehicle = "No-vehicle households (%)",
  Weighted = "Low-access residents (%)",
  Median_essentials = "Median complete essentials basket (USD)",
  Burden_pct = "52 cheapest baskets / income (%)",
  Car_switch_pct = "Tracts where car trip changes cheapest store (%)",
  No_bus_option_pct = "Tracts with no bus-reachable eligible store (%)"
)

profile_long <- county_profile |>
  select(county_fips, county_name, all_of(names(rank_measures))) |>
  pivot_longer(-c(county_fips, county_name),
               names_to = "Measure", values_to = "Value") |>
  group_by(Measure) |>
  mutate(
    Counties = sum(!is.na(Value)),
    Rank = min_rank(desc(Value)),
    Study_median = median(Value, na.rm = TRUE)
  ) |>
  ungroup() |>
  mutate(Measure = factor(rank_measures[Measure], levels = rank_measures))

report_table(
  profile_long |>
    filter(county_fips == yolo_fips) |>
```

```
    select(Measure, Value, Study_median, Rank, Counties),
  digits = 2,
  col.names = c("Measure", "Yolo", "Study median", "Yolo rank", "Counties ranked"),
  caption = "Yolo's position among the study counties (rank 1 = highest value)."
)
```

q14-comparison

```
# Rule 1: same urban/rural class, closest on standardized income, poverty, no-vehicle %
z <- function(x) (x - mean(x)) / sd(x)

scored <- county_profile |>
  mutate(zI = z(Income), zP = z(Poverty), zV = z(No_Vehicle))
yolo_z <- filter(scored, county_fips == yolo_fips)

socio_comp <- scored |>
  filter(county_fips != yolo_fips,
         urban_rural_sim == yolo_z$urban_rural_sim) |>
  mutate(Distance = sqrt((zI - yolo_z$zI)^2 + (zP - yolo_z$zP)^2 +
                           (zV - yolo_z$zV)^2)) |>
  slice_min(Distance, n = 3, with_ties = FALSE) |>
  mutate(Set = "Socioeconomic match")

# Rule 2: closest store density
density_comp <- county_profile |>
  filter(county_fips != yolo_fips) |>
  mutate(Distance = abs(D_c - yolo_z$D_c)) |>
  slice_min(Distance, n = 3, with_ties = FALSE) |>
  mutate(Set = "Density match")

comparison <- bind_rows(
  county_profile |> filter(county_fips == yolo_fips) |> mutate(Set = "Yolo"),
  socio_comp,
  density_comp
)

report_table(
  comparison |>
    select(Set, county_name, urban_rural_sim, D_c, Income, Poverty, No_Vehicle),
  digits = c(0, 0, 0, 2, 0, 1, 1),
  col.names = c("Set", "County", "Class", "Stores/10k", "Income (USD)",
                "Poverty (%)", "No vehicle (%)"),
  caption = "Comparison counties under two selection rules: profile measures."
)

report_table(
  comparison |>
```

```
    select(Set, county_name, Weighted, Complete_essentials,
           Median_essentials, Burden_pct, Car_switch_pct, No_bus_option_pct),
  digits = c(0, 0, 1, 0, 2, 1, 1, 1),
  col.names = c("Set", "County", "Low access (%)", "Complete stores",
                "Median basket (USD)", "Burden (%)", "Car switch (%)",
                "No bus option (%)"),
  caption = "Comparison counties: access, basket, and transportation outcomes."
)
```

q14-plot-distribution

```
profile_long |>
  filter(!is.na(Value)) |>
  mutate(Yolo = county_fips == yolo_fips) |>
  ggplot(aes(x = Value, y = 0)) +
  geom_point(data = \(d) filter(d, !Yolo), color = "gray55", alpha = 0.6,
             position = position_jitter(width = 0, height = 0.2, seed = 141)) +
  geom_point(data = \(d) filter(d, Yolo), color = "#D55E00", size = 3) +
  facet_wrap(vars(Measure), scales = "free_x", ncol = 2,
             labeller = label_wrap_gen(35)) +
  scale_y_continuous(limits = c(-0.4, 0.4)) +
  labs(
    title = "Where Yolo sits among the 24 study counties",
    subtitle = "Orange = Yolo; gray = other counties (jittered vertically)",
    x = NULL, y = NULL,
    caption = paste(
      "Simulated teaching data. 2024 county/tract profile; April 15, 2025 baskets.",
      "Basket and burden panels include only counties with a complete essentials store.",
      sep = "\n"
    )
  ) +
  theme(axis.text.y = element_blank(), panel.grid.major.y = element_blank(),
        panel.grid.minor.y = element_blank(),
        panel.border = element_rect(fill = NA, color = "gray80"),
        panel.spacing = unit(0.8, "lines"))
```

q14-yolo-stores

```
yolo_stores <- stores_owners |>
  filter(county_fips == yolo_fips) |>
  left_join(
    basket_costs_2025 |>
      filter(basket_id == "essentials_12") |>
      select(store_id, observed, required, basket_cost),
    by = "store_id", relationship = "one-to-one"
  ) |>
  arrange(basket_cost, store_id) |>
```

```
  transmute(
    Store = store_id, Banner = banner_name, Type = store_type,
    SNAP = snap_authorized_sim,
    Coverage = sprintf("%d/%d", observed, required),
    `Essentials (USD)` = basket_cost
  )

report_table(yolo_stores, digits = 2,
             caption = "Yolo study stores and essentials coverage, April 15, 2025.")
```

q14-yolo-tracts

```
yolo_tracts <- tract_q13 |>
  filter(county_fips == yolo_fips) |>
  left_join(
    tract_access |> select(tract_id, tract_population_2024_sim,
                           low_access_residents_2024_sim),
    by = "tract_id", relationship = "one-to-one"
  ) |>
  mutate(Low_access_pct = 100 * low_access_residents_2024_sim /
           tract_population_2024_sim)

report_table(
  yolo_tracts |>
    select(tract_id, tract_population_2024_sim, Low_access_pct,
           households_no_vehicle_2024_sim, min_basket, min_car_total,
           min_bus_total, bus_status),
  digits = c(0, 0, 1, 0, 2, 2, 2, 0),
  col.names = c("Tract", "Residents", "Low access (%)", "No-vehicle HH",
                "Basket (USD)", "+ car (USD)", "+ bus (USD)", "Bus status"),
  caption = "Yolo tracts: access and minimum essentials trip costs (NA = not evaluable)."
)
```

q14-plot-tracts

```
yolo_tracts |>
  select(tract_id, `Basket only` = min_basket,
         `Basket + car` = min_car_total, `Basket + bus` = min_bus_total) |>
  pivot_longer(-tract_id, names_to = "Scenario", values_to = "Cost") |>
  filter(!is.na(Cost)) |>
  ggplot(aes(x = Cost, y = tract_id, color = Scenario, shape = Scenario)) +
  geom_point(size = 2.5) +
  labs(
    title = "Minimum essentials trip cost for each Yolo tract",
    subtitle = "Cheapest eligible recorded candidate under each scenario, April 15, 2025",
    x = "Cost per trip (USD)", y = NULL, color = NULL, shape = NULL,
    caption = "Simulated teaching data. Missing points are scenarios that are not evaluable."
```

```
  ) +
  theme(legend.position = "bottom")
```

q15-audit

```
join_audit <- function(left, right, by, label, relationship) {
  joined <- left_join(left, right, by = by, relationship = relationship)
  tibble(
    Join = label,
    Key = paste(by, collapse = " + "),
    Type = relationship,
    `Key unique (right)` = !anyDuplicated(right[by]),
    `Rows left` = nrow(left),
    `Rows right` = nrow(right),
    `Rows after` = nrow(joined),
    `Unmatched left` = nrow(anti_join(left, right, by = by)),
    `Unmatched right` = nrow(anti_join(right, left, by = by))
  )
}

audits <- bind_rows(
  join_audit(prices, stores, "store_id",
             "Prices to stores", "many-to-one"),
  join_audit(prices, products, "product_id",
             "Prices to products", "many-to-one"),
  join_audit(county_store_counts, county_context, "county_fips",
             "Store counts to county context", "one-to-one"),
  join_audit(county_access |> select(county_fips, Weighted), county_context,
             "county_fips", "Tract summary to county context", "one-to-one"),
  join_audit(travel, tract_access, "tract_id",
             "Travel to tracts", "many-to-one"),
  join_audit(travel, stores, "store_id",
             "Travel to stores", "many-to-one")
)

report_table(audits |> select(Join, Key, Type, `Key unique (right)`),
             caption = "Join keys and intended relationships.")

report_table(audits |> select(Join, starts_with("Rows"), starts_with("Unmatched")),
             digits = 0,
             caption = "Row counts before and after each left join, and unmatched keys.")
```

q15-county-join

```
county_links <- tract_access |>
  select(tract_id, county_fips) |>
```

```
  inner_join(stores |> select(store_id, county_fips),
             by = "county_fips", relationship = "many-to-many")

recorded_links <- travel |> select(tract_id, store_id)

ess_store_ids <- basket_costs_2025 |>
  filter(basket_id == "essentials_12", coverage == 1) |>
  pull(store_id)

link_compare <- tibble(
  Measure = c(
    "Tract-store links",
    "Links that are recorded travel alternatives",
    "Recorded alternatives missing (cross-county)",
    "Mean choices per tract (tracts with any link)",
    "Tracts with at least one complete essentials store"
  ),
  `Recorded links` = c(
    nrow(recorded_links),
    nrow(recorded_links),
    0,
    mean(count(recorded_links, tract_id)$n),
    n_distinct(recorded_links$tract_id[recorded_links$store_id %in% ess_store_ids])
  ),
  `County-only join` = c(
    nrow(county_links),
    nrow(semi_join(county_links, recorded_links, by = c("tract_id", "store_id"))),
    nrow(anti_join(recorded_links, county_links, by = c("tract_id", "store_id"))),
    mean(count(county_links, tract_id)$n),
    n_distinct(county_links$tract_id[county_links$store_id %in% ess_store_ids])
  )
)

report_table(link_compare, digits = 1,
             caption = "Recorded travel links versus a county-only store-tract join.")
```

q15-double-count

```
travel_tracts <- travel |>
  left_join(tract_access, by = "tract_id", relationship = "many-to-one")

pop_compare <- tibble(
  Method = c("Sum over the tract file (one row per tract)",
             "Sum after joining tracts onto travel alternatives"),
  Rows = c(nrow(tract_access), nrow(travel_tracts)),
  Residents = c(sum(tract_access$tract_population_2024_sim),
                sum(travel_tracts$tract_population_2024_sim))
```

```
)

report_table(pop_compare, digits = 0,
             caption = "Total simulated 2024 tract population, computed two ways.")
```

q16-data

```
tract_q16 <- tract_q13 |>
  filter(evaluable) |>
  left_join(
    tract_access |> select(tract_id, tract_poverty_rate_pct_2024_sim),
    by = "tract_id", relationship = "one-to-one"
  ) |>
  left_join(
    county_context |> select(county_fips, urban_rural_sim),
    by = "county_fips", relationship = "many-to-one"
  ) |>
  left_join(
    count(cand, tract_id, name = "eligible_candidates"),
    by = "tract_id", relationship = "one-to-one"
  ) |>
  mutate(
    Poverty = tract_poverty_rate_pct_2024_sim,
    # Extra dollars the car trip adds to the cheapest basket
    Car_penalty = min_car_total - min_basket,
    No_bus_to_cheapest = bus_status %in% c(
      "Basket-cheapest store not on bus", "No bus-accessible eligible store"),
    Poverty_group = factor(ntile(Poverty, 3), levels = 1:3,
                           labels = c("Lower poverty", "Middle", "Higher poverty"))
  )

stopifnot(!anyNA(tract_q16$Poverty), !anyNA(tract_q16$Car_penalty))

variables_q16 <- tribble(
  ~Variable, ~Definition, ~Denominator,
  "Poverty group", "Tercile of tract poverty rate (file 08)", "Evaluable tracts",
  "Car penalty (USD)", "Min (basket + car) minus min basket", "Evaluable tracts",
  "Car changes store (%)", "Cheapest store differs after car trip", "Evaluable tracts in group
  "No bus to cheapest (%)", "Cheapest basket not reachable by bus, or no bus-eligible store",
  "Eligible candidates", "Recorded alternatives with a complete essentials basket", "Per tract
)
report_table(variables_q16, caption = "Variables and denominators for Question 16.")
```

q16-summary

```r
summarise_q16 <- function(d) {
  d |>
    summarise(
      Tracts = n(),
      Median_eligible = median(eligible_candidates),
      Median_poverty = median(Poverty),
      Median_penalty = median(Car_penalty),
      IQR_penalty = IQR(Car_penalty),
      Car_switch_pct = 100 * mean(car_switch),
      No_bus_pct = 100 * mean(No_bus_to_cheapest),
      .groups = "drop"
    )
}

q16_main <- tract_q16 |> group_by(Poverty_group) |> summarise_q16()

report_table(
  q16_main, digits = c(0, 0, 0, 1, 2, 2, 1, 1),
  col.names = c("Group", "Tracts", "Median eligible stores", "Median poverty (%)",
                "Median car penalty (USD)", "IQR penalty", "Car changes store (%)",
                "No bus to cheapest (%)"),
  caption = "Evaluable tracts by poverty tercile, essentials basket, April 15, 2025."
)

rho_all <- cor(tract_q16$Poverty, tract_q16$Car_penalty, method = "spearman")
```

q16-plot

```r
ggplot(tract_q16, aes(x = Poverty_group, y = Car_penalty)) +
  geom_boxplot(outlier.shape = NA, width = 0.5, fill = "gray92") +
  geom_point(aes(color = No_bus_to_cheapest),
             position = position_jitter(width = 0.15, height = 0, seed = 141),
             alpha = 0.7, size = 1.6) +
  scale_color_manual(values = c(`FALSE` = "gray45", `TRUE` = "#D55E00"),
                     labels = c("Cheapest basket reachable by bus", "Not reachable by bus")) +
  labs(
    title = "Car travel adds a similar penalty across tract poverty groups",
    subtitle = "Extra cost of the cheapest essentials trip by car, April 15, 2025",
    x = "Tract poverty tercile (simulated 2024)",
    y = "Car penalty (USD per trip)",
    color = NULL,
    caption = paste0("Simulated teaching data. ", nrow(tract_q16),
                     " tracts with at least one eligible candidate; one point per tract.")
  ) +
  theme(legend.position = "bottom")
```

q16-sensitivity

```
# Robustness: (1) metro tracts only, removing urban/rural differences;
# (2) a median split instead of terciles
q16_metro <- tract_q16 |>
  filter(urban_rural_sim == "Metro") |>
  mutate(Poverty_group = factor(ntile(Poverty, 3), levels = 1:3,
                                labels = c("Lower poverty", "Middle", "Higher poverty"))) |>
  group_by(Poverty_group) |>
  summarise_q16() |>
  mutate(Check = "Metro only (terciles)", .before = 1)

q16_split <- tract_q16 |>
  mutate(Poverty_group = if_else(Poverty >= median(Poverty),
                                 "At/above median", "Below median")) |>
  group_by(Poverty_group) |>
  summarise_q16() |>
  mutate(Check = "All tracts (median split)", .before = 1)

rho_metro <- with(filter(tract_q16, urban_rural_sim == "Metro"),
                  cor(Poverty, Car_penalty, method = "spearman"))

report_table(
  bind_rows(q16_metro, q16_split) |>
    select(Check, Poverty_group, Tracts, Median_penalty, Car_switch_pct, No_bus_pct),
  digits = c(0, 0, 0, 2, 1, 1),
  col.names = c("Check", "Group", "Tracts", "Median penalty (USD)",
                "Car changes store (%)", "No bus to cheapest (%)"),
  caption = "Sensitivity checks for Question 16."
)
```